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Oral or parenteral iron supplementation to reduce deferral, iron deficiency and/or anaemia in blood donors (Review)

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[Intervention Review]

Oral or parenteral iron supplementation to reduce deferral, iron deficiency and/or anaemia in blood donors

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ABSTRACT

Rationale

Iron deficiency is a significant cause of deferral in people wishing to donate blood. If iron removed from the body through blood donation is not replaced, then donors may become iron deficient. All donors are screened at each visit for low haemoglobin (Hb) levels. Some deferred blood donors do not return to donate. Deferred first-time donors are even less likely to return. Interventions that reduce the risk of iron deficiency and anaemia in blood donors will therefore increase the number of blood donations. Currently, iron supplementation for blood donors is not the standard of care in many blood services.

Objectives

To assess the effectiveness and safety of iron supplementation to reduce deferral, iron deficiency and/or anaemia in blood donors.

Search methods

We searched 10 electronic databases (including CENTRAL, MEDLINE, Embase, and Transfusion Evidence Library) and 4 ongoing trial registries on 28 January 2025. We did not restrict our search by language, date, or publication status.

Eligibility criteria

Randomised controlled trials (RCTs) of healthy blood donors, comparing iron supplementation versus no iron, iron in one form compared to another form (e.g. oral versus intravenous (IV)), different doses, or different formulations/preparations.

We excluded autologous blood donors and plasma donors.

Outcomes

Deferral rates due to low Hb, measures of blood indices and iron stores (Hb and ferritin as either serum or plasma ferritin), adverse events, adherence, and health-related quality of life.

Risk of bias

We used Cochrane's RoB 2 tool to assess risk of bias in all studies, including reassessing previously included studies using this updated tool.

Synthesis methods

We combined data using random-effects meta-analyses. We performed sensitivity analyses to assess the impact of trial quality on the results, and subgrouped by population and treatment characteristics to investigate heterogeneity and the impact of these characteristics on response.

Included studies

We identified 8 new RCTs, for a total of 38 RCTs (7475 participants) in the current update.

The studies were conducted in 15 different countries; however, most took place in Europe (20 RCTs, 53%) and the USA (10 RCTs, 26%).

Six studies were exclusively of male donors, and 13 were exclusively female donors. Three studies looked only at first-time donors, while 27 studies excluded them.

Various iron preparations were used, including FE2+ and FE3+ (both with/out additives like vitamin C), with doses ranging from 7 mg of elemental iron per day, up to 1800 mg, although most studies used 25 to 105 mg/day. Treatment duration varied from four days to ongoing for a year (with multiple donations in that year), but most were one to three months.

Synthesis of results

Risk of bias and certainty of the evidence

Risk of bias was often high or with some concerns across multiple domains. High risk of bias was often related to missing outcome data, and some of the potential biases in other domains (some concerns) were likely due to historical reporting standards (lack of information, especially regarding randomisation processes, blinding, and trial protocol/registration).

Overall, we downgraded the certainty of the evidence primarily for risk of bias, high heterogeneity (between-study inconsistency) that was not explained through subgrouping, and imprecision (wide confidence intervals (CIs)), largely due to very small sample sizes in some studies.

Oral iron versus no iron (placebo/standard care): 23 RCTs, N = 5518

The deferral rate due to low Hb (critical outcome) is probably reduced (risk ratio (RR) 0.39, 95% CI 0.20 to 0.76; 7 studies, 1647 participants; moderate-certainty evidence) with oral iron supplementation at the first donation since commencement of treatment when compared to no iron. This is a clinically important effect.

Hb and ferritin may also be improved with oral iron supplementation compared to no iron (Hb: mean difference (MD) 2.52, 95% CI 0.77 to 4.26 g/L higher; 14 studies, 1940 participants; ferritin: MD 12.39, 95% CI 8.16 to 16.62 ng/mL higher; 8 studies, 1335 participants), but the evidence is very uncertain.

More participants may experience adverse events with oral iron compared to no iron, including abdominal pain, constipation, and diarrhoea (RR 1.69, 95% CI 1.15 to 2.47; 8 studies, 2641 participants), but the evidence is very uncertain.

There may be no difference in reported non-gastrointestinal symptoms, except a feeling of “weakness”, which was reduced with oral iron supplements.

IV iron versus no iron (placebo/standard care): 3 RCTs, N = 569

Deferral was not reported by the IV supplementation studies, but Hb is probably higher with IV iron compared to no iron at the first donation since commencement of treatment (MD 8.02, 95% CI 3.22 to 12.83 g/L higher; 3 studies, 558 participants; moderate-certainty evidence). Serum ferritin may also be higher with IV iron at the first donation since commencement of treatment (MD 78.16, 95% CI 37.20 to 119.12 ng/mL higher; 2 studies, 479 participants), but the evidence is very uncertain.

There may be no difference between IV iron and no iron in adverse events, but the evidence is very uncertain.

Authors' conclusions

Iron supplementation may lead to improved blood indices and iron stores, but we are uncertain about the effect due to heterogeneity in the population and different methods and doses of iron supplementation.

Subgrouping (by sex, baseline iron stores, daily elemental iron dose, supplementation duration, and iron preparation) showed some 'low credibility' subgroup differences. Visual inspection of the data (forest plots) suggests more high-quality data may highlight potential differences further in some populations, to target those most in need.

Future studies should assess the impact of supplementation on potentially more vulnerable populations through the stratification of participants by both sex and baseline iron stores; investigate the impact of total and daily doses of iron; and conduct a longer-term assessment of adverse events associated with oral iron supplementation.

Funding

This review had no dedicated funding.

Registration

Protocol and previous versions available via DOIs: 10.1002/14651858.CD009532 and 10.1002/14651858.CD009532.pub2.

PLAIN LANGUAGE SUMMARY

Can giving blood donors iron supplements reduce cases of low iron that stop them from donating, and improve iron stores in their blood?

Key messages

- Taking an oral (by mouth) iron preparation may increase iron stores in the blood compared to not taking an oral iron preparation.
- The higher iron in the blood meant more people could donate that otherwise may have been deferred (stopped from donating because of low iron levels).
- More people taking iron tablets may experience unwanted effects like gut problems while taking the tablets, at least initially, compared to people not taking iron tablets.

What is iron supplementation, and how can it help with blood donation?

Low iron is a common cause of deferral (stopped from donating) in people wishing to donate blood. If iron removed from the body through blood donation is not replaced, then donors may become iron deficient. All donors are screened at each visit for low iron levels. Iron can usually be replenished through a healthy diet, but it may take a long time. Iron supplementation using tablets or injections may reduce this time and help donors return to being able to donate much sooner.

What did we want to find out?

We wanted to find out if giving iron tablets or an injection of iron to potential blood donors would improve the iron stores in their blood, and reduce the chance of them being deferred (stopped) from donating because they had too little iron in their blood.

What did we do?

We searched for studies that compared giving iron supplements (as tablets, an injection, or an infusion (into a vein)) with getting no iron supplementation, as well as comparing different types of iron and different doses of iron. We compared and summarised the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We found 8 new studies, for a total of 38 studies (7475 people). Iron was most commonly given as daily tablets (orally), with only five studies looking at other methods (injections or infusions directly into the blood).

- Iron tablets may increase the iron stored in the blood when taken over a few weeks or months compared to not taking iron tablets. This probably allowed more people to donate at their next attempt (lower deferral rates) (7 studies, 1647 people for deferral).
- More people taking iron tablets experienced unwanted effects, mostly gut problems such as stomach pain, constipation (hard stools), and diarrhoea (loose watery stools), compared to those not taking iron tablets (8 studies, 2641 people).
- There was no difference in non-gut problems (like headaches) between people who took iron tablets and those who didn't, and those that took the iron tablets found they had less "weakness" than those not taking the tablets.

What are the limitations of the evidence?

Our confidence in the evidence ranged from high to very low. In cases where our confidence was limited, this was because some of the studies were very old or very small (not many people were studied), and sometimes the studies did not report on everyone they said they would, and it was possible that people in the studies knew which treatment they were getting. We wanted to look at how different types of iron or different doses worked differently, as well as the impact on men and women separately, and whether they already had low iron stores in their blood, but not all studies gave us this information, so the answers are not as clear as we would like.

How up-to-date is the evidence?

This review updates our previous review, last published in 2014. The evidence is current to January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Oral iron versus none (placebo/standard care) at/before 1st donation since treatment

Comparison 1 (i): Oral iron versus none (placebo/standard care) at/before 1st donation since treatment

Population: blood donors

Setting: outpatient blood donor centres/hospitals

Intervention: oral iron supplementation

Comparison: no iron supplementation (standard care or placebo)

Outcomes	**Anticipated absolute effects*** (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no iron (placebo/standard care)	**Risk with oral iron**				
Deferrals **at/before 1st donation since treatment**	173 per 1000	**67 per 1000** (35 to 131)	**RR 0.39** (0.20 to 0.76)	1647 (7 RCTs)	⊕⊕⊕◯ Moderate[a,b]	Oral iron probably reduces deferrals.
Hb (g/L) **at/before 1st donation since treatment**	The mean Hb (g/L) ranged from **123 to 156** g/L.	MD **2.52 g/L higher** (0.77 higher to 4.26 higher)	-	1940 (14 RCTs)	⊕◯◯◯ Very low[c,d,e]	Oral iron may increase Hb (g/L), but the evidence is very uncertain.
Ferritin (ng/mL) **at/before 1st donation since treatment**	The mean ferritin (ng/mL) ranged from **12.9 to 60.02** ng/mL.	MD **12.39 ng/mL higher** (8.16 higher to 16.62 higher)	-	1335 (8 RCTs)	⊕◯◯◯ Very low[d,f]	Oral iron may increase ferritin (ng/mL), but the evidence is very uncertain.
Serum ferritin (ng/mL) – reported as geometric mean/log-transformed data **at/before 1st donation since treatment**	The mean serum ferritin (ng/mL) – reported as geometric mean/log-transformed data – ranged from **10 to 15** ng/mL.	Ratio of geometric means **4.20 ng/mL higher** (3.34 higher to 5.28 higher)	-	371 (3 RCTs)	⊕◯◯◯ Very low[g,h]	Oral iron may increase serum ferritin (ng/mL) (reported as geometric mean/log-transformed data), but the evidence is very uncertain.
Any adverse events	217 per 1000	**367 per 1000** (250 to 537)	**RR 1.69** (1.15 to 2.47)	2641 (8 RCTs)	⊕◯◯◯ Very low[i,j]	Oral iron may increase any adverse events, but the evidence is very uncertain.

at/before 1st donation since treatment						
Adherence (n/N) **at/before 1st donation since treatment**	835 per 1000	**776 per 1000** (751 to 810)	**RR 0.93** (0.90 to 0.97)	1685 (5 RCTs)	⊕⊕◯◯ Low[k]	Oral iron may result in a slight reduction in adherence (n/N).
Adherence (mean % of tablets consumed) **at/before 1st donation since treatment**	The mean adherence (mean % of tablets consumed) was **96%**.	MD **0%** (5.04 lower to 5.04 higher)	-	145 (1 RCT)	⊕⊕⊕⊕ High	Oral iron results in little to no difference in adherence (mean % of tablets consumed).

***The risk in the intervention group (and its 95% confidence interval) is based on the assumed risk in the comparison group and the relative effect of the intervention (and its 95% CI).**

CI: confidence interval; **Hb:** haemoglobin; **MD:** mean difference; **RCT:** randomised controlled trial; **RR:** risk ratio.

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Comparison 1 (ii) (at/before second donation since commencement of treatment) and Comparison 1 (iii) (after multiple donations) are available in Supplementary material 10*;* Supplementary material 11.

[a]Downgraded once for risk of bias: all but two contributing studies were at high risk of bias overall, and in at least one domain. The two non-high risk of bias studies contributed nearly 50% weight to the result when Cable 1988 was removed (due to non-specific deferral thresholds).

[b]As all heterogeneity is eliminated (from I^2 = 78%) if Cable 1988 is excluded, we have not downgraded for inconsistency, as Cable 1988 used a non-specific deferral guideline, compared to all other studies that specified a threshold. This likely explains the inconsistency in deferral (for other reasons) compared to their results for Hb and ferritin.

[c]Downgraded twice for risk of bias: only two contributing studies were at low risk of bias overall, and four more had some concerns. All others were at high risk of bias, especially with regard to missing outcome data – excluding those who did not complete, or dropped out.

[d]Downgraded once for inconsistency: I^2 = 75% and large variation visible in the forest plot.

[e]Publication bias assessed: visual inspection of funnel plot shows balance around the estimated effect (Figure 1).

[f]Downgraded twice for risk of bias: only two of the contributing studies (27% weight) were at low risk of bias; all other studies had high risk of bias in at least one domain, most commonly missing outcome data where exclusions and dropouts had not been explained.

[g]Downgraded twice for risk of bias: three of the four contributing studies had high risk of bias in at least one domain (commonly missing outcome data) and some concerns in the fourth study.

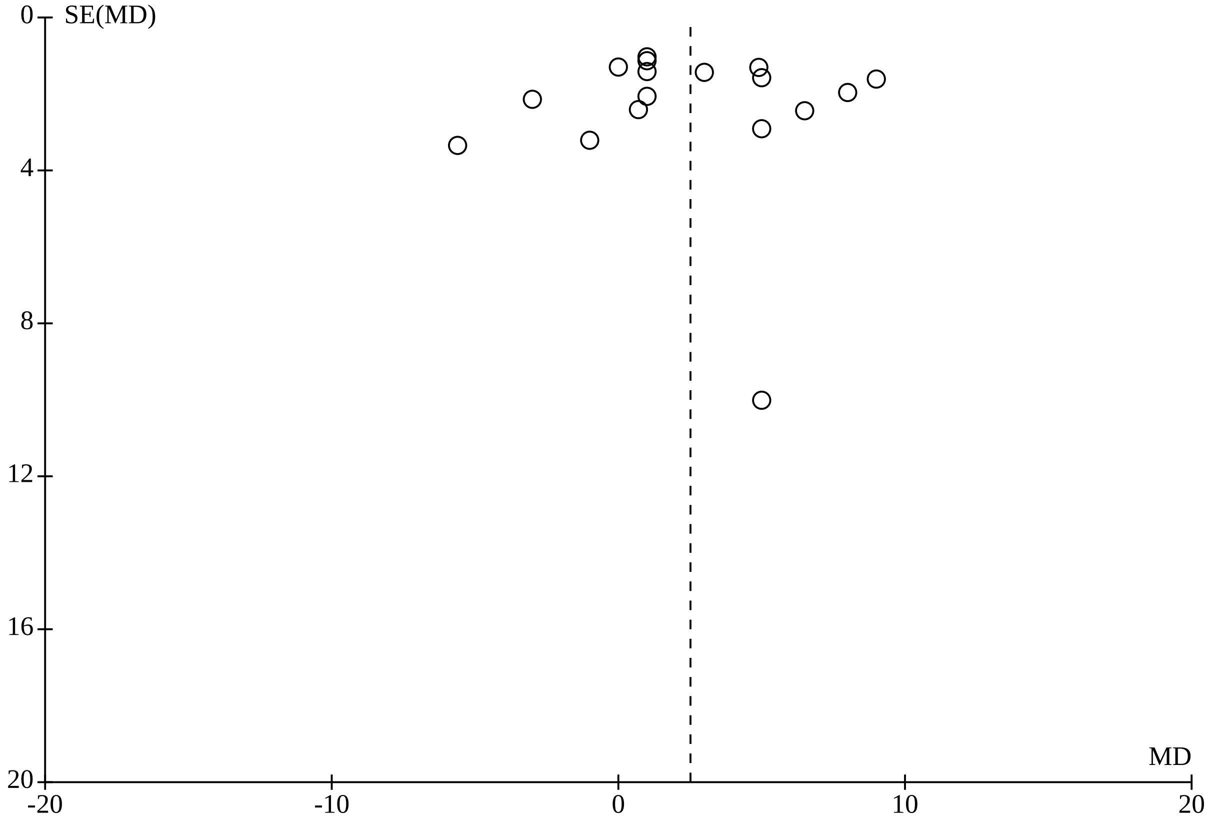
[h]Downgraded once for inconsistency: I^2 = 66%, although plot appears consistent.

[i]Downgraded twice for risk of bias: only two of the contributing studies (approximately 30% weight) were not at high risk of bias overall. Risk of bias was largely due to reporting of the result, as adverse events were often not defined, and no information was provided on how these were collected.

[j]Downgraded twice for inconsistency: I^2 = 86% with no obvious single cause – all studies appear to favour the same direction of effect (although some cross the line of no effect).

[k]Downgraded twice for risk of bias: three of the five contributing studies had some concerns overall, and the other two were at high risk of bias. None of the studies had an adequate trial registration/protocol, and there was a lack of information regarding the randomisation process in four of five studies.

Figure 1. Oral iron vs none: at/before 1st donation since treatment: Number of people deferred from donating due to low Hb. Funnel plot assesses potential publication bias where there are at least 10 studies included in an analysis for a single outcome and time point: there is no suggestion here of publication bias, as data appear to be clustered around the effect estimate, mostly of similarly weighted studies.



Summary of findings 2. Intravenous (IV) iron versus none (placebo/standard care) at/before 1st donation since treatment

Comparison 2 (i): Intravenous (IV) iron versus none (placebo/standard care) at/before 1st donation since treatment
Population: blood donors

Setting: outpatient blood donor centres/hospitals

Intervention: IV iron supplementation

Comparison: no iron supplementation (standard care or placebo)

Outcomes	**Anticipated absolute effects*** (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no iron (placebo/standard care)	**Risk with IV iron**				
Deferrals **at/before 1st donation since treatment** – not reported	-	-	-	-	-	No data were available for this outcome.
Hb (g/L) **at/before 1st donation since treatment**	The mean Hb (g/L) ranged from **121 to 137** g/L.	MD **8.02 g/L higher** (3.22 higher to 12.83 higher)	-	558 (3 RCTs)	⊕⊕⊕◯ Moderate[a]	IV iron probably results in a slight increase in Hb (g/L).
Serum ferritin (ng/mL) **at/before 1st donation since treatment**	The mean serum ferritin (ng/mL) ranged from **13.98 to 28.6** ng/mL.	MD **78.16 ng/mL higher** (37.20 higher to 119.12 higher)	-	479 (2 RCTs)	⊕◯◯◯ Very low[b,c]	IV iron may increase serum ferritin (ng/mL), but the evidence is very uncertain.
Any adverse events **at/before 1st donation since treatment**	132 per 1000	**112 per 1000** (22 to 547)	**RR 0.85** (0.17 to 4.14)	484 (2 RCTs)	⊕◯◯◯ Very low[d,e]	IV iron may have little to no effect on any adverse events, but the evidence is very uncertain.
Adherence (n/N or %) **at/before 1st donation since treatment** – not reported	-	-	-	-	-	No data were available for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **Hb:** haemoglobin; **IV:** intravenous; **MD:** mean difference; **RCT:** randomised controlled trial; **RR:** risk ratio; **SD:** standard deviation.

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Comparison 2 (ii) (at/before second donation since commencement of treatment) is available in Supplementary material 12.
[a]Downgraded once for inconsistency: I^2 = 78%, but visual inspection shows all studies having the same direction of effect, just differences in magnitude and significance of the effect.
[b]Downgraded twice for inconsistency: I^2 = 95%, but visual inspection shows consistency in direction of effect, but not the magnitude of the effect.
[c]Downgraded thrice for imprecision: very wide 95% CI: crosses the expected upper and lower boundaries, calculated as mean +/− 0.5*SD in the control group (MD +/−12.5).
[d]Downgraded twice for inconsistency: I^2 = 59% and visual assessment show different directions of effect, although both cross the null (both absolute and relative effect).
[e]Downgraded twice for imprecision: very wide 95% CI incorporating both significant harm and benefit, and the null.

Summary of findings 3. Oral iron versus intravenous (IV) iron at/before 1st donation since treatment

Comparison 3 (i): Oral iron versus intravenous (IV) iron at/before 1st donation since treatment

Population: blood donors

Setting: outpatient blood donor centres/hospitals

Intervention: oral iron supplementation

Comparison: IV iron supplementation

Outcomes	**Anticipated absolute effects*** (95% CI)		Relative effect (95% CI)	№ of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with IV iron	**Risk with oral iron**				
Deferrals **at/before 1st donation since treatment** – not reported	-	-	-	-	-	No data were available for this outcome.
Hb (g/L) **at/before 1st donation since treatment** – not reported	-	-	-	-	-	No data were available for this outcome.

Plasma ferritin (ng/mL) **at/before 1st donation since treatment**	The mean plasma ferritin (ng/mL) ranged from **32.3 to 33.2** ng/mL.	MD **1.18 ng/mL lower** (9.12 lower to 6.77 higher)	-	120 (1 RCT)	⊕⊕⊕◯ Moderate[a]	Oral iron probably results in little to no difference in plasma ferritin (ng/mL) compared to IV iron.
Any adverse events **at/before 1st donation since treatment**	384 per 1000	**384 per 1000** (261 to 560)	**RR 1.00** (0.68 to 1.46)	172 (1 RCT)	⊕◯◯◯ Very low[b,c]	Oral iron may have little to no effect on any adverse events compared to IV iron, but the evidence is very uncertain.
Adherence (n/N) **at/before 1st donation since treatment**	1000 per 1000	**860 per 1000** (790 to 940)	**RR 0.86** (0.79 to 0.94)	172 (1 RCT)	⊕⊕◯◯ Low[d]	Oral iron may result in a slight reduction in adherence (n/N) compared to IV iron.
Adherence (%) **at/before 1st donation since treatment** – not reported	-	-	-	-	-	No data were available for this outcome.

*__The risk in the intervention group__ (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **Hb:** haemoglobin; **IV:** intravenous; **MD:** mean difference; **RCT:** randomised controlled trial; **RR:** risk ratio.

GRADE Working Group grades of evidence
High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Comparison 3 (ii) (at/before second donation since commencement of treatment) and Comparison 3 (iii) (after multiple donations) are available in Supplementary material 13; Supplementary material 14.

[a]Downgraded once for risk of bias: some concerns regarding randomisation and lack of protocol/trial registration.

[b]Downgraded twice for risk of bias: high risk of bias due to lack of information regarding how adverse events were collected in an unblinded study.

[c]Downgraded twice for imprecision: very wide 95% CI incorporating both significant benefit and harm, and the null.

[d]Downgraded twice for risk of bias: high risk of bias due to different ways of assessing adherence by group: pill counting in oral iron group. No way to assess comparatively in IV iron group, assumed to be compliant once injected.

BACKGROUND

Approximately 120 million units of blood are donated globally each year to meet clinical demand, yet this supply is often insufficient [1]. The World Health Organization (WHO) recommends minimum haemoglobin (Hb) thresholds for whole blood donation of 130 g/L for men and 120 g/L for women [2].

In the United States, the minimum levels are 130 g/L for men and 125 g/L for women. Women with Hb levels between 120 and 125 g/L may be permitted to donate under enhanced protocols approved by the US Food and Drug Administration (FDA) [3]. In Europe, the thresholds are 135 g/L for men and 125 g/L for women [4].

Description of the condition

Voluntary blood donation forms an integral part of the overall healthcare system worldwide in maintaining a steady and safe blood and blood components supply [5, 6, 7] according to needs related to often life-saving treatments in patients undergoing surgery, victims of trauma, patients with leukaemia, sickle cell anaemia, or thalassaemia, and in premature babies [8, 9]. A blood donor is defined as any healthy person who donates blood, plasma, or other blood components voluntarily and without any form of payment, either monetary or otherwise. In this context, it is clear that blood donation is a public health priority [8]. Therefore, it is vital to safeguard donor health.

Healthy blood donors can donate whole blood (a standard donation usually consists of a donation of 450 mL) or can donate plasma or platelets via the aphaeresis procedure (referred to as plasmaphaeresis or plateletphaeresis). To be able to donate blood or blood components, donors must answer a questionnaire regarding their medical history, general health, lifestyle, risk of infection and travel prior to donation, as well as exceed minimum physical requirements such as weight and Hb level [10]. Testing Hb concentration is a routine part of the donor selection process, both to ensure adequate quality of red cell concentrates collected and to protect the potential donor's health. In fact, low Hb is globally the most common reason for deferral of prospective blood donors [11], affecting between 2% to 15% of female compared to 0.5% to 5% of male returning donors in Europe [11], with comparable rates reported in some studies from India [12], but very much higher rates in donors from Africa [13, 14]. In some countries the Hb threshold has been reduced to 110 g/L (Ivory Coast) [15], and in the Caribbean [16]. Donors deferred because of low Hb levels not only reduce the efficiency of collection but are very much less likely to return to donate [17].

One of the most significant physiological impacts of blood donation is iron loss. Each whole blood donation depletes approximately 200 mg to 250 mg of iron bound in Hb [10, 18], representing 8% to 13% of total iron stores in men and non-menstruating women, and up to 81% in menstruating women. Without adequate recovery time or supplementation, frequent donors – particularly premenopausal women and adolescents – are at increased risk of developing iron deficiency anaemia or iron deficiency without anaemia [19]. Data indicate that iron deficiency affects up to 39% of frequently donating premenopausal women, 22% of postmenopausal women, and 9% of male donors [20].

The clinical manifestations of iron deficiency are often non-specific but significant, including fatigue, reduced physical endurance, impaired cognitive function, pica, and restless leg syndrome [21]. Ideally, donors would need to absorb approximately 4.5 mg of iron daily to fully replenish the iron lost in the 56-day minimum interval between donations. However, this requirement exceeds the typical maximum daily iron absorption of 3 mg to 4 mg, posing a challenge for recovery through diet alone (Zalpuri 2019 [22]).

To mitigate these risks, blood services mandate Hb screening at each donation session. The threshold for eligibility in most countries in Europe is 135 g/L for males and 125 g/L for females. Donors who fail to meet these levels are deferred and rescheduled for a future appointment after the standard inter-donation interval (12 weeks for men and 16 weeks for women in England) to allow for physiological iron recovery [23].

Studies show that deferral has a negative impact on future donor returns [24, 25], particularly on first-time donors and those deferred for longer than a year. Deferred donors are less likely to return to donate if unclear or unsatisfactory information is given about the reason for deferral. Many temporarily deferred donors may need to be recalled after the deferral period is over [24, 25]. Understanding and addressing the main reasons for deferrals will help promote the blood donor's well-being and improve the security and resilience of the blood supply [26].

Description of the intervention and how it might work

Strategies to reduce deferral may include ferritin measurements, iron supplementation, and extended donation intervals, either directed to all donors or to specific risk groups.

Iron supplementation aims to increase iron stores in blood donors by increasing the amount of iron available for absorption. Iron stores are regulated through the absorption of iron, and so the supplementation of iron may either directly or indirectly increase iron available for absorption. Recent studies and donor management policies have emphasised the need for individualised approaches, considering both the frequency of donation and baseline ferritin levels [27].

Iron supplementation is available in oral forms (e.g. ferrous sulphate, ferrous fumarate, ferrous gluconate) and intravenous preparations (e.g. iron sucrose, ferric carboxymaltose). Oral iron is typically first-line due to its convenience, over-the-counter availability, and low cost. Effective repletion generally requires 50 mg to 200 mg/day of elemental iron for 3 to 12 weeks. However, only 10% to 20% is absorbed, leaving excess iron in the gastrointestinal tract, which often causes side effects such as constipation, nausea, and abdominal discomfort [28].

Why it is important to do this review

Considering the deferral rates for low Hb and the frequency of shortages, maintaining the health of donors and their availability to donate becomes essential, therefore a systematic review of the efficacy and safety of iron supplementation in the blood donor population is vital.

This review was last updated in 2014, and there have since been a number of randomised controlled trials (RCTs) published assessing the use of iron supplementation for donors.

OBJECTIVES

To assess the effectiveness and safety of iron supplementation to reduce deferral, iron deficiency and/or anaemia in blood donors.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) when conducting the review [29] and PRISMA 2020 for the reporting [30].

Differences between the protocol, previous review, and review update are available in Supplementary material 9.

Criteria for considering studies for this review

Types of studies

RCTs only (including cross-over and cluster-randomised trials, but excluding quasi-randomised trials), irrespective of language or publication status.

Types of participants

Healthy prospective first-time or repeat blood donors: whole blood donors (standard one-unit collection), double-dose red cell donors, and platelet donors (referred to as plateletphaeresis).

Autologous blood donors (donation of blood for the donor's own use) and plasma donors (plasmaphaeresis) were excluded.

Types of interventions

- Iron supplementation versus placebo/control.
- Iron supplementation: oral versus parenteral iron supplementation.
- Iron supplementation versus iron-rich food supplements (fortified foods with a quantifiable amount of iron).
- Iron supplementation, dose A versus iron supplementation, dose B.
- Iron supplementation, treatment duration A versus iron supplementation, treatment duration B.
- Iron supplementation, preparation A versus iron supplementation, preparation B.

Outcome measures

Critical outcomes

- Deferral of blood donors (number of prospective blood donors who are at least temporarily rejected from blood donation) due to low haemoglobin.

The low Hb deferral threshold differs across studies according to the population and sex of the donor studied (see Supplementary material 2).

Important outcomes

- Blood indices and iron stores: including mean levels of Hb, ferritin (as serum or plasma ferritin).
- Adverse events (AEs) from interventions received.
- Adherence (previously Compliance).
- Health-related quality of life (HRQoL), especially changes in cognitive function, 'mood' disturbances, aerobic power, fatigue score, physical activity.

Analyses of blood indices and iron stores were restricted to Hb, serum/plasma ferritin, mean cell volume (MCV), serum/plasma iron, total iron binding capacity (TIBC), and transferrin saturation, although only Hb and ferritin were assessed for the certainty of the evidence (and reported in the summary of findings tables). We noted other reported blood indices and described these in the Supplementary material 2 tables.

Serum and plasma ferritin measurements are effectively equivalent for assessing iron stores, showing a strong correlation and nearly identical results. Serum ferritin represents the ferritin circulating in the blood, while plasma ferritin refers to the same protein within the liquid component of blood. In individuals without inflammation, both reliably reflect total body iron levels [31, 32, 33]. For the purposes of this review, the two are considered interchangeable (although for clarity, we have footnoted any contributing plasma ferritin data in the analyses).

Search methods for identification of studies

The Systematic Review Initiative's (SRI) Information Specialist (CD) formulated the search strategies in collaboration with the Cochrane Injuries Group.

In order to reduce publication and retrieval bias, we did not restrict our search by language, date, or publication status.

Electronic searches

We searched the following for RCTs and systematic reviews from database inception to 28 January 2025. Search strategies used to search the registers are listed in Supplementary material 1.

- Cochrane Central Register of Controlled Trials (CENTRAL; 2024, Issue 12), in the Cochrane Library
- PubMed (NLM, for in process and e-publications ahead of print only)
- MEDLINE (Ovid, 1946 to 28 January 2025)
- Embase (Ovid, 1974 to 28 January 2025)
- CINAHL (Cumulative Index to Nursing and Allied Health Literature) (EBSCOhost, 1981 to 28 January 2025)
- British Nursing Index (NHS Evidence, 1992 to 28 January 2025)
- Transfusion Evidence Library (Evidentia, 1950 to 28 January 2025)
- LILACS (Latin American and Caribbean Health Science Information database) (BIREME, 1980 to 28 January 2025)
- KoreaMed (KAMJE, 1934 to 28 January 2025)
- Web of Science Conference Proceedings Citation Index – Science (CPCI-S, Clarivate, 1990 to 28 January 2025).

We searched the following clinical trials registers.

- ClinicalTrials.gov (www.clinicaltrials.gov)
- ISRCTN register (www.isrctn.com)
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (apps.who.int/trialsearch/)
- Hong Kong Clinical Trials Registry (www.hkclinicaltrials.com)

In MEDLINE, we combined the search strategy with an adaptation of the Cochrane highly sensitive filter for identifying RCTs, as detailed in Chapter 6 of the *Cochrane Handbook for Systematic Reviews of Interventions* [34]. We combined searches in Embase and CINAHL with adaptations of the relevant SIGN RCT filters (www.sign.ac.uk/methodology/filters.html).

Database changes

Two of the main databases previously searched were not searched for this update: IndMed (no longer available) and PakMediNet (not fully functional). One of the ongoing trial databases (UMIN CTR) was not searched for this update as it is not fully functional, while two of the ongoing trial databases listed in the protocol (the Chinese Clinical Trials Registry and the Sri Lanka Clinical Trials Registry) are now included within the WHO ICTRP database.

Searching other resources

Handsearching of reference lists

We checked the references of all identified trials, relevant review articles, and current treatment guidelines for further literature. These searches were limited to the 'first-generation' reference lists. On 12 November 2025, we also searched MEDLINE, Embase, and the Retraction Watch Database (via EndNote 2025: support.clarivate.com/Endnote/s/article/EndNote-20-Retraction-Alerts) for post-publication amendments (including errata, retractions, corrigenda, and expressions of concern) to all included studies and their related secondary published papers.

Personal contacts

We contacted lead authors of relevant studies, study groups, and international experts active in the field to request any unpublished material, missing data, or further information on ongoing trials.

Data collection and analysis

We performed the systematic review using the methods described in Chapter 5 of the *Cochrane Handbook for Systematic Reviews of Interventions* [35]. We ran analyses using RevMan software [36].

Selection of studies

Two review authors (LJG, CK) independently screened the titles and abstracts of citations identified by the electronic searches for potential relevance. We retrieved the full-text reports of citations deemed to be potentially relevant, and the same review authors then independently screened the full texts for inclusion in the review, and recorded the reasons for exclusion of excluded studies. We resolved disagreements through discussion. Arbitration with another review author was unnecessary for this update.

We recorded the study selection process and presented the information in a PRISMA flowchart [37].

Translations of data published in languages other than English were provided by colleagues or individuals responding to calls made via Cochrane resources such as Task Exchange (now Cochrane Engage (engage.cochrane.org/)) for the previous version of this review. None of the newly included studies required translation for this update.

Data extraction and management

Dual extraction was performed independently by review authors (LJG, WM, CK) into a customised form. We piloted these forms with two included RCTs and made subsequent changes to the data extraction form where appropriate and agreed upon by the authors. Throughout the data extraction process, we resolved any disagreements by consensus. The review authors were not blinded to the names of authors, institutions, journals, or the outcomes of the trials. Non-English language publications were translated into English prior to data extraction.

Where data were reported graphically, and we considered the graphs to be of sufficient quality, we estimated values from the graph and used the average value across both estimates. These data have been clearly marked as estimates in all analyses (only two studies provided clear enough graphs, including a clear line of variance for blood measures: Devasthali 1991 [38]; Simon 1984 [39, 40]).

Where data were reported using different scales, we converted these to a standardised scale (e.g. Hb from g/dL or mmol/L to g/L).

We used all relevant sources of data for each study, including full-text publications (with or without supplements), trial registration documents, published protocols, and preliminary results released in the form of abstracts. We used one data extraction form for each unique study. We contacted the study authors for additional details as needed.

Furthermore, one review author (LJG) compared data extraction against the previously published review [41]. We contacted trial authors to request the provision of missing data where possible.

One review author (LJG) entered data into RevMan [36] and performed GRADE assessments [42].

Risk of bias assessment in included studies

Two review authors (LJG, KA) assessed risk of bias in the included studies using Cochrane's RoB 2 tool. Previously published risk of bias assessments using the RoB 1 tool were disregarded, and the updated RoB 2 tool was used for reassessment. We assessed the assignment to the intervention, as we planned to analyse data using intention-to-treat (ITT).

We evaluated the design, conduct, and analysis of the trial using answers to the questions: yes, probably yes, no, probably no, and no information. This results in an assessment for each domain of low risk of bias, high risk of bias, or some concerns. Due to the age of many of the studies, they were often deemed as having some concerns, likely due to historical reporting standards and the lack of explicit information regarding their methods.

- D1: Bias arising from the randomisation process
- D2: Bias due to deviations from the intended interventions
- D3: Bias due to missing outcome data
- D4: Bias in measurement of the outcome
- D5: Bias in selection of the reported result

We explored the impact of bias through sensitivity analyses (see Sensitivity analysis).

Due to the large number of outcomes, with similarities in testing procedures for all blood tests of iron stores, we grouped outcomes as follows.

- Deferral (critical outcome)
- Blood indices and iron stores (measured by standard laboratory tests)
- Adverse events/side effects
- Adherence (compliance)
- HRQoL

When assessing each outcome, we noted any differences in assessment at different time points within the study as appropriate to our analyses, or in different ways adverse events may have been reported/tested.

Measures of treatment effect

We presented dichotomous outcomes as risk ratios (RR) with 95% confidence intervals (CI). Where zero cases were reported in both groups, we have reported these using risk difference (RD) with 95% CI.

For continuous outcomes, we recorded the mean and standard deviation (SD). For continuous outcomes reported on the original scale, the effect measure is the mean difference (MD) with 95% CI.

Where continuous outcomes were measured on a logarithmic scale, we recorded the mean and variance as reported on a log scale; for outcomes measured on a logarithmic scale, the effect measure is the ratio of geometric means with 95% CI. This was only the case for serum/plasma ferritin.

Unit of analysis issues

Within each comparison of interventions of this review, for studies with more than two treatment arms, we avoided multiple pair-wise comparisons of treatment groups by pooling treatment groups as appropriate. For dichotomous variables, we summed count data across groups; for continuous variables, we calculated the mean and SD of the combined group from the mean and SDs of each subgroup.

Thus, for the comparison of iron supplementation versus placebo, we combined multiple iron supplementation trial arms for an overall comparison with the control or placebo arm. Similarly, in the comparison of different iron preparations, we combined different doses of an identical preparation for comparison with an alternative iron preparation.

For studies where results were reported separately for males and females, we used these data to assess the impact of sex by subgrouping. For all other analyses, male and female data were combined. Likewise, where data were reported by high or low serum ferritin (or other measure of baseline iron status), we used these data to assess them separately when investigating that subgroup, and combined when not.

We converted standard errors, P values, and CIs to SDs where necessary. We excluded from analyses those studies in which continuous variables were reported as medians (and range or interquartile range), or geometric means without any measure of variation.

Where data were presented using different units for the same outcome, we converted to a standard measure using widely accepted conversion rates.

Dealing with missing data

Given the time that has elapsed since the publication of the majority of studies, we made no attempt to contact individual study authors or institutions regarding missing data for studies included in the previous iteration of this review. We did attempt to contact trialists for newly included studies.

We recorded the number of participants lost to follow-up for each trial. Where possible, we used data reported on an ITT basis, but if insufficient data were available, we used the reported per-protocol data. Often this was unclear, and this has been reflected in the risk of bias assessment.

We handled missing data using the approach discussed in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [43].

Reporting bias assessment

We made every effort to identify unpublished studies through searching of conference abstracts and ongoing trial databases as described in the Search methods for identification of studies.

We intended to assess publication bias using funnel plots for the critical outcome of deferral due to low Hb for each comparison where there were sufficient data. However, the number of included studies was lower than the minimum suggested for the evaluation of funnel plot asymmetry [44].

We were instead able to assess publication bias for Hb before/at the first donation since treatment, for the oral iron versus no iron comparison, as this is the only outcome that provided a sufficient number of studies for formal evaluation (Figure 1).

However, due to the potential for there to appear to be asymmetry when publication bias does not exist when using funnel plots for continuous outcomes [45], we have interpreted these results with caution.

We did not formally test publication bias using Egger's test for asymmetry [46] because of the same issues surrounding its use with continuous outcomes, especially where there is potential for baseline risk differences. In future reviews, we may be able to perform a modified assessment of funnel asymmetry adjusting for baseline risk using regression residuals, though we have not been able to perform these tests in this update.

Synthesis methods

We performed all meta-analyses using RevMan [36], and testing for subgroup differences. We used the random-effects model for all analyses due to the expected heterogeneity in populations and study designs.

Where possible, we have used restricted maximum likelihood (REML) over DerSimonian and Laird (D-L) for estimating heterogeneity, especially as we noted high heterogeneity in the observed treatment across studies [47]. In RevMan this is only possible when using inverse variance [36]. Mantel-Haenszel (MH) is preferred for the calculation of ratios (RR or odds ratio) when there are many studies to combine but the within-study sample size in each study is small, as when studies have small sample sizes and the number of events is small in these studies, the inverse variance method may not be appropriate [48]. We have thus used D-L for approximating heterogeneity for our dichotomous outcomes using the MH method.

We have used the Wald-type method for calculating CIs, as described in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [43].

Forest plots illustrating these results are shown with 95% CIs for all analyses.

Dichotomous outcomes

We have presented analyses using RR, RD where there were zero cases in both arms, or Peto odds ratio (POR) for rare events (< 5% in each arm), always with 95% CIs. We performed analysis of dichotomous outcomes using MH (random-effects model), heterogeneity using D-L, and CIs using the Wald-type method.

Continuous outcomes

When reported as mean and SD, using the same scale, we analysed using MD with a 95% CI. We performed analysis of continuous outcomes using inverse variance (random-effects model), heterogeneity using REML, and CIs calculated using the Wald-type method.

For serum ferritin, which is likely to be skewed from a normal distribution, some studies reported data on a logarithmic scale. We analysed the data using the generic inverse variance method implemented in RevMan. We analysed studies that did not report on a logarithmic scale separately. While we stated that we were interested in serum ferritin, many studies now also assess plasma ferritin. These have been combined as "ferritin" (and some studies only reported it as this). Where studies stated whether they measured either serum or plasma ferritin, this has been footnoted, or where all data contributing to a single analysis all reported the same measure, we have named it as such.

We assessed all outcomes at three defined time points:

- at/before the first donation since commencement of treatment (this also included studies where there was no follow-up donation, but there was testing);
- at/before the second donation since commencement of treatment;
- after multiple donations.

In addition to the quantitative synthesis described above, we made an overall interpretation of the data based on a qualitative summary of the included studies.

Investigation of heterogeneity and subgroup analysis

We assessed the statistical heterogeneity of treatment effects between trials using Chi^2 tests with a significance level at $P < 0.1$. We used the I^2 statistic to quantify the amount of possible heterogeneity (where $I^2 > 75\%$ denotes considerable heterogeneity). We assessed uncertainty in I^2 values with 95% CIs calculated using the test-based method [49]. We explored potential causes of heterogeneity through sensitivity and subgroup analyses.

We assessed clinical heterogeneity based on individual study characteristics (e.g. by examining differences in study quality, in the donation history and donor characteristics, and in the definition or measurement of outcomes of each study).

We investigated the following subgroups, where there were sufficient data.

- Sex
  - Male
  - Female
  - Not reported (NR)/mixed – *not included when assessing subgroup differences, but included in the overall effect*
- Baseline iron status, as defined by each study
  - Healthy
  - Low Hb/ferritin, iron deficiency anaemia, or non-anaemic iron deficiency
  - Not reported (NR)/mixed – *not included when assessing subgroup differences, but included in the overall effect*
- Daily dose of elemental iron
  - Low: < 25 mg/day (based on doses available as over-the-counter supplements [50])
  - Normal: 25 mg/day to < 105 mg/day (upper limit based on National Institute for Health and Care Excellence (NICE) guidelines for supplementation in iron deficiency anaemia [51])
  - High: 105 mg/day to < 195 mg/day (upper limit based on [52, 53, 54])
  - Very high: 195+ mg/day
- Treatment duration
  - Very short: < 2 weeks
  - Short: 2 weeks to 1 month
  - Normal: 1 to 3 months
  - Long: > 3 months
- Iron preparation, with or without additives (e.g. ascorbic acid, glycerophosphate, bicarbonate)
  - FE2+ (ferrous salts)
  - FE3+ (ferric complex)
  - FE2+ (ferrous salts) with additives
  - FE3+ (ferric complex) with additives
  - Haem iron/combination containing haem iron

Where a subgroup difference was suggested for data where a defined group was possible (e.g. male only compared to female only, not "mixed population" or "not reported", with $P < 0.1$ for the random-effects models), we then assessed the credibility using the ICEMAN criteria for assessing the credibility of subgroups in meta-analysis [55].

Equity-related assessment

We did not investigate equity-related, or indeed ethnic, characteristics in this review. We had not planned such an investigation for this version of the review, as this was not a predefined outcome from our protocol or previous versions of the review. Inspection of the trials included in this review showed no breakdown of outcomes by socio-economic or ethnic group.

However, if sufficient data become available in the future, we may potentially subgroup (or perform sensitivity analyses) according to population and systems demographics, such as a nation's membership of the Organisation for Economic Co-operation and Development (OECD), or by the United Nations (UN) definition of high-, middle-, or lower-income countries. This may be of interest, as some developing countries have lowered the Hb threshold for deferral in the Ivory Coast [15] and Caribbean [16].

If iron supplementation were to be implemented by health services or blood donor centres, we do not believe an individual's economic or other demographic situation would be important, as the focus would be on health indicators over demographics that could be impacted by socio-economic status.

Sensitivity analysis

We assessed the robustness of findings for the RR for low Hb deferral (critical outcome) and adverse events, and MD for Hb using sensitivity analysis, including only those trials at low risk of bias in the randomisation process (Cochrane RoB 2 tool, domain 1).

Information size

For the critical outcome of deferral of blood donors due to low Hb and the important outcome of mean Hb level, we calculated the information size (number of participants required in a meta-analysis to reliably detect an intervention effect) required to detect a significant effect with 90% power. The information size was calculated using trial sequential analysis (TSA) [56]. For deferral due to low Hb (dichotomous outcome), the required information size was based on a D-L random-effects model for a relative risk reduction of 68% (consistent with our observed effect size). We assumed an incidence rate in the control group equal to that observed in our control data (10.6%). The required information size for deferral due to low Hb under these assumptions was 526 participants.

For mean Hb (continuous outcome), we calculated the required information size using a D-L random-effects model with a model variance-based heterogeneity correction assuming an a priori absolute MD of 2.5 g/L. These model assumptions led to an information size for comparing mean Hb levels of 1199 participants.

Certainty of the evidence assessment

We used the GRADE approach to generate summary of findings tables, as suggested in the *Cochrane Handbook for Systematic Reviews of Interventions* [57]. Using GRADEpro GDT software [42], we employed the GRADE approach to rate the certainty of the evidence as high, moderate, low, or very low, according to the five GRADE considerations:

- risk of bias (serious or very serious);
- inconsistency (serious or very serious);
- indirectness (serious or very serious);
- imprecision (serious, very serious, or extremely serious);
- publication bias (suspected or undetected).

We have only presented data in the summary of findings tables for the three main comparisons, where data were available for at least one of the outcomes.

Comparisons included in summary of findings tables

- Oral iron versus none
  - 1.1 at/before first donation since treatment (Summary of findings 1)
  - 1.2 at/before second donation since treatment (Supplementary material 10)
  - 1.3 after multiple donations since treatment (Supplementary material 11)
- Intravenous (IV) iron versus none
  - 2.1 at/before first donation since treatment (Summary of findings 2)
  - 2.2 at/before second donation since treatment (Supplementary material 12)
- Oral iron versus IV iron
  - 3.1 at/before first donation since treatment (Summary of findings 3)
  - 3.2 at/before second donation since treatment (Supplementary material 13)
  - 3.3 after multiple donations since treatment (Supplementary material 14)

Outcomes included in summary of findings tables

- Deferral rate due to low Hb (dichotomous outcome: RR with 95% CI)
- Blood indices and iron stores: Hb (g/L) (continuous outcome: MD with 95% CI)
- Blood indices and iron stores: serum/plasma ferritin (ng/mL) (continuous outcomes: MD with 95% CI or Ratio of Geometric Means (RoGM) with 95% CI for log-transformed data)
- Any adverse events (number of people experiencing any adverse event; dichotomous outcome: RR with 95% CI)
- Adherence (number of people compliant to study/prescribed supplementation; dichotomous outcome: RR with 95% CI)
- Adherence (percentage of tablets consumed/completing prescribed supplementation; continuous outcome: MD with 95% CI)

Assessment of subgroup differences

Where the analyses suggested a subgroup difference ($P < 0.1$ for subgroup differences for the random-effects model), we assessed the credibility that this was a true effect using the ICEMAN criteria [55]. This was not incorporated into the summary of findings tables, and these data have been reported elsewhere.

Consumer involvement

Consumers were not involved in this review, although the review authors did use core outcome sets for the review's outcomes, which were developed with consumer involvement.

RESULTS

Description of studies

See also Supplementary material 2; Supplementary material 3; Supplementary material 4; Supplementary material 5.

Results of the search

A study flow diagram is presented in Figure 2.

Figure 2. PRISMA flow diagram showing study flow, including inclusion of studies from the previous version of review.

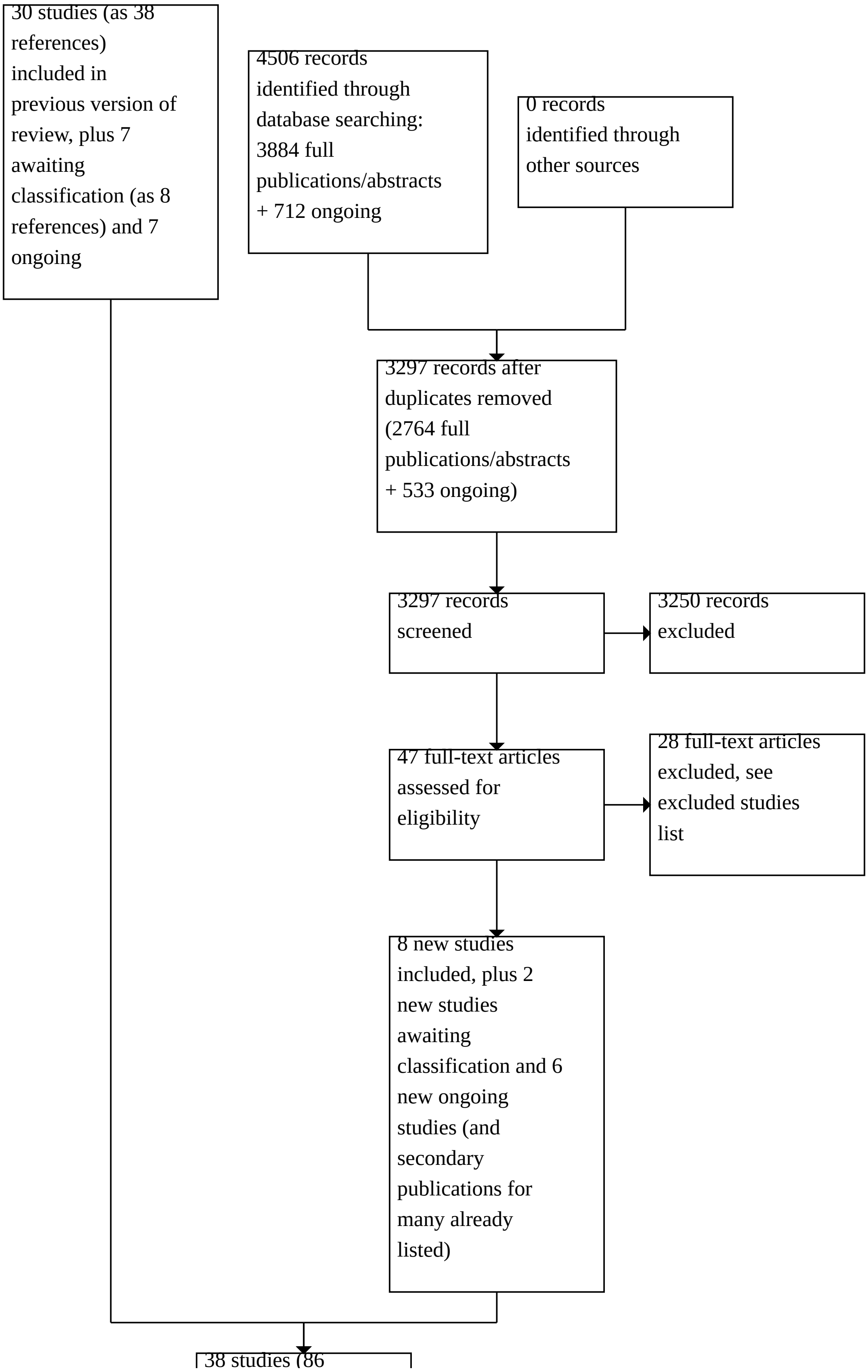
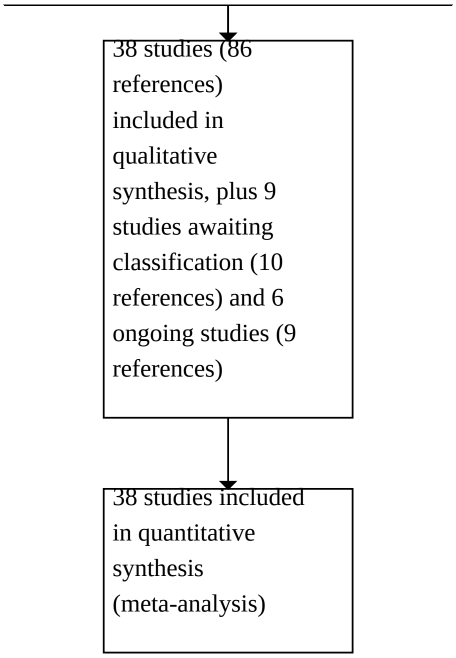


Figure 2. (Continued)



After de-duplication, the update search (January 2025) identified 3297 new references for assessment. We excluded 3250 as irrelevant based on title and abstract, assessing 47 references as full-text publications.

We also checked the included and excluded studies identified in the previous version of this review, and reassessed the studies previously listed as ongoing or awaiting classification.

We finally included eight new studies (38 studies included in total), plus two new studies awaiting classification and six new ongoing studies. There were also additional references that were secondary publications for studies already listed (as included, excluded, ongoing, and awaiting classification).

The search for errata, retractions, corrigenda, and expressions of concern, for all new and previously included studies, revealed an additional publication related to Jacobs 1993 [58, 59]. The corrected data were checked and incorporated into our analyses, where appropriate.

Included studies

See Table 1 for more information, by study and comparison.

Seven references describing six independent studies were translated into the English language for the previous version of the review; we returned to these translations to reassess risk of bias and to check extracted data (Blot 1980 [60]; Bucher 1973 [61]; Busch 1972 [62]; Buzi 1980 [63]; Ehn 1968 [64, 65, 66]; Lindholm 1981 [67]).

Additional data or clarification on the methods were provided by the authors of seven studies (see Acknowledgements).

Study selection

We included 38 RCTs (7475 participants). Some studies had multiple arms, and so contributed to more than one comparison.

- Oral versus none: 23 RCTs, N = 5518
- IV versus none: 3 RCTs, N = 569
- Oral versus IV: 2 RCTs, N = 296
- Oral (different durations): 1 RCT, N = 80
- Oral (different doses): 10 RCTs, N = 1499
- Oral (different preparations): 11 RCTs, N = 2087

Trial registration

Seven trials were prospectively registered on a database prior to commencement of the trial (before randomisation occurred): Gybel-Brask 2018 [68, 69, 70, 71]; Hod 2022 (DIDS) [72, 73, 74, 75, 76]; Kiss 2015 (REDS-III HEIRS) [77, 78, 79, 80, 81, 82, 83]; Macher 2016 (IronWoMan) [84, 85, 86, 87, 88, 89, 90, 91, 92]; Pachikian 2020 [93, 94]; Rosvik 2010 [95, 96]; Waldvogel 2012 [97, 98, 99, 100].

Three trials were registered retrospectively (after randomisation, or later): Fontana 2014a (ISUB) [101, 102, 103]; Marks 2013 (FIRST) [104, 105, 106, 107]; Mast 2016 (STRIDE) [108, 109, 110, 111, 112, 113, 114, 115, 116].

One trial appeared to have a trial registration (Birgegard 2010 [117, 118, 119]), although if it is correctly linked, it changed from IV (50 mg ferric carboxymaltose/Ferinject) as the intervention, to IV (200 mg Venofer). The registration (2009) is not mentioned in the publication, and an earlier abstract (2006) suggests there were no changes in this period, and with no other confirmation that this is the correct registration.

The remaining 27 trials did not have a trial registration, although this is unsurprising for any published before 2005, as it was only then that registration became mandated for RCTs by the International Committee of Medical Journal Editors [120]. Since 2005, only three of our included trials were unregistered: Dara 2017 [121, 122]; Maghsudlu 2008 [123, 124]; Mirrezaie 2008 [125].

Funding

Twelve studies did not report the source of funding or other support for the trial (Blot 1980; Borch-Iohnsen 1993 [126]; Brittenham 1996 [127]; Busch 1972; Buzi 1980; Dara 2017; Devasthali 1991; Frykman 1994 [128]; Gordeuk 1987b [129]; Landucci 1987 [130]; Lindholm 1981; Rybo 1971 [131]).

Of those who did report some assistance, 10 had assistance from commercial/pharmaceutical companies, through the provision of iron supplements only (Borch-Iohnsen 1993; Macher 2016 (IronWoMan); Simon 1984) or more widely involved/unspecified involvement (Birgegard 2010; Bucher 1973; Gybel-Brask 2018; Jacobs 2000 [132]; Lieden 1975 [133]; Radtke 2004a [134, 135]; Radtke 2004b [136]). However, the majority of included studies (17) were funded through charities, academia, or state support, or reported that they had no specific funding (Cable 1988 [137]; Ehn 1968; Fontana 2014a (ISUB); Gordeuk 1987a [138]; Gordeuk 1990 [139, 140]; Hod 2022 (DIDS); Jacobs 1993; Kiss 2015 (REDS-III HEIRS); Linpisarn 1986 [141]; Mackintosh 1988 [142]; Maghsudlu 2008; Marks 2013 (FIRST); Mast 2016 (STRIDE); Mirrezaie 2008; Pachikian 2020; Rosvik 2010; Waldvogel 2012).

Setting

The included studies were published from the 1960s (Ehn 1968) and 70s (Bucher 1973; Busch 1972; Lieden 1975; Rybo 1971) to most recently in the 2020s (Hod 2022 (DIDS); Pachikian 2020).

Most of the included studies were conducted in Europe (20 RCTs, 53%) and the USA (10 RCTs, 26%).

Studies were conducted in 15 different countries:

- Africa: South Africa (3 RCTs);
- Asia: India (1 RCT), Thailand (1 RCT);
- Australia: (1 RCT);
- Europe: Austria (1), Belgium (1), Denmark (1), France (1), Germany (3), Italy (1), Norway (2), Sweden (6), Switzerland (4);
- Middle East: Iran (2 RCTs);
- North America: the USA (10 RCTs).

Participants

Age and sex

Except for those studies of women of child-bearing age, only three studies reported an age restriction on participants, which was from 18 to 56 years (Landucci 1987), from 18 to 65 years (Macher 2016 (IronWoMan)), and from 18 to 25 years (Lieden 1975). Participants in one study were exclusively military service recruits (Ehn 1968).

Two studies did not report the sex of the participants (Busch 1972; Jacobs 2000).

Six studies included exclusively men (Ehn 1968; Lieden 1975; Lindholm 1981; Linpisarn 1986; Mackintosh 1988; Pachikian 2020). For purposes of analysis, we also included two studies that included 95.5% (Dara 2017) and 98.8% (Radtke 2004b) men.

Thirteen studies included exclusively women (Borch-Iohnsen 1993; Brittenham 1996; Cable 1988; Devasthali 1991; Gordeuk 1987a; Gordeuk 1987b; Gordeuk 1990; Gybel-Brask 2018; Maghsudlu 2008; Marks 2013 (FIRST); Mirrezaie 2008; Simon 1984; Waldvogel 2012). For purposes of analysis, we also included the study that had only 96.9% women in the all-female group (Buzi 1980).

Seven studies used a mixed population and did not report data separately for men and women (Bucher 1973; Fontana 2014a (ISUB); Frykman 1994; Jacobs 1993; Landucci 1987; Macher 2016 (IronWoMan); Rybo 1971).

The remaining seven studies stratified or reported data from their mixed population by sex (Birgegard 2010; Blot 1980; Hod 2022 (DIDS); Kiss 2015 (REDS-III HEIRS); Mast 2016 (STRIDE); Radtke 2004a; Rosvik 2010).

Pre-treatment donation and baseline iron status

Seven studies recruited participants who were deferred at an attempted pre-treatment donation (Brittenham 1996; Buzi 1980; Cable 1988; Devasthali 1991; Gordeuk 1987b; Jacobs 1993; Jacobs 2000); six studies did not report whether a pre-treatment donation was attempted (Bucher 1973; Busch 1972; Landucci 1987; Lieden 1975; Linpisarn 1986; Mackintosh 1988) or was specifically not part of the study (Dara 2017; Frykman 1994). The remaining 23 studies recruited people with a successful pre-treatment donation.

Data were reported for those with 'low' baseline iron status in 15 studies. This includes studies that were exclusively targeting low iron, and those that stratified by this. This varied as low Hb, low haematocrit (Hct) or ferritin (with or without low Hb), iron deficiency anaemia, or non-anaemic iron deficiency. When it was not reported, but the study used deferred participants (as above), these were also classed as low baseline iron status.

Baseline iron status (healthy or low) was not reported, and could not be inferred, for five studies (Brittenham 1996; Busch 1972; Lindholm 1981; Marks 2013 (FIRST); Mast 2016 (STRIDE)).

First-time and repeat donors

Three studies included exclusively first-time donors (Ehn 1968; Gybel-Brask 2018; Pachikian 2020); five studies did not report this information (Busch 1972; Buzi 1980; Cable 1988; Devasthali 1991; Frykman 1994); three studies included a mixture of first and repeat donors (Lindholm 1981; Maghsudlu 2008; Waldvogel 2012); and the remaining 27 completely excluded first-time donors, focusing only on repeat donors.

Interventions

Iron preparation

Multiple preparations were used. All IV infusions were a ferric complex (FE3+).

Oral supplementation used FE2+ or FE3+, with or without additives, or haem iron. Additives were most commonly ascorbic acid (vitamin C), but also included bicarbonate, sugar, glycerophosphate, and other undefined additives as part of a "complex".

Ferrous salts (FE2+) were most commonly assessed, used for supplementation in 27 studies (Birgegard 2010; Bucher 1973; Busch 1972; Buzi 1980; Cable 1988; Dara 2017; Devasthali 1991; Ehn 1968; Frykman 1994; Gordeuk 1987a; Gordeuk 1987b; Jacobs 1993; Jacobs 2000; Kiss 2015 (REDS-III HEIRS); Landucci 1987; Lieden 1975; Lindholm 1981; Macher 2016 (IronWoMan); Maghsudlu 2008; Mast 2016 (STRIDE); Mirrezaie 2008; Pachikian 2020; Radtke 2004a; Radtke 2004b; Rosvik 2010; Rybo 1971; Simon 1984).

Ferrous complex (FE3+) was used in 14 studies (Brittenham 1996; Devasthali 1991; Fontana 2014a (ISUB); Gordeuk 1987a; Gordeuk 1987b; Gordeuk 1990; Gybel-Brask 2018; Hod 2022 (DIDS); Jacobs 1993; Jacobs 2000; Landucci 1987; Macher 2016 (IronWoMan); Mackintosh 1988; Marks 2013 (FIRST)).

Haem iron was used in one study only (Frykman 1994: 1.2 mg haem iron from porcine blood mixed with 8 mg FE2+), compared to non-haem iron (60 mg FE2+), meaning it was assessed more as differing doses of elemental iron (18 mg/day versus 60 mg/day) than a comparison of the preparation.

Daily and total iron dose (elemental iron)

IV iron supplementation was offered as a single infusion of varying dosages: 200 mg (Birgegard 2010), 800 mg (Fontana 2014a (ISUB)), or 1000 mg (Gybel-Brask 2018; Hod 2022 (DIDS); Macher 2016 (IronWoMan)).

Oral daily (elemental) iron doses varied from a very low 6.6 mg/day (total 990 mg across the 150-day study period; Borch-Iohnsen 1993), to a very high 1800 mg/day (total dose 37,800 mg over 21 days, Gordeuk 1987a; or 12,600 mg over 7 days, Gordeuk 1987b). However, the very high values were used only in these two studies; otherwise, the highest daily dose was 180 mg/day for 7 days (total dose 1260 mg, Gordeuk 1987a), and 200 mg/day for 14 days (total dose 2800 mg, Rybo 1971), 56 days (total dose 11,200 mg, Mackintosh 1988), or 84 days (total dose 16,800 mg, Jacobs 1993; Jacobs 2000).

Most (26 of 38) studies used daily doses in the 'normal range' of 25 mg/day to 105 mg/day (Birgegard 2010; Blot 1980; Brittenham 1996; Busch 1972; Buzi 1980; Cable 1988; Dara 2017; Devasthali 1991; Ehn 1968; Frykman 1994; Gordeuk 1990; Jacobs 1993; Kiss 2015 (REDS-III HEIRS); Landucci 1987; Lieden 1975; Lindholm 1981; Macher 2016 (IronWoMan); Marks 2013 (FIRST); Mast 2016 (STRIDE); Mirrezaie 2008; Pachikian 2020; Radtke 2004a; Radtke 2004b; Rosvik 2010; Simon 1984; Waldvogel 2012).

The lowest total dose of elemental iron given was 444 mg (111 mg/day for only 4 days, Bucher 1973), 560 mg (20 mg/day for 28 days, Pachikian 2020), and 990 mg (6.6 mg/day for 150 days, Borch-Iohnsen 1993).

Treatment duration

For studies of IV iron supplementation, treatment duration was a single infusion after each donation.

For oral iron supplementation, the treatment course varied from the shortest (four days) to ongoing for a year (study period). Most studies assessed daily supplementation (once or twice per day) for one to three months of treatment.

Outcomes and follow-up

Deferral

Our critical outcome (deferral for low Hb) was reported by only nine studies (Cable 1988; Gordeuk 1990; Jacobs 1993; Kiss 2015 (REDS-III HEIRS); Lindholm 1981; Maghsudlu 2008; Marks 2013 (FIRST); Radtke 2004a; Radtke 2004b).

Blood indices and iron stores: haemoglobin (Hb) and ferritin (serum/plasma)

Hb was the most commonly reported outcome (25 studies: Blot 1980; Borch-Iohnsen 1993; Bucher 1973; Buzi 1980; Cable 1988; Dara 2017; Devasthali 1991; Fontana 2014a (ISUB); Gordeuk 1987a; Gordeuk 1990; Gybel-Brask 2018; Hod 2022 (DIDS); Jacobs 1993; Jacobs 2000; Kiss 2015 (REDS-III HEIRS); Landucci 1987; Lindholm 1981; Linpisarn 1986; Maghsudlu 2008; Marks 2013 (FIRST); Mast 2016 (STRIDE); Pachikian 2020; Rosvik 2010; Simon 1984; Waldvogel 2012), plus two studies where data were unusable due to their being presented graphically with no variance (Ehn 1968), and as median only (Frykman 1994).

Ferritin was reported as plasma ferritin in three studies (Birgegard 2010; Kiss 2015 (REDS-III HEIRS); Mast 2016 (STRIDE); Pachikian 2020). Where data were available, these have been footnoted in the analysis, but were combined as the same measure with serum ferritin data. Serum/plasma ferritin also varied in the manner of reporting: using log-transformed data (or geometric mean) in six studies (Blot 1980; Gordeuk 1987a; Marks 2013 (FIRST); Mast 2016 (STRIDE); Rosvik 2010; Simon 1984).

Six studies reported serum/plasma ferritin in an unusable form for our analyses: change data only, with no baseline (Borch-Iohnsen 1993); scale seems inappropriate when converted to a standard unit, and would not be in a 'normal' range (Devasthali 1991); only available as medians (Frykman 1994; Macher 2016 (IronWoMan)); variance only available as a range (Gordeuk 1990); or no variance reported (Blot 1980).

The remaining studies that reported serum/plasma ferritin were reported using the arithmetic mean, or this was assumed as they did not state otherwise (14 studies in total reported non-log transformed serum/plasma ferritin data in a usable form: Birgegard 2010; Cable 1988; Dara 2017; Fontana 2014a (ISUB); Hod 2022 (DIDS); Jacobs 1993; Jacobs 2000; Kiss 2015 (REDS-III HEIRS); Landucci 1987; Maghsudlu 2008; Mirrezaie 2008; Pachikian 2020; Radtke 2004a; Waldvogel 2012).

Adverse events (AEs)

Any AEs were reported in a usable form (number of people experiencing any AE per group) in 22 studies (Buzi 1980; Dara 2017; Devasthali 1991; Fontana 2014a (ISUB); Frykman 1994; Gordeuk 1987a; Gordeuk 1987b; Gybel-Brask 2018; Hod 2022 (DIDS); Jacobs 1993; Kiss 2015 (REDS-III HEIRS); Landucci 1987; Lindholm 1981; Macher 2016 (IronWoMan); Mackintosh 1988; Maghsudlu 2008; Marks 2013 (FIRST); Mast 2016 (STRIDE); Pachikian 2020; Radtke 2004a; Rybo 1971; Waldvogel 2012). Two studies reported AEs overall, not per intervention group, and so could not be used (Birgegard 2010; Blot 1980).

More detailed AEs were also reported by several studies (e.g. gastrointestinal symptoms, headaches, weakness), although often the data could not be used, as AEs were reported across the entire cohort, not by intervention group (Bucher 1973; Jacobs 2000; Rosvik 2010). The single cross-over RCT unfortunately reported AEs by combining both periods, combining both supplementation and placebo together, so we were unable to extract data showing the impact of iron supplementation on AEs (Radtke 2004b).

Adherence

Adherence (sometimes known as compliance) was reported in multiple ways: percentage of pills consumed, mean number of pills consumed per day, or number of people "compliant" to a minimum requirement. The last included the number of people completing the study period, returning to at least 2/3/4 attempted donations (or testing days), or taking a given percentage of the medication.

Multiple studies did not fully report adherence, or reported it in an unclear manner, including only reporting compliance for the intervention (iron) arm, with no comparator data (when they had no iron/placebo, Kiss 2015 (REDS-III HEIRS); Maghsudlu 2008); giving an overall rate of adherence for the entire cohort (not by group, Blot 1980); giving a range over multiple periods but no mean or other average measure (Birgegard 2010); or providing a percentage of tablets taken per group but with no variance (Bucher 1973).

Health-related quality of life (HRQoL)

HRQoL was only reported in six studies: three compared oral iron supplementation to no iron (or placebo) (Mast 2016 (STRIDE); Pachikian 2020; Waldvogel 2012), and three compared IV iron supplementation to no iron (Fontana 2014a (ISUB); Gybel-Brask 2018; Hod 2022 (DIDS)). Data could not be combined due to the difference in scales used, including varying objective physical function tests (e.g. maximal oxygen consumption (VO_2 max) tests), fatigue, sleep, anxiety, depression, restless leg, and general wellness questionnaires.

Excluded studies

For details, see Supplementary material 3.

We excluded a total of 48 studies (50 references), including those excluded in the previous version of this review, for the reasons listed below.

- Non-RCT: ACTRN12614000430639 [143]; Brugnara 1993 [144]; Bryant 2012b [145, 146]; Chansky 2017 [147]; Eder 2017 [148]; Fontana 2014b [149]; Ghaith 1970 [150]; Gordeuk 1986 [151]; Hoppe 2003 [152]; Ikeda 2021 [153]; Lieden 1975b [154]; Magnussen 2008 [155]; NCT04949165 [156]; Nielsen 1973 [157]; Nielsen 1976 [158]; Remy 1956 [159]; Spencer 2016 [160]; Tobias 1981 [161]; UMIN000047925 [162]
- Ineligible study design: Hoppe 2006 [163]; Rigas 2018 [164]; Russell 2021 [165]; Vhanda 2021 [166]
- Review or commentary (references checked for inclusion): Brittenham 2011 [167]; Sun 2024 [168]; Thijsen 2024 [169]; White 2014 [170]; Zalpuri 2019
- Non-red cell/whole blood donor population: Bier-Ulrich 2003 [171]; Borch-Iohnsen 1989 [172]; Lopes 1999 [173]; TCTR20200614001 [174]
- Intervention not iron supplementation: Baart 2020 [175]; Chiamchanya 2013 [176]; France 2021 [177]; Hasan 2022 [178]; McDonagh 2016 [179]; NCT04514978 [180]; Nwagha 2021 (Ranferon study) [181, 182]; Sweegers 2020 [183]; Tsamesidis 2022 [184]; Xu 2020 [185]; Zanella 1989 [186]
- No relevant comparator: ACTRN12614001078640 [187]; Gordeuk 1986; NCT04250298 [188]; Radtke 2005 [189]
- Unavailable: Aramburu 1991 [190]; Blagoevska 2005 [191]

Studies awaiting classification

We categorised 9 studies (10 references) as awaiting classification (see Table 2; Supplementary material 4). All were available as abstracts or trial registrations only, with insufficient information to include or exclude. One study was terminated prematurely with no results posted and no further information.

Ongoing studies

We listed 6 studies (9 references) as ongoing, despite some expected end dates having passed, as no publications have been identified for these trials, and no results posted (see Table 3 and Supplementary material 5 for more information). All listed studies should theoretically be completed by the end of 2026: five are assessing oral iron supplementation (N = 3700), and one is assessing IV supplementation (N = 19).

Risk of bias in included studies

For further details, see Supplementary material 6.

Deferral

See Figure 3 for more information.

Figure 3. RoB 2: Deferral D1: Randomisation process D2: Deviations from the intended interventions D3: Missing outcome data D4: Measurement of the outcome D5: Selection of the reported result Green: low risk of bias; red: serious (or high) risk of bias; yellow: some concerns (or unclear) risk of bias for each study and domain

Study ID	Experimental	Comparator	Outcome	D1	D2	D3	D4	D5	Overall
Cable 1988	oral ferrous gluconate	oral placebo	Deferral	!	-	+	+	!	-
Gordeuk 1990	Oral carbonyl iron	oral placebo	Deferral	!	-	-	+	!	-
Gybel-Brask 2018	IV iron oligosaccharide complex	IV placebo	Deferral	+	!	-	+	!	-
Jacobs 1993	oral ferric polymatose (daily)	oral ferric polymatose (twice daily)	Deferral	-	!	+	+	!	-
Kiss 2015 (RED-III HEIRS)	Oral ferrous gluconate	oral placebo	Deferral	+	+	+	+	+	+
Lindholm 1981	Oral ferrous sulphate	Oral iron (diff prep)	Deferral	!	-	-	+	!	-
Maghsudlu 2008	Oral ferrous sulphate	oral placebo	Deferral	!	+	-	-	!	-
Marks 2013 (FIRST)	Oral carbonyl iron	oral placebo	Deferral	+	+	+	+	!	!
Mast 2016 (STRIDE)	Oral ferrous gluconate	oral placebo	Deferral	+	+	-	+	!	-
Radtke 2004a	Oral ferrous gluconate	oral placebo	Deferral	!	!	-	+	!	-
Radtke 2004b	Oral glycine-sulphate-complex	oral placebo	Deferral	+	!	-	+	!	-

Eleven RCTs reported deferral rates at at least one time point. Nine of these were assessed as high risk of bias overall, with at least one domain at high risk.

We assessed only one study as low risk across all five domains (Kiss 2015 (REDS-III HEIRS); one other study was low risk across four domains (Marks 2013 (FIRST)), with some concerns only regarding selection of the reported result due to the presence of a trial registration, but lacking in detail.

Flaws arising in the randomisation process (domain 1) were often due to a lack of information regarding how participants were randomised, and what was done to conceal the process. This is likely due to historical reporting standards, except in the case of Jacobs 1993, which used a "computer-generated chart", raising questions as to whether this chart was visible ahead of or during the randomisation. This was the only study assessed as high risk of bias for this domain. Jacobs 1993 also did not blind (unable to) anyone to the allocation.

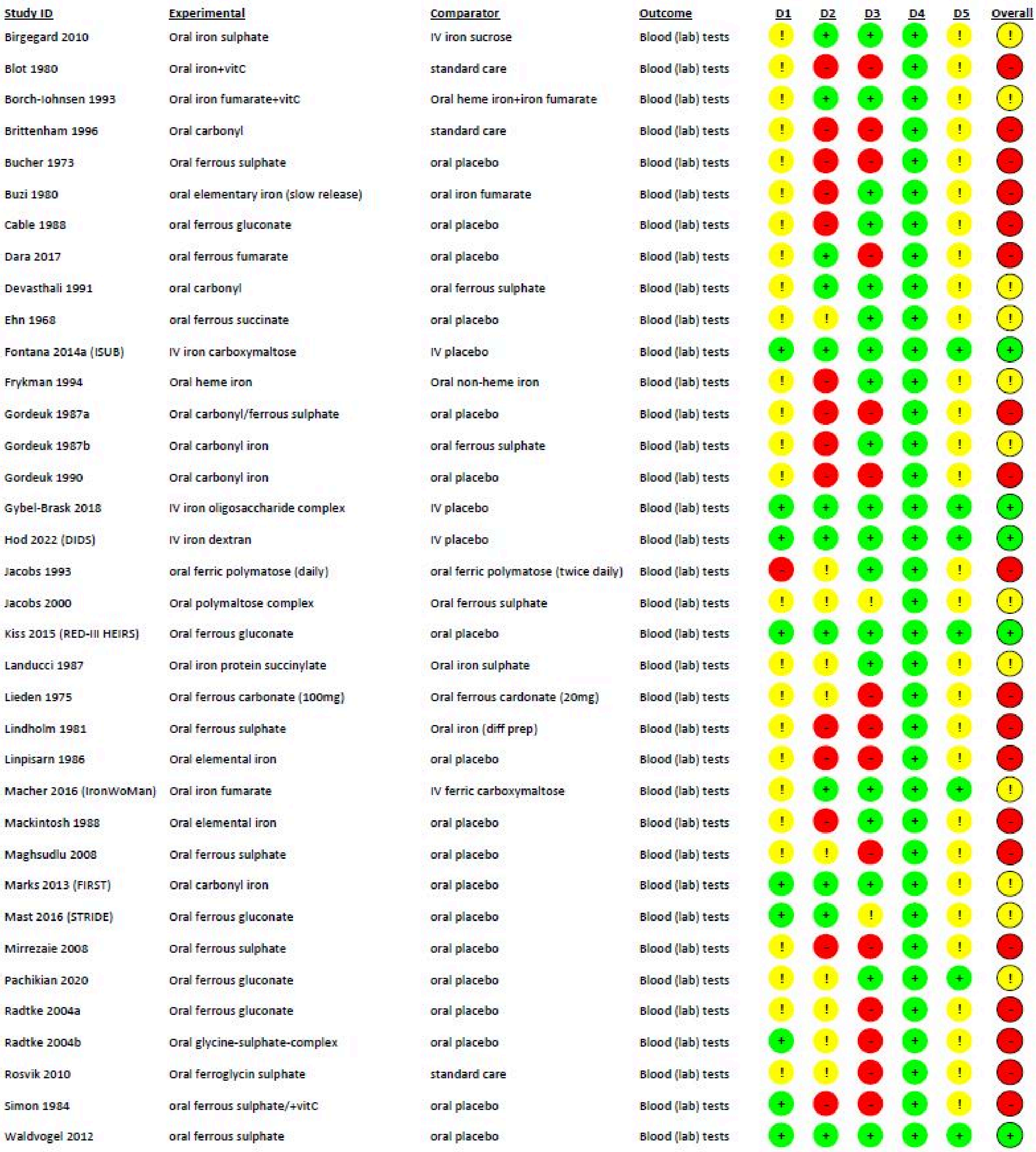
Measurement of the outcome (domain 4) was low risk in all studies except one (Maghsudlu 2008). It was unclear in this study how they had decided to defer: whether they used the same screening test (subjective, visual inspection of a colour scale) or an objective Hb test. The remaining studies stated that they used an objective Hb test (blood test), and so were deemed objectively measured.

All 11 studies (except Kiss 2015 (REDS-III HEIRS)) had limited information in the trial registration, or no trial registration or protocol was identified, causing issues in assessing the selection of the reported result, leading to a rating of some concerns (domain 5).

Blood indices and iron stores: blood (lab) tests

See Figure 4 for more information.

Figure 4. RoB 2: Blood (lab) tests D1: Randomisation process D2: Deviations from the intended interventions D3: Missing outcome data D4: Measurement of the outcome D5: Selection of the reported result Green: low risk of bias; red: serious (or high) risk of bias; yellow: some concerns (or unclear) risk of bias for each study and domain



Study ID	Experimental	Comparator	Outcome	D1	D2	D3	D4	D5	Overall
Birgegard 2010	Oral iron sulphate	IV iron sucrose	Blood (lab) tests	!	+	+	+	!	!
Blot 1980	Oral iron+vitC	standard care	Blood (lab) tests	!	-	-	+	!	-
Borch-Iohnsen 1993	Oral iron fumarate+vitC	Oral heme iron+iron fumarate	Blood (lab) tests	!	+	+	+	!	!
Brittenham 1996	Oral carbonyl	standard care	Blood (lab) tests	!	-	-	+	!	-
Bucher 1973	Oral ferrous sulphate	oral placebo	Blood (lab) tests	!	-	-	+	!	-
Buzi 1980	oral elementary iron (slow release)	oral iron fumarate	Blood (lab) tests	!	-	+	+	!	-
Cable 1988	oral ferrous gluconate	oral placebo	Blood (lab) tests	!	-	+	+	!	-
Dara 2017	oral ferrous fumarate	oral placebo	Blood (lab) tests	!	+	-	+	!	-
Devasthali 1991	oral carbonyl	oral ferrous sulphate	Blood (lab) tests	!	+	+	+	!	!
Ehn 1968	oral ferrous succinate	oral placebo	Blood (lab) tests	!	!	+	+	!	!
Fontana 2014a (ISUB)	IV iron carboxymaltose	IV placebo	Blood (lab) tests	+	+	+	+	+	+
Frykman 1994	Oral heme iron	Oral non-heme iron	Blood (lab) tests	!	-	+	+	!	!
Gordeuk 1987a	Oral carbonyl/ferrous sulphate	oral placebo	Blood (lab) tests	!	-	-	+	!	-
Gordeuk 1987b	Oral carbonyl iron	oral ferrous sulphate	Blood (lab) tests	!	-	+	+	!	!
Gordeuk 1990	Oral carbonyl iron	oral placebo	Blood (lab) tests	!	-	-	+	!	-
Gybel-Brask 2018	IV iron oligosaccharide complex	IV placebo	Blood (lab) tests	+	+	+	+	+	+
Hod 2022 (DIDS)	IV iron dextran	IV placebo	Blood (lab) tests	+	+	+	+	+	+
Jacobs 1993	oral ferric polymatose (daily)	oral ferric polymatose (twice daily)	Blood (lab) tests	-	!	+	+	!	-
Jacobs 2000	Oral polymaltose complex	Oral ferrous sulphate	Blood (lab) tests	!	!	!	+	!	!
Kiss 2015 (RED-III HEIRS)	Oral ferrous gluconate	oral placebo	Blood (lab) tests	+	+	+	+	+	+
Landucci 1987	Oral iron protein succinylate	Oral iron sulphate	Blood (lab) tests	!	!	+	+	!	!
Lieden 1975	Oral ferrous carbonate (100mg)	Oral ferrous cardonate (20mg)	Blood (lab) tests	!	!	-	+	!	-
Lindholm 1981	Oral ferrous sulphate	Oral iron (diff prep)	Blood (lab) tests	!	-	-	+	!	-
Linpisarn 1986	Oral elemental iron	oral placebo	Blood (lab) tests	!	-	-	+	!	-
Macher 2016 (IronWoMan)	Oral iron fumarate	IV ferric carboxymaltose	Blood (lab) tests	!	+	+	+	+	!
Mackintosh 1988	Oral elemental iron	oral placebo	Blood (lab) tests	!	-	+	+	!	-
Maghsudlu 2008	Oral ferrous sulphate	oral placebo	Blood (lab) tests	!	!	-	+	!	-
Marks 2013 (FIRST)	Oral carbonyl iron	oral placebo	Blood (lab) tests	+	+	+	+	!	!
Mast 2016 (STRIDE)	Oral ferrous gluconate	oral placebo	Blood (lab) tests	+	+	!	+	!	!
Mirrezaie 2008	Oral ferrous sulphate	oral placebo	Blood (lab) tests	!	-	-	+	!	-
Pachikian 2020	Oral ferrous gluconate	oral placebo	Blood (lab) tests	!	!	+	+	+	!
Radtke 2004a	Oral ferrous gluconate	oral placebo	Blood (lab) tests	!	!	-	+	!	-
Radtke 2004b	Oral glycine-sulphate-complex	oral placebo	Blood (lab) tests	+	!	-	+	!	-
Rosvik 2010	Oral ferroglycin sulphate	standard care	Blood (lab) tests	!	!	-	+	!	-
Simon 1984	oral ferrous sulphate/+vitC	oral placebo	Blood (lab) tests	+	-	-	+	!	-
Waldvogel 2012	oral ferrous sulphate	oral placebo	Blood (lab) tests	+	+	+	+	+	+

Thirty-six RCTs reported some form of blood test for iron levels. The most commonly reported blood tests were Hb and serum (or plasma) ferritin. As objective measures, all blood tests have been combined here.

Overall risk of bias was high in 19 studies, and had some concerns in 12 studies, leaving only five studies at low risk of bias overall (Fontana 2014a (ISUB); Gybel-Brask 2018; Hod 2022 (DIDS); Kiss 2015 (REDS-III HEIRS); Waldvogel 2012).

Given the objective nature of the tests, the outcome could not be affected even with no blinding, therefore all studies were rated at low risk of bias for measurement of the result (domain 4).

Where studies were rated as high risk of bias, this was due to issues with one or both of domains 2 (deviation from the intended intervention) and 3 (missing outcome data) arising from unclear analyses (ITT or per-protocol (PP)), large exclusions or dropouts with no or insufficient explanation, or unclear participant flow, so occasionally the final sample size for analysis (N) had to be assumed.

We largely assessed domains 1 (randomisation process) and 5 (selection of the reported result) as having some concerns due to lack of information (e.g. no protocol or detailed trial registration, and no details on the randomisation process). Often these limitations were due to historical reporting standards.

Adverse events (AEs)

See Figure 5 for more information.

Figure 5. RoB 2: Adverse events D1: Randomisation process D2: Deviations from the intended interventions D3: Missing outcome data D4: Measurement of the outcome D5: Selection of the reported result Green: low risk of bias; red: serious (or high) risk of bias; yellow: some concerns (or unclear) risk of bias for each study and domain

Study ID	Experimental	Comparator	Outcome	D1	D2	D3	D4	D5	Overall
Birgegard 2010	Oral iron sulphate	IV iron sucrose	Adverse events	!	+	+	-	-	-
Bucher 1973	Oral ferrous sulphate	oral placebo	Adverse events	!	-	-	-	!	-
Busch 1972	Oral iron sulphate	oral placebo	Adverse events	-	-	-	+	-	-
Buzi 1980	oral elementary iron sulphate (slow release)	oral iron fumarate	Adverse events	!	-	+	-	!	-
Dara 2017	oral ferrous fumarate	oral placebo	Adverse events	!	+	-	-	!	-
Devasthali 1991	oral carbonyl	oral ferrous sulphate	Adverse events	!	+	+	+	!	!
Fontana 2014a (ISUB)	IV iron carboxymaltose	IV placebo	Adverse events	+	+	+	+	+	+
Frykman 1994	Oral heme iron	Oral non-heme iron	Adverse events	!	-	+	+	!	-
Gordeuk 1987a	Oral carbonyl/ferrous sulphate	oral placebo	Adverse events	!	+	+	+	!	!
Gordeuk 1987b	Oral carbonyl iron	oral ferrous sulphate	Adverse events	!	+	+	+	-	-
Gordeuk 1990	Oral carbonyl iron	oral placebo	Adverse events	!	-	-	+	!	-
Gybel-Brask 2018	IV iron oligosaccharide complex	IV placebo	Adverse events	+	+	+	+	+	+
Hod 2022 (DIDS)	IV iron dextran	IV placebo	Adverse events	+	+	+	+	+	+
Jacobs 1993	oral ferric polymatose (daily)	oral ferric polymatose (twice daily)	Adverse events	-	+	-	-	!	-
Jacobs 2000	Oral polymaltose complex	Oral ferrous sulphate	Adverse events	!	!	-	-	!	-
Kiss 2015 (RED-III HEIRS)	Oral ferrous gluconate	oral placebo	Adverse events	+	+	+	-	+	-
Landucci 1987	Oral iron protein succinylate	Oral iron sulphate	Adverse events	!	!	+	-	!	-
Lindholm 1981	Oral ferrous sulphate	Oral iron (diff prep)	Adverse events	!	-	+	-	!	-
Macher 2016 (IronWoMan)	Oral iron fumarate	IV ferric carboxymaltose	Adverse events	!	+	+	-	+	-
Mackintosh 1988	Oral elemental iron	oral placebo	Adverse events	!	-	-	-	!	-
Maghsudlu 2008	Oral ferrous sulphate	oral placebo	Adverse events	!	!	-	-	!	-
Marks 2013 (FIRST)	Oral carbonyl iron	oral placebo	Adverse events	+	+	+	-	!	-
Mast 2016 (STRIDE)	Oral ferrous gluconate	oral placebo	Adverse events	+	+	+	+	!	!
Mirrezaie 2008	Oral ferrous sulphate	oral placebo	Adverse events	!	+	+	-	!	-
Pachikian 2020	Oral ferrous gluconate	oral placebo	Adverse events	!	!	+	-	!	-
Radtke 2004a	Oral ferrous gluconate	oral placebo	Adverse events	!	!	-	-	!	-
Radtke 2004b	Oral glycine-sulphate-complex	oral placebo	Adverse events	+	!	-	-	!	-
Rybo 1971	oral ferrous sulphate/sustained release	oral placebo	Adverse events	!	-	+	-	!	-
Waldvogel 2012	oral ferrous sulphate	oral placebo	Adverse events	+	+	+	+	+	+

Twenty-nine RCTs reported AEs or side effects, 23 of which were deemed as at high risk of bias overall. High risk of bias was most commonly identified in the measurement of the outcome (domain 4) due to reporting by unblinded participants, interpretation of AEs by an unblinded investigator, or potential issues regarding long recall, or non-standard reporting forms.

AEs were often not clearly defined, or reporting was done through interview with an outcome assessor aware of the participant's assignment. It is possible that some interviewers held either unconscious or conscious bias in their recording. Studies at low risk of bias for measurement of the outcome provided standardised questionnaires/surveys/diaries for participants to complete, with clear definitions provided for the participants.

High risk of bias for missing outcome data (domain 3) was often due to lack of information about exclusions and dropouts, with issues arising from not fully reporting when people left the study due to side effects, thereby preventing these numbers from being recorded and analysed as such.

As with other outcomes, selection of the reported result (domain 5) was largely deemed as having some concerns as no trial registration or protocol was identified, or was lacking in detail.

Adherence

See Figure 6 for more information.

Figure 6. RoB 2: Adherence D1: Randomisation process D2: Deviations from the intended interventions D3: Missing outcome data D4: Measurement of the outcome D5: Selection of the reported result Green: low risk of bias; red: serious (or high) risk of bias; yellow: some concerns (or unclear) risk of bias for each study and domain

Study ID	Experimental	Comparator	Outcome	D1	D2	D3	D4	D5	Overall
Birgegard 2010	Oral iron sulphate	IV iron sucrose	Compliance	!	+	+	-	!	-
Bucher 1973	Oral ferrous sulphate	oral placebo	Compliance	!	-	-	+	!	-
Cable 1988	oral ferrous gluconate	oral placebo	Compliance	!	-	+	+	!	-
Dara 2017	oral ferrous fumarate	oral placebo	Compliance	!	+	-	+	!	-
Gordeuk 1987a	Oral carbonyl/ferrous sulphate	oral placebo	Compliance	!	+	+	+	!	!
Gordeuk 1987b	Oral carbonyl iron	oral ferrous sulphate	Compliance	!	+	+	+	!	!
Gordeuk 1990	Oral carbonyl iron	oral placebo	Compliance	!	+	+	+	!	!
Kiss 2015 (RED-III HEIRS)	Oral ferrous gluconate	oral placebo	Compliance	+	!	+	-	+	-
Lindholm 1981	Oral ferrous sulphate	Oral iron (diff prep)	Compliance	!	-	+	+	!	-
Macher 2016 (IronWoMan)	Oral iron fumarate	IV ferric carboxymaltose	Compliance	!	+	+	-	+	-
Maghsudlu 2008	Oral ferrous sulphate	oral placebo	Compliance	!	+	-	-	!	-
Marks 2013 (FIRST)	Oral carbonyl iron	oral placebo	Compliance	+	+	+	-	!	-
Mirrezaie 2008	Oral ferrous sulphate	oral placebo	Compliance	!	+	-	!	!	-
Radtke 2004a	Oral ferrous gluconate	oral placebo	Compliance	!	!	-	+	!	-
Radtke 2004b	Oral glycine-sulphate-complex	oral placebo	Compliance	+	!	-	+	!	-
Rybo 1971	oral ferrous sulphate/sustained release	oral placebo	Compliance	!	!	+	+	!	!
Waldvogel 2012	oral ferrous sulphate	oral placebo	Compliance	+	+	+	+	+	+

Seventeen RCTs reported adherence. This was usually as number of people completing the study, or performing a minimum number of follow-up appointments. Alternatively, with oral supplementation, adherence was assessed by pill counting, which was deemed at high risk of bias in unblinded studies, as there is the potential for pills to be discarded and not actually consumed.

Only one study utilised pill counting in an effective way, employing the use of MEMS (Medication Event Monitoring System) bottles that recorded when they were opened (Waldvogel 2012).

We assessed 12 studies as at high risk of bias overall, due to being at high risk in at least one of the five domains. Only one study was at low risk of bias across all domains (and therefore overall) (Waldvogel 2012).

The randomisation process (domain 1) had some concerns due to lack of information available in 13 studies. We deemed four studies as at low risk of bias, as they usually used off-site central randomisation, with a well-described process of allocation concealment (Kiss 2015 (REDS-III HEIRS); Marks 2013 (FIRST); Radtke 2004b; Waldvogel 2012).

As with other outcomes, selection of the reported result (domain 5) was largely deemed as having some concerns, as no trial registration or protocol was identified, or was lacking in detail.

Health-related quality of life (HRQoL)

See Figure 7 and Table 4 for more information.

Figure 7. RoB 2: HRQoL D1: Randomisation process D2: Deviations from the intended interventions D3: Missing outcome data D4: Measurement of the outcome D5: Selection of the reported result Green: low risk of bias; red: serious (or high) risk of bias; yellow: some concerns (or unclear) risk of bias for each study and domain

Study ID	Experimental	Comparator	Outcome	D1	D2	D3	D4	D5	Overall
Fontana 2014a (ISUB)	IV iron carboxymaltose	IV placebo	HRQOL	+	+	+	+	+	+
Gybel-Brask 2018	IV iron oligosaccharide complex	IV placebo	HRQOL	+	+	+	+	+	+
Hod 2022 (DIDS)	IV iron dextran	IV placebo	HRQOL	+	+	+	+	+	+
Mast 2016 (STRIDE)	Oral ferrous gluconate	oral placebo	HRQOL	+	+	+	+	!	!
Pachikian 2020	Oral ferrous gluconate	oral placebo	HRQOL	!	!	+	-	!	-
Waldvogel 2012	oral ferrous sulphate	oral placebo	HRQOL	+	+	+	+	+	+

Six RCTs reported data for HRQoL, including functional exercise tests and questionnaires on mental and physical health.

We assessed four studies as at low risk of bias across all five domains (Fontana 2014a (ISUB); Gybel-Brask 2018; Hod 2022 (DIDS); Waldvogel 2012). One study had some concerns for selection of the reported result (domain 5) due to the trial registration not including a detailed analysis plan (Mast 2016 (STRIDE)).

One study had some concerns in multiple domains, and high risk of bias for measurement of the outcome (domain 4) (Pachikian 2020), as it was only single-blinded, and knowledge of allocation by the outcome assessors may have impacted how they 'encouraged' participants during exercise testing (which can impact the motivation of untrained participants to truly reach maximum exertion).

Synthesis of results

All analyses can be found in Supplementary material 7.

Sensitivity analyses

Before undertaking analyses for all outcomes, we performed sensitivity analyses on our critical outcome (deferral rate), as well as the most commonly reported outcome (Hb), and AEs, to assess the impact of study conduct (risk of bias) (as described in Sensitivity analysis).

Initial analyses included all studies reporting each outcome, for each comparison and time point. We then observed the impact of removing those studies at high risk of bias for the randomisation process (no difference, as none of the studies contributing data to these analyses were at high risk of bias), and then removing those with some concerns, leaving only those studies at low risk of bias for the randomisation process (Table 5).

Comparison 1: Oral iron versus none

We noted some differences in the direction and magnitude of the effect for deferral (time point 3), Hb (time points 2 and 3), and AEs (time points 1 and 2).

For both deferral and Hb, the evidence changed from no difference, with wide CIs, to favouring oral iron.

For AEs, the change was from favouring no iron to there being no statistical difference between iron and no iron, where data remained, but with very wide CIs.

Comparison 2: IV iron versus none

No differences, as all contributing studies were at low risk of bias for the randomisation process.

Comparison 3: Oral iron versus IV iron

As the only contributing study was excluded due to some concerns for randomisation, no data were available for the single outcome and time point reported.

Comparison 1: Oral iron versus no iron (placebo/standard care)

Twenty-three RCTs compared oral iron to no iron (placebo/standard care).

See Table 6, Table 7, and Table 8 for an overview of all the analyses of the outcomes and subgroups available for this comparison. We have discussed here only those outcomes presented in the summary of findings tables. See Summary of findings 1, Supplementary material 10, and Supplementary material 11 for more information on the three time points reported for this comparison (at/before the first donation since commencement of treatment, at/before the second donation since commencement of treatment, and after multiple donations since commencement of treatment.

Deferral

Oral iron supplementation probably improves deferral rates (reduced deferral), seen at the first donation since commencement of treatment (Analysis 1.1: RR 0.39, 95% CI 0.20 to 0.76; I^2 = 78%; 7 studies, 1647 participants; moderate-certainty evidence), and may improve deferral at the second donation since commencement of treatment (Analysis 2.1: RR 0.38, 95% CI 0.16 to 0.91; I^2 = 74%; 4 studies, 936 participants; low-certainty evidence) compared to no iron supplementation. There may be no difference after multiple donations (Analysis 3.1).

The certainty of the evidence was low to moderate, downgraded for risk of bias (commonly missing outcome data and lack of information about randomisation). Heterogeneity was due to a single study in all analyses, likely because of the use of non-specific reasons for deferral (American Red Cross Guidelines 1985 which include other reasons, not associated with Hb, ferritin, or other measure of iron stores in the blood), compared to other studies that specified an Hb threshold.

Blood indices and iron stores

There may be an improvement (increase) in blood indices and iron stores at the first donation since treatment (Hb: Analysis 1.2: MD 2.52, 95% CI 0.77 to 4.26; I^2 = 76%; 14 studies, 1940 participants; very low-certainty evidence; ferritin: Analysis 1.3: MD 12.39, 95% CI 8.16 to 16.62; I^2 = 76%; 8 studies, 1335 participants; very low-certainty evidence; and Analysis 1.4: Ratio of Geometric Means 4.20, 95% CI 3.34 to 5.28; I^2 = 66%; 3 studies, 371 participants; very low-certainty evidence) and after multiple donations (Hb: Analysis 3.2: MD 4.91, 95% CI 0.66 to 9.16; I^2 = 86%; 5 studies, 622 participants; very low-certainty evidence; ferritin: Analysis 3.3: MD 8.28, 95% CI 4.76 to 11.81; I^2 = 0%; 4 studies, 524 participants; low-certainty evidence). There were less consistent results between Hb and ferritin at second donation since treatment: there may be no difference in measures of Hb between iron and no iron (Analysis 2.2), and there may be a difference in measures of ferritin (Analysis 2.3: MD 8.92, 95% CI 3.86 to 13.98; I^2 = 68%; 4 studies, 694 participants; very low-certainty evidence; Analysis 2.4: Ratio of Geometric Means 7.70, 95% CI 5.76 to 10.28; 1 study, 63 participants; very low-certainty evidence).

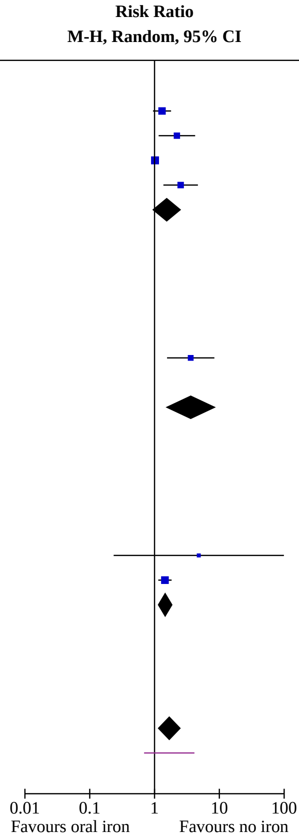
The certainty of the evidence was very low to low, downgraded for risk of bias (commonly missing outcome data and lack of information about randomisation) and high heterogeneity (inconsistency) between studies, sometimes with no obvious cause. Evidence was rarely downgraded for imprecision.

Adverse events (AEs)

AEs were often reported as the number of people experiencing any side effect, or a more specific study definition was given. We pooled these data, finding that there may be a difference in favour of no iron (more of the iron group experienced AEs) at the first donation since commencement of treatment (Analysis 1.9: RR 1.69, 95% CI 1.15 to 2.47; I^2 = 86%; 8 studies, 2641 participants; very low-certainty evidence; Figure 8) and the second donation since the commencement of treatment (Analysis 2.9: RR 3.15, 95% CI 1.54 to 6.46; 1 study, 314 participants; low-certainty evidence). There may be no difference after multiple donations (Analysis 3.9).

Figure 8. Forest plot subgrouped by participant sex assessing whether any side effects were experienced (and reported) by participants in the contributing studies.

Study or Subgroup	oral iron Events	oral iron Total	no iron Events	no iron Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F
1.9.1 Female												
Gordeuk 1987a[a]	40	47	15	23	18.9%	1.30 [0.95 , 1.80]	?	+	+	+	?	?
Maghsudlu 2008[a]	27	185	12	182	13.4%	2.21 [1.16 , 4.23]	?	?	−	−	?	−
Marks 2013 (FIRST)[b]	116	141	114	141	21.5%	1.02 [0.91 , 1.14]	+	+	+	−	?	−
Waldvogel 2012[c]	29	74	11	71	13.9%	2.53 [1.37 , 4.67]	+	+	+	+	+	+
Subtotal (Wald[d])		**447**		**417**	**67.8%**	**1.54 [0.96 , 2.48]**						
Total events:	212		152									
Test for overall effect: Z = 1.78 (P = 0.08)												
Heterogeneity: Tau² (DL[e]) = 0.19; Chi² = 21.65, df = 3 (P < 0.0001); I² = 86%												
1.9.2 Male												
Dara 2017[f]	22	72	6	71	10.6%	3.62 [1.56 , 8.38]	?	+	−	−	?	−
Mackintosh 1988[g]	0	22	0	24		Not estimable	?	−	−	−	?	−
Subtotal		**94**		**95**	**10.6%**	**3.62 [1.56 , 8.38]**						
Total events:	22		6									
Test for overall effect: Z = 3.00 (P = 0.003)												
Heterogeneity: Not applicable												
1.9.3 Mixed												
Kiss 2015 (REDS-III HEIRS)[h]	2	108	0	104	1.5%	4.82 [0.23 , 99.14]	+	+	+	−	+	−
Rybo 1971[i]	219	920	75	456	20.2%	1.45 [1.14 , 1.83]	?	−	+	−	?	−
Subtotal (Wald[d])		**1028**		**560**	**21.7%**	**1.46 [1.15 , 1.85]**						
Total events:	221		75									
Test for overall effect: Z = 3.13 (P = 0.002)												
Heterogeneity: Tau² (DL[e]) = 0.00; Chi² = 0.60, df = 1 (P = 0.44); I² = 0%												
Total (Wald[d])		**1569**		**1072**	**100.0%**	**1.69 [1.15 , 2.47]**						
95% prediction interval						[0.69 , 4.13]						
Total events:	455		233									
Test for overall effect: Z = 2.70 (P = 0.007)												
Test for subgroup differences: Chi² = 4.15, df = 2 (P = 0.13), I² = 51.9%												
Heterogeneity: Tau² (DL[e]) = 0.17; Chi² = 41.63, df = 6 (P < 0.00001); I² = 86%												



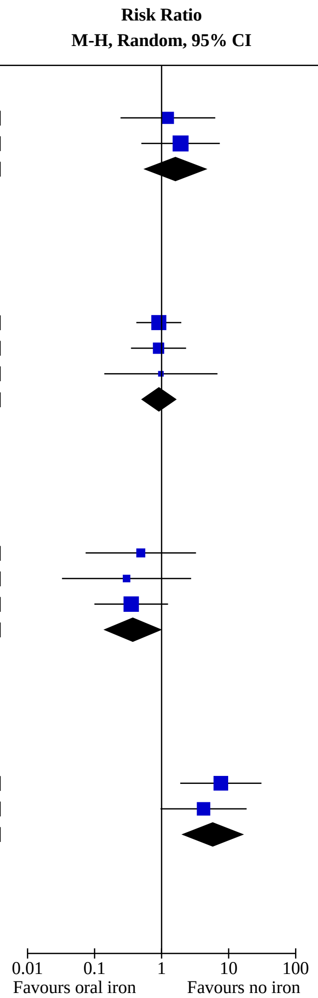
Footnotes
[a]number of pts experiencing any side effects
[b]gastrointestinal complaints in participants as determined by a self-administered diary, phone call at weeks 1,4,8,12
[c]any adverse event
[d]CI calculated by Wald-type method.
[e]Tau² calculated by DerSimonian and Laird method.
[f]number of pts experiencing side effects in first month
[g]Gastrointestinal tract toxicity reflected in anorexia or nausea was specifically enquired for
[h]GI side effects causing discontinuation
[i]subjects with side effects

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Additionally, a number of studies reported more specific side effects. Often, an individual participant experienced more than one type of AE, preventing data pooling. However, we have presented these graphically as non-gastrointestinal side effects (Figure 9; Analysis 1.12) and gastrointestinal side effects (Figure 10; Analysis 1.13).

Figure 9. Oral iron vs none: non-gastrointestinal side effects, at any point in the study period. Forest plot subgrouped by type of non-gastrointestinal (non-GI) side effects experienced (and reported) by contributing studies. An overall effect size was not calculated due to the reporting of multiple side effects by single studies.

	oral iron		no iron			Risk Ratio	Risk of Bias					
Study or Subgroup	Events	Total	Events	Total	Weight	M-H, Random, 95% CI	A	B	C	D	E	F
1.12.1 Any non-GI symptoms												
Mast 2016 (STRIDE)	5	278	2	138	40.7%	1.24 [0.24 , 6.32]	+	+	+	+	?	?
Waldvogel 2012	6	74	3	71	59.3%	1.92 [0.50 , 7.38]	+	+	+	+	+	+
Subtotal (Wald[a])		**352**		**209**	**100.0%**	**1.61 [0.57 , 4.54]**						
Total events:	11		5									
Test for overall effect: Z = 0.90 (P = 0.37)												
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.16, df = 1 (P = 0.69); I² = 0%												
1.12.2 Headache												
Gordeuk 1987a	13	47	7	23	54.8%	0.91 [0.42 , 1.97]	?	+	+	+	?	?
Gordeuk 1990	7	40	7	36	36.5%	0.90 [0.35 , 2.32]	?	−	−	+	?	−
Maghsudlu 2008[c]	2	122	2	119	8.6%	0.98 [0.14 , 6.81]	?	?	−	−	?	−
Subtotal (Wald[a])		**209**		**178**	**100.0%**	**0.91 [0.51 , 1.61]**						
Total events:	22		16									
Test for overall effect: Z = 0.32 (P = 0.75)												
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.01, df = 2 (P = 1.00); I² = 0%												
1.12.3 Weakness												
Gordeuk 1987a	2	47	2	23	25.1%	0.49 [0.07 , 3.26]	?	+	+	+	?	?
Gordeuk 1990	1	40	3	36	18.4%	0.30 [0.03 , 2.76]	?	−	−	+	?	−
Mirrezaie 2008	3	49	8	46	56.5%	0.35 [0.10 , 1.25]	?	+	+	−	?	−
Subtotal (Wald[a])		**136**		**105**	**100.0%**	**0.37 [0.14 , 0.96]**						
Total events:	6		13									
Test for overall effect: Z = 2.04 (P = 0.04)												
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.12, df = 2 (P = 0.94); I² = 0%												
1.12.4 Unpleasant taste												
Gordeuk 1990	17	40	2	36	52.9%	7.65 [1.90 , 30.84]	?	−	−	+	?	−
Mirrezaie 2008	9	49	2	46	47.1%	4.22 [0.96 , 18.53]	?	+	+	−	?	−
Subtotal (Wald[a])		**89**		**82**	**100.0%**	**5.78 [2.10 , 15.95]**						
Total events:	26		4									
Test for overall effect: Z = 3.39 (P = 0.0007)												
Heterogeneity: Tau² (DL[b]) = 0.00; Chi² = 0.33, df = 1 (P = 0.56); I² = 0%												



Test for subgroup differences: Chi² = 16.21, df = 3 (P = 0.001), I² = 81.5%

Footnotes
[a]CI calculated by Wald-type method.
[b]Tau² calculated by DerSimonian and Laird method.
[c]at visit 3

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Figure 10. Oral iron vs none: gastrointestinal side effects, at any point in the study period. Forest plot subgrouped by type of gastrointestinal (GI) side effects experienced (and reported) by contributing studies. An overall effect size was not calculated due to the reporting of multiple side effects by single studies.

Study or Subgroup	oral iron Events	oral iron Total	no iron Events	no iron Total	Weight	Risk Ratio M-H, Random, 95% CI	A	B	C	D	E	F
1.13.1 Any GI symptoms												
Gordeuk 1987a	39	47	8	23	47.5%	2.39 [1.34 , 4.24]	?	+	+	+	?	?
Mast 2016 (STRIDE)	13	278	1	138	12.8%	6.45 [0.85 , 48.83]	+	+	+	+	?	?
Radtke 2004a[a]	18	351	9	175	39.6%	1.00 [0.46 , 2.17]	?	?	−	−	?	−
Subtotal (Wald[b])		**676**		**336**	**100.0%**	**1.92 [0.85 , 4.32]**						
Total events:	70		18									
Test for overall effect: Z = 1.57 (P = 0.12)												
Heterogeneity: Tau² (DL[c]) = 0.28; Chi² = 4.64, df = 2 (P = 0.10); I² = 57%												
1.13.2 Abdominal pain												
Dara 2017[d]	0	72	3	71	3.6%	0.14 [0.01 , 2.68]	?	+	−	−	?	−
Gordeuk 1990[d]	5	40	2	36	12.7%	2.25 [0.46 , 10.89]	?	−	−	+	?	−
Maghsudlu 2008[e]	3	122	2	119	10.1%	1.46 [0.25 , 8.60]	?	?	−	−	?	−
Marks 2013 (FIRST)[f]	19	141	7	141	45.4%	2.71 [1.18 , 6.25]	+	+	+	−	?	−
Mirrezaie 2008[d]	2	49	1	46	5.6%	1.88 [0.18 , 20.01]	?	+	+	−	?	−
Pachikian 2020[g]	0	22	0	11		Not estimable	?	?	+	−	?	−
Waldvogel 2012[f]	7	74	4	71	22.5%	1.68 [0.51 , 5.49]	+	+	+	+	+	+
Subtotal (Wald[b])		**520**		**495**	**100.0%**	**1.97 [1.12 , 3.45]**						
Total events:	36		19									
Test for overall effect: Z = 2.36 (P = 0.02)												
Heterogeneity: Tau² (DL[c]) = 0.00; Chi² = 3.93, df = 5 (P = 0.56); I² = 0%												
1.13.3 Constipation												
Dara 2017[h]	2	72	0	71	1.1%	4.93 [0.24 , 100.94]	?	+	−	−	?	−
Gordeuk 1990	10	40	5	36	10.8%	1.80 [0.68 , 4.77]	?	−	−	+	?	−
Maghsudlu 2008	2	122	1	119	1.8%	1.95 [0.18 , 21.23]	?	?	−	−	?	−
Marks 2013 (FIRST)[i]	16	141	6	141	12.4%	2.67 [1.07 , 6.62]	+	+	+	−	?	−
Mirrezaie 2008	1	49	1	46	1.4%	0.94 [0.06 , 14.57]	?	+	+	−	?	−
Pachikian 2020	2	22	0	11	1.2%	2.61 [0.14 , 50.09]	?	?	+	−	?	−
Rybo 1971	88	920	30	456	64.4%	1.45 [0.98 , 2.17]	?	−	+	−	?	−
Waldvogel 2012[j]	13	74	3	71	7.0%	4.16 [1.24 , 13.98]	+	+	+	+	+	+
Subtotal (Wald[b])		**1440**		**951**	**100.0%**	**1.76 [1.28 , 2.42]**						
Total events:	134		46									
Test for overall effect: Z = 3.46 (P = 0.0005)												
Heterogeneity: Tau² (DL[c]) = 0.00; Chi² = 4.36, df = 7 (P = 0.74); I² = 0%												
1.13.4 Diarrhoea												
Gordeuk 1990	8	40	1	36	4.6%	7.20 [0.95 , 54.79]	?	−	−	+	?	−
Marks 2013 (FIRST)	14	141	5	141	19.3%	2.80 [1.04 , 7.57]	+	+	+	−	?	−
Mirrezaie 2008	0	49	0	46		Not estimable	?	+	+	−	?	−
Pachikian 2020	1	22	1	11	2.7%	0.50 [0.03 , 7.26]	?	?	+	−	?	−
Rybo 1971	57	920	15	456	61.4%	1.88 [1.08 , 3.29]	?	−	+	−	?	−
Waldvogel 2012[k]	9	74	3	71	11.9%	2.88 [0.81 , 10.20]	+	+	+	+	+	+
Subtotal (Wald[b])		**1246**		**761**	**100.0%**	**2.20 [1.42 , 3.40]**						
Total events:	89		25									
Test for overall effect: Z = 3.53 (P = 0.0004)												
Heterogeneity: Tau² (DL[c]) = 0.00; Chi² = 3.20, df = 4 (P = 0.52); I² = 0%												
1.13.5 Flatulence												
Marks 2013 (FIRST)[l]	9	141	7	141	100.0%	1.29 [0.49 , 3.36]	+	+	+	−	?	−
Subtotal		**141**		**141**	**100.0%**	**1.29 [0.49 , 3.36]**						
Total events:	9		7									
Test for overall effect: Z = 0.51 (P = 0.61)												
Heterogeneity: Not applicable												
1.13.6 Heartburn/acid reflux/epigastric pain												
Dara 2017[m]	6	72	2	71	9.4%	2.96 [0.62 , 14.17]	?	+	−	−	?	−
Gordeuk 1990[n]	6	40	2	36	9.8%	2.70 [0.58 , 12.54]	?	−	−	+	?	−
Marks 2013 (FIRST)[o]	2	141	1	141	4.0%	2.00 [0.18 , 21.81]	+	+	+	−	?	−
Mirrezaie 2008[m]	5	49	3	46	12.2%	1.56 [0.40 , 6.18]	?	+	+	−	?	−
Rybo 1971[p]	40	920	14	456	64.5%	1.42 [0.78 , 2.58]	?	−	+	−	?	−

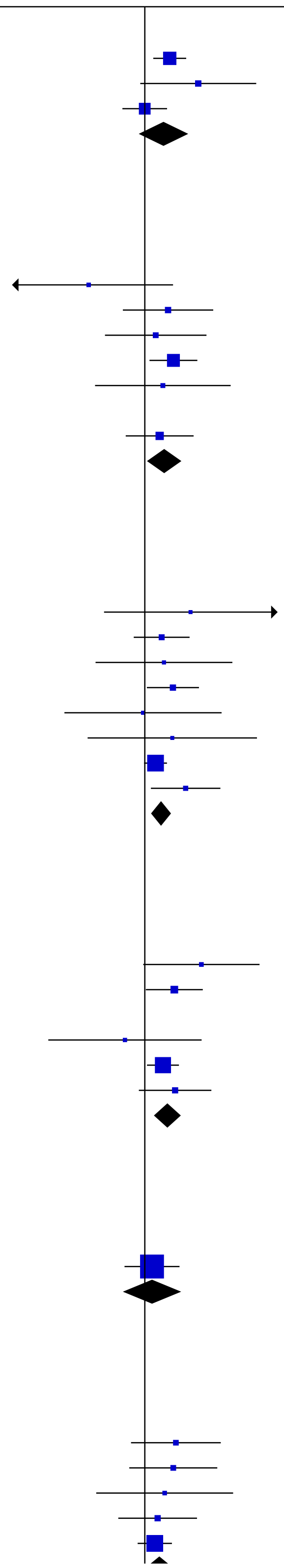
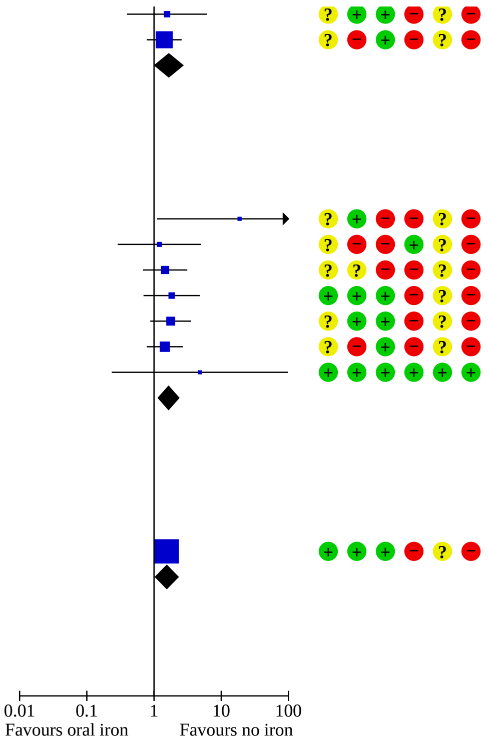


Figure 10. (Continued)

Mirrezaie 2008[m]	5	49	3	46	12.2%	1.56 [0.40 , 6.18]
Rybo 1971[p]	40	920	14	456	64.5%	1.42 [0.78 , 2.58]
Subtotal (Wald[b])		**1222**		**750**	**100.0%**	**1.66 [1.03 , 2.68]**
Total events:	59		22			
Test for overall effect: Z = 2.07 (P = 0.04)						
Heterogeneity: Tau^2 (DL[c]) = 0.00; Chi^2 = 1.21, df = 4 (P = 0.88); I^2 = 0%						
1.13.7 Nausea						
Dara 2017	9	72	0	71	1.5%	18.74 [1.11 , 316.01]
Gordeuk 1990	4	40	3	36	6.0%	1.20 [0.29 , 5.00]
Maghsudlu 2008[q]	15	122	10	119	21.2%	1.46 [0.68 , 3.13]
Marks 2013 (FIRST)[r]	11	141	6	141	13.1%	1.83 [0.70 , 4.82]
Mirrezaie 2008	17	49	9	46	24.9%	1.77 [0.88 , 3.57]
Rybo 1971	38	920	13	456	31.9%	1.45 [0.78 , 2.69]
Waldvogel 2012	2	74	0	71	1.3%	4.80 [0.23 , 98.27]
Subtotal (Wald[b])		**1418**		**940**	**100.0%**	**1.65 [1.16 , 2.33]**
Total events:	96		41			
Test for overall effect: Z = 2.79 (P = 0.005)						
Heterogeneity: Tau^2 (DL[c]) = 0.00; Chi^2 = 4.03, df = 6 (P = 0.67); I^2 = 0%						
1.13.8 Other GI symptoms						
Marks 2013 (FIRST)[s]	48	141	31	141	100.0%	1.55 [1.05 , 2.28]
Subtotal		**141**		**141**	**100.0%**	**1.55 [1.05 , 2.28]**
Total events:	48		31			
Test for overall effect: Z = 2.22 (P = 0.03)						
Heterogeneity: Not applicable						

Test for subgroup differences: Chi^2 = 2.20, df = 7 (P = 0.95), I^2 = 0%



Footnotes

[a]any GI complaint leading to dropout
[b]CI calculated by Wald-type method.
[c]Tau^2 calculated by DerSimonian and Laird method.
[d]abdominal cramps
[e]at visit 3: abdominal cramps
[f]abdominal pain
[g]GI discomfort
[h]constipation leading to dropout
[i]hardened stools also reported in 14/141 vs 7/141
[j]hard stools
[k]liquid stools
[l]abdominal bloating also reported: 14/141 vs 7/141
[m]heartburn
[n]epigastric pain also reported in 3/40 vs 4/36
[o]acid reflux
[p]epigastric pain
[q]at visit 3: nausea and vomiting
[r]vomiting also reported in 5/141 vs 3/141
[s]darkened bowel motions; rectal bleeding also reported by one per group

Risk of bias legend

(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Adherence

Adherence was defined as either the number of people completing the study, returning for a minimum number of visits, or taking a minimum number of tablets. We pooled these data, resulting in low-certainty evidence that there may be greater adherence in the no-iron group at the first donation since treatment (Analysis 1.10: RR 0.93, 95% CI 0.90 to 0.97; I^2 = 0%; 5 studies, 1685 participants; low-certainty evidence), downgraded for high risk of bias. When adherence was assessed as a percentage of pills consumed (1 study), there was high-certainty evidence of no difference between groups at the first donation since treatment (Analysis 1.11) and after multiple donations (Analysis 3.10).

Health-related quality of life (HRQoL)

Only three RCTs reported a measure of HRQoL (Mast 2016 (STRIDE); Pachikian 2020; Waldvogel 2012). These data could not be pooled due to the variation in measures used. This information is presented in Table 9.

There may be no difference in exercise capacity and patient-reported fatigue scores. Patient-reported physical and mental health was reported by one study, and varied depending on the scale used.

Subgrouping by participant (donor) characteristics

Evidence subgrouped by sex is in Analyses 1 to 3 and by baseline iron status in Analyses 4 to 6; both are displayed in Table 6.

Potential subgroup differences were noted in deferrals and measures of blood indices and iron stores (ferritin, MCV, TIBC, and transferrin saturation) when subgrouped by sex or baseline iron status ($P < 0.1$), as well as AEs (subgrouped by sex, $P < 0.1$).

Visual assessment suggests that females (compared to males) and those with low baseline iron stores (compared to healthy baseline stores) may benefit more from supplementation, whereas males may have a greater adverse reaction to oral iron supplementation (Figure 8). These subgroup differences were investigated for credibility (Table 10) and found to be low credibility, largely due to the small number of studies contributing to each subgroup, and not having stratification within each of those studies.

Subgrouping by treatment characteristics

Evidence subgrouped by daily elemental iron dose is in Analyses 7 to 9, by treatment duration in Analyses 10 to 12, and by iron preparation in Analyses 13 to 15, and are displayed in Table 7 and Table 8.

Potential subgroup differences were noted in deferrals (when subgrouped by iron preparation only); some measures of blood indices and iron stores (Hb, ferritin, serum iron, transferrin saturation) when subgrouped by daily dose, duration, and preparation; and AEs when subgrouped by treatment duration, as follows.

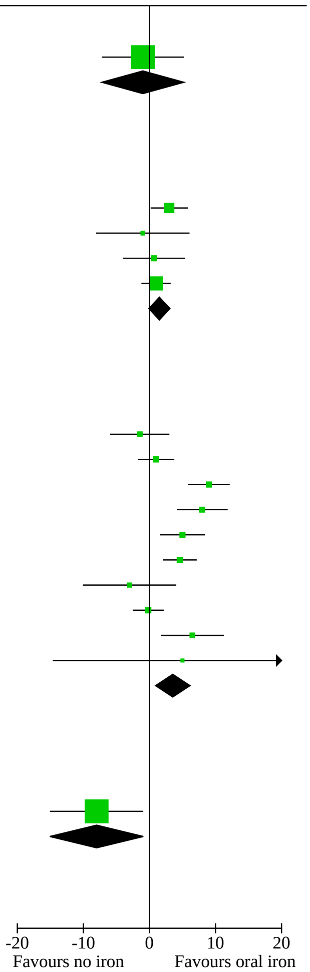
- **Daily dose:** visual assessment suggests there may be the greatest benefit to measures of blood indices and iron stores (Hb, ferritin, transferrin saturation) with supplementation within the normal range (25 mg to 105 mg elemental iron per day; Figure 11). This subgroup also had the largest sample size. There were no potential subgroup differences for AEs.
- **Treatment duration:** visual assessment suggests there may be the greatest benefit to measures of blood indices and iron stores (ferritin, serum iron, transferrin saturation) when supplementation was consumed for 1 to 3 and 3+ months. For AEs, data for up to one month suggested there may be a detriment (increased side effects) with oral iron supplementation compared to no iron, but there may be no difference between groups for supplementation durations longer than one month (1 to 3 and 3+ months; Figure 12), although most AE data were available for studies lasting only two weeks to one month.
- **Iron preparation:** visual assessment suggests there may be a greater benefit to deferral rates and Hb for FE2+ with additives, compared to FE2+ alone. For FE2+ versus FE3+, transferrin saturation appeared to be higher with FE2+, although both formulations had an impact. There were no potential subgroup differences for AEs.

Figure 11. Forest plot subgrouped by the daily dose of iron consumed, and whether this impacted the difference in haemoglobin between intervention and control. Doses were defined as very high (> 195 mg/day), high (105 to 195 mg/day), normal (25 to 195 mg/day), low (< 25 mg/day).

	oral iron			no iron				Mean Difference
Study or Subgroup	**Mean**	**SD**	**Total**	**Mean**	**SD**	**Total**	**Weight**	**IV, Random, 95% CI**
7.2.1 Very high daily dose								
Gordeuk 1987a[a]	125	3.87	15	126	13.08	19	100.0%	-1.00 [-7.20 , 5.20]
Subtotal			**15**			**19**	**100.0%**	**-1.00 [-7.20 , 5.20]**
Test for overall effect: Z = 0.32 (P = 0.75)								
Heterogeneity: Not applicable								
7.2.2 High daily dose								
Bucher 1973	129	5.519	81	126	8	38	32.0%	3.00 [0.19 , 5.81]
Gordeuk 1987a[a]	125	8.25	17	126	13.08	19	5.1%	-1.00 [-8.07 , 6.07]
Linpisarn 1986	151.9	12.2	47	151.2	11.6	51	11.4%	0.70 [-4.02 , 5.42]
Maghsudlu 2008	132.9	9.19	132	131.9	8.76	120	51.6%	1.00 [-1.22 , 3.22]
Subtotal (Wald[b])			**277**			**228**	**100.0%**	**1.50 [-0.09 , 3.10]**
Test for overall effect: Z = 1.85 (P = 0.06)								
Heterogeneity: Tau^2 (REML[c]) = 0.00; Chi^2 = 1.88, df = 3 (P = 0.60); I^2 = 0%								
7.2.3 Normal daily dose								
Blot 1980	146.2	14.056	73	147.67	13.348	70	10.0%	-1.47 [-5.96 , 3.02]
Cable 1988	133	7.55	57	132	8	64	12.3%	1.00 [-1.77 , 3.77]
Dara 2017	137	10	64	128	8	62	11.8%	9.00 [5.84 , 12.16]
Gordeuk 1990	131	8	40	123	9	36	10.9%	8.00 [4.15 , 11.85]
Kiss 2015 (REDS-III HEIRS)	141	11	95	136	13	97	11.5%	5.00 [1.60 , 8.40]
Marks 2013 (FIRST)	134.6	8.7	109	130	9.9	97	12.6%	4.60 [2.04 , 7.16]
Pachikian 2020[a]	153	6.63	11	156	9.95	11	7.0%	-3.00 [-10.07 , 4.07]
Rosvik 2010	135.961	10.246	153	136.149	10.985	161	12.8%	-0.19 [-2.54 , 2.16]
Simon 1984[d]	140.5	9.304	44	134	8.7	19	9.6%	6.50 [1.72 , 11.28]
Waldvogel 2012	135	67	74	130	53	71	1.6%	5.00 [-14.62 , 24.62]
Subtotal (Wald[b])			**720**			**688**	**100.0%**	**3.53 [0.93 , 6.13]**
Test for overall effect: Z = 2.66 (P = 0.008)								
Heterogeneity: Tau^2 (REML[c]) = 12.29; Chi^2 = 40.43, df = 9 (P < 0.00001); I^2 = 79%								
7.2.4 Low daily dose								
Pachikian 2020[a]	148	6.93	12	156	9.95	11	100.0%	-8.00 [-15.07 , -0.93]
Subtotal			**12**			**11**	**100.0%**	**-8.00 [-15.07 , -0.93]**
Test for overall effect: Z = 2.22 (P = 0.03)								
Heterogeneity: Not applicable								

Test for subgroup differences: Chi^2 = 9.90, df = 3 (P = 0.02), I^2 = 69.7%

Mean Difference
IV, Random, 95% CI

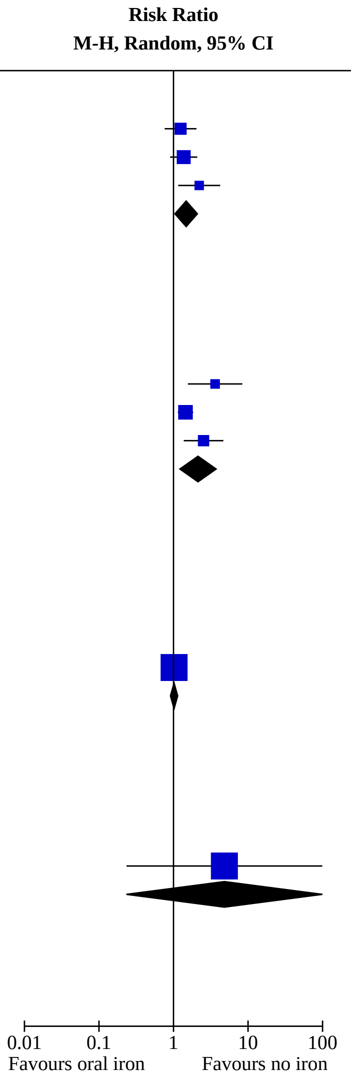


Footnotes

[a]control group NOT split across subgroups, as using subtotals only
[b]CI calculated by Wald-type method.
[c]Tau^2 calculated by Restricted Maximum-Likelihood method.
[d]Data extracted from visual approximation of graphs

Figure 12. Forest plot subgrouped by duration of the iron supplementation assessing whether any side effects were experienced (and reported) by participants in the contributing studies.

Study or Subgroup	oral iron Events	oral iron Total	no iron Events	no iron Total	Weight	Risk Ratio M-H, Random, 95% CI
10.9.1 <2 weeks						
Gordeuk 1987a[a]	19	24	7	11	34.6%	1.24 [0.76 , 2.03]
Gordeuk 1987a[b]	21	23	8	12	42.6%	1.37 [0.90 , 2.08]
Maghsudlu 2008[c]	27	185	12	182	22.8%	2.21 [1.16 , 4.23]
Subtotal (Wald[d])		**232**		**205**	**100.0%**	**1.48 [1.05 , 2.09]**
Total events:	67		27			
Test for overall effect: Z = 2.21 (P = 0.03)						
Heterogeneity: Tau² (DL[e]) = 0.03; Chi² = 2.81, df = 2 (P = 0.25); I² = 29%						
10.9.2 2weeks - 1month						
Dara 2017[f]	22	72	6	71	23.4%	3.62 [1.56 , 8.38]
Rybo 1971[g]	219	920	75	456	45.6%	1.45 [1.14 , 1.83]
Waldvogel 2012[h]	29	74	11	71	31.1%	2.53 [1.37 , 4.67]
Subtotal (Wald[d])		**1066**		**598**	**100.0%**	**2.13 [1.22 , 3.73]**
Total events:	270		92			
Test for overall effect: Z = 2.65 (P = 0.008)						
Heterogeneity: Tau² (DL[e]) = 0.16; Chi² = 6.39, df = 2 (P = 0.04); I² = 69%						
10.9.3 1-3 months						
Mackintosh 1988[i]	0	22	0	24		Not estimable
Marks 2013 (FIRST)[j]	116	141	114	141	100.0%	1.02 [0.91 , 1.14]
Subtotal		**163**		**165**	**100.0%**	**1.02 [0.91 , 1.14]**
Total events:	116		114			
Test for overall effect: Z = 0.31 (P = 0.76)						
Heterogeneity: Not applicable						
10.9.4 3+ months						
Kiss 2015 (REDS-III HEIRS)[k]	2	108	0	104	100.0%	4.82 [0.23 , 99.14]
Subtotal		**108**		**104**	**100.0%**	**4.82 [0.23 , 99.14]**
Total events:	2		0			
Test for overall effect: Z = 1.02 (P = 0.31)						
Heterogeneity: Not applicable						
Test for subgroup differences: Chi² = 10.89, df = 3 (P = 0.01), I² = 72.4%						



Footnotes

[a]180mg/day; number of pts experiencing any side effects; control group split by dose, not across subgroups
[b]1800mg/day; number of pts experiencing any side effects; control group split by dose, not across subgroups
[c]150mg/day; number of pts experiencing any side effects
[d]CI calculated by Wald-type method.
[e]Tau² calculated by DerSimonian and Laird method.
[f]98.6mg/day; number of pts experiencing side effects in first month
[g]200mg/day; subjects with side effects
[h]80mg/day; any adverse event
[i]200mg/day; Gastrointestinal tract toxicity reflected in anorexia or nausea was specifically enquired for
[j]45mg/day; gastrointestinal complaints in participants as determined by a self-administered diary, phone call at weeks 1,4,8,12
[k]37.5mg/day; GI side effects causing discontinuation

As with all potential subgroup assessments for credibility (Table 10), all of these potential subgroup differences were found to have low credibility, largely due to the small number of studies contributing to each subgroup, and not having stratification within each of those studies.

Comparison 2: IV iron versus no iron (placebo/standard care)

Only three RCTs compared IV iron to no iron (placebo/standard care) (Fontana 2014a (ISUB); Gybel-Brask 2018; Hod 2022 (DIDS)).

See Table 11 for an overview of all data and subgroups available for this comparison, and Summary of findings 2 and Supplementary material 12 for more information about the two time points reported for this comparison.

Evidence subgrouped by sex are in Analysis 16.1; Analysis 16.2; Analysis 16.3; Analysis 16.4; Analysis 16.5; Analysis 16.6; Analysis 16.7; Analysis 16.8; Analysis 17.1; Analysis 17.2; Analysis 17.3; Analysis 17.4, and evidence subgrouped by baseline iron status are in Analysis 18.1; Analysis 18.2; Analysis 18.3; Analysis 18.4; Analysis 18.5; Analysis 18.6; Analysis 18.7; Analysis 19.1; Analysis 19.2; Analysis 19.3; Analysis 19.4.

Deferral rates

Deferral was not reported by any IV studies.

Blood indices and iron stores

All blood iron outcomes showed there is probably a benefit with IV iron compared to no iron (Hb: at the first donation since treatment (Analysis 16.1: MD 8.02, 95% CI 3.22 to 12.83; I^2 = 78%; 3 studies, 558 participants; moderate-certainty evidence) and second donation since treatment (Analysis 17.1: MD 13.90, 95% CI 9.50 to 18.30; 1 study, 74 participants; high-certainty evidence); and ferritin at the first donation since treatment (Analysis 16.2: MD 78.16, 95% CI 37.20 to 119.12; I^2 = 95%; 2 studies, 479 participants; very low-certainty evidence)). The certainty of the evidence ranged from very low to high, downgraded due to the heterogeneity (inconsistency) between studies and imprecision (wide CIs).

Potential subgroup differences were noted in Hb and serum ferritin when subgrouped by sex or baseline iron status (P < 0.1). Visual assessment was inconsistent, with no greater benefit in one group or another. These subgroup differences were investigated for credibility (Table 12) and found to be low credibility, largely due to the small number of studies contributing to each subgroup, and not having stratification within each of those studies.

Adverse events (AEs)

AEs were often reported as the number of people experiencing any side effect, or a more specific study definition was given. These data were pooled (showing that there may be no difference between IV iron and no iron at any time point; Analysis 16.7; Analysis 17.4) and could be subgrouped by participant and treatment characteristics. The certainty of the evidence was very low, downgraded for risk of bias (often a subjective outcome, with very few guidelines on how and when to report AEs) and imprecision (wide CIs).

Additionally, a number of studies reported more specific side effects. Often, an individual participant would experience more than one type of AE, preventing data pooling. However, we have presented these graphically as specific side effects (Figure 13; Analysis 16.8).

Figure 13. IV iron vs none: specified side effects, at any point in the study period Forest plot subgrouped by different types of side effects experienced (and reported) by contributing studies. An overall effect size was not calculated due to the reporting of multiple side effects by single studies.

Study or Subgroup	IV iron Events	IV iron Total	no iron Events	no iron Total	Weight	Risk Ratio M-H, Random, 95% CI
16.8.1 Allergic reaction or bronchospasm						
Fontana 2014a (ISUB)[a]	2	203	0	202	100.0%	4.98 [0.24 , 102.99]
Subtotal		**203**		**202**	**100.0%**	**4.98 [0.24 , 102.99]**
Total events:	2		0			
Test for overall effect: Z = 1.04 (P = 0.30)						
Heterogeneity: Not applicable						
16.8.2 Fatigue or insomnia						
Fontana 2014a (ISUB)[a]	4	203	4	202	100.0%	1.00 [0.25 , 3.92]
Subtotal		**203**		**202**	**100.0%**	**1.00 [0.25 , 3.92]**
Total events:	4		4			
Test for overall effect: Z = 0.01 (P = 0.99)						
Heterogeneity: Not applicable						
16.8.3 GI symptoms						
Fontana 2014a (ISUB)[b]	2	203	1	202	44.5%	1.99 [0.18 , 21.77]
Gybel-Brask 2018[c]	0	41	3	41	29.9%	0.14 [0.01 , 2.68]
Hod 2022 (DIDS)[d]	0	39	1	40	25.6%	0.34 [0.01 , 8.14]
Subtotal (Wald[e])		**283**		**283**	**100.0%**	**0.58 [0.11 , 2.90]**
Total events:	2		5			
Test for overall effect: Z = 0.67 (P = 0.50)						
Heterogeneity: Tau^2 (DL[f]) = 0.04; Chi^2 = 2.04, df = 2 (P = 0.36); I^2 = 2%						
16.8.4 Headache						
Gybel-Brask 2018[g]	2	41	4	41	78.9%	0.50 [0.10 , 2.58]
Hod 2022 (DIDS)	0	39	1	40	21.1%	0.34 [0.01 , 8.14]
Subtotal (Wald[e])		**80**		**81**	**100.0%**	**0.46 [0.11 , 1.98]**
Total events:	2		5			
Test for overall effect: Z = 1.04 (P = 0.30)						
Heterogeneity: Tau^2 (DL[f]) = 0.00; Chi^2 = 0.04, df = 1 (P = 0.83); I^2 = 0%						
16.8.5 Neuropathy or taste alterations						
Fontana 2014a (ISUB)[a]	3	203	3	202	74.8%	1.00 [0.20 , 4.87]
Hod 2022 (DIDS)[h]	1	39	1	40	25.2%	1.03 [0.07 , 15.83]
Subtotal (Wald[e])		**242**		**242**	**100.0%**	**1.00 [0.25 , 3.96]**
Total events:	4		4			
Test for overall effect: Z = 0.00 (P = 1.00)						
Heterogeneity: Tau^2 (DL[f]) = 0.00; Chi^2 = 0.00, df = 1 (P = 0.99); I^2 = 0%						
16.8.6 Palpitations						
Fontana 2014a (ISUB)[a]	1	203	0	202	100.0%	2.99 [0.12 , 72.85]
Subtotal		**203**		**202**	**100.0%**	**2.99 [0.12 , 72.85]**
Total events:	1		0			
Test for overall effect: Z = 0.67 (P = 0.50)						
Heterogeneity: Not applicable						
16.8.7 Rash						
Fontana 2014a (ISUB)[a]	1	203	1	202	100.0%	1.00 [0.06 , 15.80]
Subtotal		**203**		**202**	**100.0%**	**1.00 [0.06 , 15.80]**
Total events:	1		1			
Test for overall effect: Z = 0.00 (P = 1.00)						
Heterogeneity: Not applicable						
16.8.8 Respiratory symptoms						
Fontana 2014a (ISUB)[a]	2	203	0	202	100.0%	4.98 [0.24 , 102.99]
Subtotal		**203**		**202**	**100.0%**	**4.98 [0.24 , 102.99]**
Total events:	2		0			
Test for overall effect: Z = 1.04 (P = 0.30)						
Heterogeneity: Not applicable						

Test for subgroup differences: Chi^2 = 4.11, df = 7 (P = 0.77), I^2 = 0%

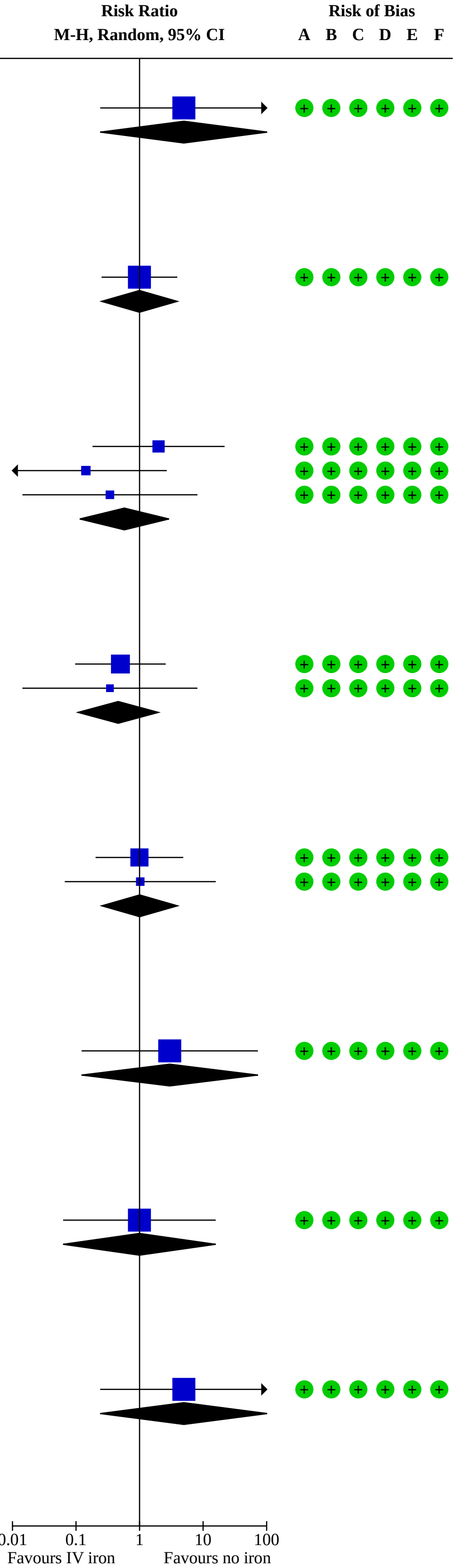


Figure 13. (Continued)

Test for subgroup differences: Chi² = 4.11, df = 7 (P = 0.77), I² = 0%

0.01 0.1 1 10 100
Favours IV iron Favours no iron

Footnotes
(a) drug-associated determined by PI
(b) any GI; drug-associated determined by PI
(c) vomiting
(d) diarrhoea
(e) CI calculated by Wald-type method.
(f) Tau² calculated by DerSimonian and Laird method.
(g) headache; also reported migraine as 1/41 vs 3/41; dizziness 2/41 vs 4/41
(h) paresthesia

Risk of bias legend
(A) Bias arising from the randomization process
(B) Bias due to deviations from intended interventions
(C) Bias due to missing outcome data
(D) Bias in measurement of the outcome
(E) Bias in selection of the reported result
(F) Overall bias

Adherence

Adherence was not reported by any of the IV iron studies.

Health-related quality of life (HRQoL)

All three RCTs reported a measure of HRQoL (Fontana 2014a (ISUB); Gybel-Brask 2018; Hod 2022 (DIDS)). These data could not be pooled due to the variation in measures used and are not included in the summary of findings tables or the overview table. Data for HRQoL are presented in Table 13. There may be no differences in patient-reported fatigue, physical health, and mental health scores, as well as pain and general well-being.

Comparison 3: Oral iron versus IV iron

Only two RCTs compared oral and IV iron (Birgegard 2010; Macher 2016 (IronWoMan)).

See Table 14 for an overview of all data and subgroups available for this comparison, and Summary of findings 3, Supplementary material 13, and Supplementary material 14 for more information about each of the three time points.

Evidence subgrouped by sex are in Analysis 20.1; Analysis 20.2; Analysis 20.3; Analysis 20.4; Analysis 20.5; Analysis 20.6; Analysis 21.1; Analysis 22.1, and evidence subgrouped by baseline iron status are in Analysis 23.1; Analysis 23.2; Analysis 23.3; Analysis 23.4; Analysis 24.1; Analysis 25.1.

Deferral rates

Deferral was not reported by any of the studies assessing oral versus IV iron.

Blood indices and iron stores

The blood iron outcomes showed no statistical difference between oral and IV iron (Analysis 20.1: MD −1.18, 95% CI −9.12 to 6.77; 1 study, 120 participants; moderate-certainty evidence; Analysis 21.1: MD −6.62, 95% CI −13.57 to 0.33; 1 study, 120 participants; moderate-certainty evidence), until the latest time point (after multiple donations): plasma ferritin favoured the use of IV iron after every donation (Analysis 22.1: MD −7.89, 95% CI −15.17 to −0.62; 1 study, 120 participants; moderate-certainty evidence). The certainty of the evidence for the only blood iron measure (plasma ferritin) was moderate, downgraded for risk of bias (lack of information regarding randomisation).

Adverse events (AEs)

There may be no differences in AEs between oral and IV iron. Data were only available from one study, at one time point for AEs (Analysis 20.3: RR 1.00, 95% CI 0.68 to 1.46; 1 study, 172 participants; very low-certainty evidence). The certainty of the evidence was very low, downgraded due to risk of bias (lack of information on how data were collected in an unblinded study) and imprecision (wide CIs).

Adherence

Adherence rates were higher with IV iron, although this was a single infusion after each donation, and did not require daily adherence, as with oral tablets. Data were only available from one study, at one time point for adherence (Analysis 20.4: RR 0.86, 95% CI 0.79 to 0.94; 1 study, 172 participants; low-certainty evidence). The certainty of the evidence was low, downgraded due to risk of bias (lack of information on how data were collected in an unblinded study, and different ways of assessing compliance) and imprecision (wide CIs).

Health-related quality of life (HRQoL)

Neither study reported HRQoL.

Other comparisons

Some studies directly compared different treatment methods of delivering oral iron. This included different treatment durations, different elemental iron doses, and different iron preparations.

Different durations

Only one RCT directly compared different durations of oral iron treatment (28 days versus 4 days of iron supplementation). Evidence was available for Hb (Analysis 26.1; MD 1.00, 95% CI −1.19 to 3.19; 1 study, 80 participants), plasma iron (Analysis 26.2; MD

22.00, 95% CI 3.70 to 40.30), and transferrin saturation (Analysis 26.3; MD 4.60, 95% CI 1.00 to 8.20). All three outcomes favoured the longer (28-day) supplementation period.

Different doses

Ten RCTs directly compared different doses of oral iron supplementation. We grouped these doses as very high, high, normal, and low daily doses of elemental iron. These data could not be pooled as comparisons were too heterogeneous, and often resulted in only one or two studies contributing to a comparison. We did not attempt to subgroup by any donor or intervention characteristics.

Evidence is available at each time point:

- at/before first donation since treatment: 7 RCTs, Analysis 27.1; Analysis 27.10; Analysis 27.11; Analysis 27.12; Analysis 27.2; Analysis 27.3; Analysis 27.4; Analysis 27.5; Analysis 27.6; Analysis 27.7; Analysis 27.8; Analysis 27.9;
- at/before second donation since treatment: 2 RCTs, Analysis 28.1; Analysis 28.2; Analysis 28.3; Analysis 28.4; Analysis 28.5;
- after multiple donations: 4 RCTs, Analysis 29.1; Analysis 29.2; Analysis 29.3; Analysis 29.4; Analysis 29.5; Analysis 29.6; Analysis 29.7; Analysis 29.8.

Different preparations

Eleven RCTs directly compared different preparations of oral iron supplementation. We grouped these preparations as FE2+, FE3+, FE2+ with additives, FE3+ with additives, haem iron (combination), and one non-specific "iron" with additives. These data could not be pooled as comparisons were too heterogeneous, and often resulted in only one or two studies contributing to a comparison. We did not attempt to subgroup by any characteristics.

Evidence is available at each time point:

- at/before first donation since treatment: 9 RCTs, Analysis 30.1; Analysis 30.10; Analysis 30.2; Analysis 30.3; Analysis 30.4; Analysis 30.5; Analysis 30.6; Analysis 30.7; Analysis 30.8; Analysis 30.9;
- at/before second donation since treatment: 1 RCT, Analysis 31.1; Analysis 31.2; Analysis 31.3;
- after multiple donations: 1 RCT, Analysis 32.1; Analysis 32.2; Analysis 32.3.

Equity assessment

We did not investigate equity-related characteristics in this review. We had not planned to investigate this for this version of the review, but if in future sufficient data become available, we could potentially subgroup (or perform sensitivity analyses) according to high- versus lower-income countries at population level.

As noted in the Included studies section, 31 of the 38 studies included in this review were from high-income countries (Europe, North America, Australia), with only seven studies from lower-income countries (India, Iran, South Africa, Thailand), all with very diverse socio-economic and ethnic backgrounds. The evidence was therefore insufficient to assess potential equity-related differences across our various comparisons and outcomes. Likewise, the included trials did not break down outcome by ethnicity or individual socio-economic status.

DISCUSSION

Iron deficiency is a significant cause of deferral in people wishing to donate blood. Deferral of donation for a period of six months through failure to pass the Hb threshold is associated with failure of donors to return to give blood. Avoiding iron deficiency is therefore essential not only to minimise symptoms and morbidity in donors and hence increase retention of donors in the long term, but also to maintain the efficiency of a donor session, where deferral is costly and disruptive. Iron supplementation for blood donors has been considered, and in some settings has been implemented for certain groups of 'at risk' donors. Rigorous evidence of the cost and benefits of iron supplementation is essential to guide policy.

In this review, we have considered supplementation through oral (tablets, capsules, tinctures) or IV delivery methods compared to either no supplementation (placebo or no treatment) or against another form (different dose, preparation, duration, or delivery method). Follow-up was considered at/before the first donation since supplementation commenced (most common time point), at/before the second donation since supplementation commenced, or after multiple donations.

Within each of these comparisons and time points, we have attempted to subgroup the evidence according to the following.

- Participant (donor) characteristics:
  - Donor sex (male, female, or mixed population)
  - Donor baseline iron status (healthy, iron deficient, or mixed population)
- Intervention/comparator characteristics:
  - Daily elemental iron dosage (calculated to standardise iron dose and remove variation in formulations): low (< 25 mg/day), normal (25 to 105 mg/day), high (105 to 195 mg/day), very high (195+ mg/day)
  - Duration of supplementation/treatment (< 2 weeks, 2 weeks to 1 month, 1 to 3 months, 3+ months)
  - Iron formulation (different preparations, where iron dose was equivalent: FE2+, FE3+, with additives, haem iron).

Summary of main results

The search was conducted on 28 January 2025. We identified 8 new RCTs, for a total of 38 RCTs with 7475 participants included in this update.

Our main comparisons of interest were as follows.

- Oral iron versus no iron (placebo/standard care): 23 RCTs, N = 5518
- IV iron versus no iron (placebo/standard care): 3 RCTs, N = 569
- Oral iron versus IV iron: 2 RCTs, N = 296

Risk of bias was often high or with some concerns across multiple domains. High risk of bias was often related to missing outcome data and/or less stringent historical reporting standards (lack of information especially regarding randomisation processes, blinding, and trial protocol/registration availability).

Oral iron supplementation probably reduces the deferral rate due to low Hb compared to no supplementation (moderate-certainty evidence). There may also be an improvement (increase) in the blood indices and iron stored (Hb and ferritin) after oral iron supplementation, and overall these changes were clinically significant (very low-certainty evidence). For both deferral and the measures of blood indices and iron stores, the evidence of these effects in specific subgroups of donors was rated as very low certainty due to risk of bias and heterogeneity. See Summary of findings 1; Supplementary material 10; Supplementary material 11.

Deferral was not reported in the IV iron studies, but the blood indices and iron stores (Hb and ferritin) may be improved (higher) as a result of IV iron supplementation compared to no iron supplementation (moderate- and very low-certainty evidence). See Summary of findings 2; Supplementary material 12.

More adverse events (AEs or side effects) may be experienced with oral iron compared to no iron, and there may be no difference in AEs when comparing IV iron to no iron (very low-certainty evidence).

Directly comparing oral iron to IV iron, there is probably no difference between the two methods of administration for the only measure of iron stores reported (ferritin) (moderate-certainty evidence). See Summary of findings 3; Supplementary material 13; Supplementary material 14.

Subgrouping (by sex, baseline iron stores, daily elemental iron dose, supplementation duration, and iron preparation) showed some low credibility subgroup differences for some outcomes when comparing oral iron supplementation to no iron. However, visual inspection suggests more high-quality data from additional studies may highlight potential differences in some populations, which could be used to target those most in need in the future.

Limitations of the evidence included in the review

Study design and sample sizes

All included studies were randomised trials. One study was a cross-over RCT (Radtke 2004b), although because of the way the data were reported, we were only able to use the first period as a parallel RCT. One study had an unclear study design (Busch 1972), as it appeared to use two separate randomisations; however, the data were then combined into three groups, instead of the randomised four groups, thus the type of randomisation was undetermined and the data could not be used.

Many of the included studies had very small sample sizes with fewer than 20 participants per arm. These studies were largely underpowered for our outcomes of interest, often resulting in imprecise effect sizes with wide CIs. Small sample sizes are more problematic in cases of multi-arm studies (e.g. multi-dose) with a single, small control group, where the control group is split to assess subgroups. We attempted to overcome this by not calculating totals when subgrouping by dose.

For deferral (our critical outcome), our data suggest that an expected deferral rate in the control (no iron) groups ranged from 17.3% (at/before first donation, Summary of findings 1), 19.9% (at/before second donation, Supplementary material 10), and 25.6% (after multiple donations, Supplementary material 11), thus suggesting a minimum baseline recruitment requirement of 220 to 300 people per study comparison (110 to 150 per group).

Population

Stratification by sex and baseline iron stores is very important. The current dataset allowed subgrouping by these characteristics into separate analyses (i.e. sex, or baseline iron status), although even these had limited power to detect any difference. There were suggestions of potential subgroup differences, with some groups more likely to benefit from supplementation, but were deemed 'low credibility' (using the ICEMAN criteria) due to the number of studies contributing to each subgroup, as well as the lack of intra-study data (where stratification within the study is reported, as opposed to an entire study cohort contributing to a single subgroup).

Some of the more recent studies presented data that could be used with greater granularity: healthy males, healthy females, low-iron males, low-iron females. However, this meant the analyses were severely underpowered to detect a subgroup difference, so larger studies with this stratification would aid assessment of more specific populations, and see who would benefit most, and where supplementation may not be required (or in lower/shorter doses).

Interventions and comparators

Most of the data (23 RCTs) compared oral iron supplementation to no iron (placebo or standard care). We combined the placebo and standard-care trials as the no-iron comparator and did not explore the impact of using placebo-controlled data (compared to standard care), as only a few studies were open-label, and this limitation primarily affected non-objective (subjective) measures (e.g. AEs, adherence, HRQoL, where reported), which were instead addressed in the risk of bias assessment.

Route of administration (oral, IV, etc.)

Only a small portion of the included studies compared IV iron to no iron (3 RCTs), or oral iron to IV iron (2 RCTs), and none of the IV studies reported on deferral. We did not identify any studies examining iron-rich food or intramuscular iron administration. Intramuscular iron supplementation (similar to IV administration) is less likely to be used in standard practice for blood donors for a variety of reasons, including the additional time and staffing requirement required to manage and monitor donors [192], greater cost and inconvenience [193], and the potential for increased risk of infection compared to oral iron administration (tablets) [194]. The volume of evidence for oral supplementation (compared to other forms, like intramuscular and IV) likely reflects the apparent ease of use and accessibility, and the minimally invasive nature of oral iron supplements (tablets or liquid form).

Daily and total dose (elemental iron)

Some of the studies are quite old (from the 1960s and 70s), and they were somewhat unclear regarding the iron dose administered (elemental iron dose or preparation dose). More recent studies have generally provided clearer information, reporting the elemental iron dose per administration, per day, and the target total dose. Some studies also reported the expected absorbed dose rather than the administered dose, particularly to align dosages between oral and IV iron.

The majority of included studies fell within the subgroup administering elemental iron doses between 25 and 105 mg/day. Very few studies evaluated lower doses (< 25 mg/day of elemental iron), which are commonly available over the counter, or higher doses (> 105 mg/day). As a result, the subgrouping system lacked sufficient power to assess any dose-dependent effects, although the dose thresholds were based on the literature (using prespecified thresholds), rather than available data (splitting the data into equally weighted groupings), which is more robust.

A small number of studies of oral iron (8 RCTs) had a total iron dose of less than 2850 mg, and all five studies assessing IV iron (3 RCTs IV iron versus none, 2 RCTs IV versus oral iron) employed total doses of 2000 mg or less. We did not perform analyses based on the total dose (or target total dose) due to the significant between-study heterogeneity (daily dose and duration) that contributed to the total dose, and instead tried to look at these factors separately. It may be of interest to research the same total dose, but delivered over different durations, to assess whether there is an optimal schedule for efficient iron repletion, weighed against the risk profile.

Iron preparation/formulation and use of additives

The included studies used various formulations/preparations of iron: FE2+, FE3+, and haem iron (all with or without additives). The vast majority of included studies used FE2+ (27 RCTs). This may be due to its high bioavailability compared to FE3+, as well as its relatively low cost [195]. However, the side effect profile appears to be greater with FE2+ [195]. Although assessment was limited as so few studies directly compared FE2+ to FE3+, we were able to compare the size of the effect through subgrouping for AE data (Analysis 13.9; Analysis 13.9; Analysis 13.9): both preparations favoured no iron (fewer side effects). With only one study contributing data to the FE3+ group, we could not draw any conclusions on the impact of iron preparation, and where potential subgroup differences were indicated, they were deemed of low credibility due to the small number of studies per subgroup, and no within-study comparisons.

Haem iron was very underrepresented in our analyses, with only two studies employing it in their comparisons. This may be because, despite the apparent higher bioavailability and absorption rate [196] and potentially lower side effect profile [197], it is more expensive per tablet, and cannot be used by those who do not eat meat derived from land animals (vegans, vegetarians, and pescatarians).

Some studies of oral iron supplementation used additives in their iron preparation/formulation, although there is limited evidence to allow any strong conclusions as to their impact. Often the description of the preparation was unclear, and we were unable to ascertain whether additives were used. The placebo group formulation was also rarely described, and so it was unclear whether they contained additives equivalent to those within the iron supplement, especially in older studies.

Outcomes, timing, and follow-up

Deferral and blood indices and iron stores

Low-Hb deferral is defined as a donor's Hb falling below the threshold required for blood donation. Most of the included studies adopted the European threshold of 135 g/L for males and 125 g/L for females. Notably, one study involving only male donors used a more stringent threshold of 140 g/L. Given that iron deficiency is a principal concern in repeat blood donors, higher Hb thresholds (even above normal clinical ranges) are often employed as a safety measure to reduce the risk of iron depletion.

Deferral was not reported as commonly as Hb (Hb was the most reported outcome). However, it would be relatively easy for trialists to report the number of people achieving Hb of 125 or 135 g/L (or another defined threshold) alongside their reports of mean Hb.

Relying solely on Hb levels in regular donors is insufficient to detect early or pre-clinical iron deficiency. Serum ferritin measurement provides a more sensitive indicator of iron depletion in this population [4, 198, 199], assuming no underlying inflammation or liver disease.

As some blood services use ferritin-guided deferral [200, 201, 202, 203, 204], it may be useful in future for studies to report deferral numbers based on both Hb and ferritin, to allow ongoing comparison with already well-established data. We did not identify any studies that reported deferring donors based on ferritin measures.

Serum and plasma ferritin measurements are effectively equivalent for assessing iron stores, showing a strong correlation and nearly identical results, and both reliably reflect total body iron levels [31, 32, 33]. For the purposes of this review, the two are considered interchangeable (although for clarity, we have footnoted any contributing plasma ferritin data in the analyses).

Additionally, knowing the expectation that ferritin data would likely be skewed, some studies reported using geometric mean or log-transformed data, compared to other studies that reported using a standard scale (arithmetic mean, non-log transformed). We were unable to combine these data into a single analysis, thereby weakening the potential power of the data.

Measures of iron in the blood were generally well-reported, and were more likely to be reported within subgroups (sex, baseline iron status) than other measures.

Adverse events (AEs), adherence, and health-related quality of life (HRQoL)

Despite the high rate of side effect reporting, most studies lack structured monitoring protocols for adherence and AEs. AEs were often not well-defined, and protocols for how they were reported and recorded were often unclear. AEs were often reported as at any time throughout the entire study period, even if there were multiple donations and supplementation periods. When AEs were well-reported, they often incorporated the use of a checklist and daily diaries, with clear descriptions (or definitions) of each side effect to also permit an assessment of severity.

Self-reported adherence is prone to recall bias and often leads to overestimation. More accurate but costlier methods to assess adherence include electronic medication monitoring systems (MEMS) and pill counts [205].

Additionally, reporting of AEs often lacks stratification by sex, iron dosage, and baseline ferritin levels, which limits the interpretability of safety data, which is vital if future studies or guidelines plan to target specific groups of people.

AEs largely focused on symptoms of gastrointestinal disturbances. However, there was no evidence in the included studies regarding the impact of oral iron tablets on the gut microbiome. Oral iron supplementation can significantly alter the gut microbiota by increasing luminal iron availability, which affects bacterial growth in the gut and may lead to the overgrowth of pathogenic organisms [206, 207]. This imbalance can cause intestinal inflammation, increased gut permeability, and reduced efficiency of iron absorption.

Furthermore, the majority of studies did not incorporate HRQoL assessments, potentially underestimating the symptomatic benefits of iron supplementation. Common symptoms such as fatigue and low energy may improve with treatment, even in the absence of significant changes in ferritin or Hb levels.

Timing and follow-up

Most trials lasted between one and three months, with some as short as 28 days, durations that may be insufficient to fully assess effectiveness and side effect profiles. Longer-term follow-up should be implemented to ascertain any continuing issues as a result of iron supplementation, looking beyond the supplementation period.

Limitations of the review processes

We have attempted to minimise bias in the review process. We conducted a comprehensive search of multiple data sources (including multiple databases and clinical trial registries) to ensure that we captured all relevant studies. We imposed no language restrictions on study reports. We carefully assessed the relevance of each publication, and performed all screening and data extractions in duplicate. We prespecified all outcomes and subgroups prior to analysis. We are therefore confident that we have included all relevant trials. Based on our sensitivity analysis assessing the impact of including trials with potential bias in the randomisation process, we were correct to include these studies, especially as most of the issues surrounding the randomisation process were likely due to historical reporting standards (lack of information), and no study was at high risk of bias in this domain. However, it appears that the estimate of effect for some outcomes may be more conservative when including all studies, irrespective of their risk of bias assessment.

For consistency across all included and excluded studies from the previous versions and this update, we reassessed previously included and excluded studies, and applied our definitions of outcomes to all studies. This included checking all extracted data for all outcomes, extracting additional information, and performing more in-depth risk of bias assessments (all studies were reassessed using Cochrane's RoB 2 tool).

We pooled data that appeared to have high statistical heterogeneity without investigating the causes beyond the subgroups and sensitivity analyses we have mentioned above. However, we do not think this has biased the results, as the certainty of the evidence for all outcomes has been downgraded accordingly.

Agreements and disagreements with other studies or reviews

We found the deferral rate due to low Hb (critical outcome) was reduced (RR 0.39, 95% CI 0.20 to 0.76) with oral iron supplementation at the first donation since commencement of treatment. This is a clinically important effect, and our findings are consistent not only with previous systematic reviews but also with the known physiology of iron collection, where loss of 250 mg of elemental iron with each donation can deplete iron stores and lead to iron-deficient erythropoiesis and then deferral due to failure to reach the prescribed Hb threshold for donation. Indeed, our review shows that Hb and serum ferritin were also improved with oral iron supplementation (Hb: MD 2.52, 95% CI 0.77 to 4.26 g/L higher; serum ferritin: MD 12.39, 95% CI 8.16 to 16.62 ng/mL higher).

This Cochrane review of the effect of iron supplementation in donors agrees with main findings from the previous (2014) version showing that iron supplementation helps iron stores [41], and findings from the 2020 update (non-Cochrane review) of factors causing low-Hb deferral [208]. However, the current version of the review examined the effect of iron supplementation in relation to the sex and baseline iron stores of donors and the dosage, duration, and formulation of iron preparation. This review also separated out studies examining the effect of oral iron from IV iron in detail by analysing the studies comparing oral iron versus no iron, IV iron versus no iron, and IV versus oral iron.

Unfortunately, and in common with previous reviews, the current review shows that many studies have considerable risk of bias, and that heterogeneity was considerable between included studies. This does not allow for a fine definition of the effect of iron supplementation in subgroups of data, and further, large-scale, well-powered studies that provide evidence for iron supplementation in donor subgroups are clearly required.

Nevertheless, our overall findings of iron supplementation improving the iron stores of donors are supported by key trials. Kiss and colleagues showed that the median recovery time of baseline Hb after blood donation is approximately 76 days with one daily iron tablet (37.5 mg elemental iron), compared to 168 days without supplementation [77]. Furthermore, for blood donors with normal Hb levels at baseline, low-dose iron supplementation, compared with no supplementation, reduced time to 80% recovery in Hb concentration irrespective of baseline iron stores.

The findings from this review showing more adverse events (or side effects) with oral iron compared to no iron, including increased abdominal pain, constipation, and diarrhoea, are entirely consistent with studies in other groups of patients. Ferrous salts, particularly ferrous sulphate, are more likely to cause gastrointestinal AEs than placebo or IV iron. A meta-analysis of 43 RCTs involving 6831 adults found that ferrous sulphate approximately doubled the risk of gastrointestinal symptoms than placebo or IV iron (odds ratio 2.32 versus placebo; odds ratio 3.05 versus IV iron) [28]. Clearly, more detailed studies of the side effects of the formulation, dose, and duration of oral iron supplementation on short-term symptoms and on the long-term gut physiology and microbiome are required for detailed guidelines on iron supplementation.

In the absence of detailed evidence on the efficacy and effectiveness of iron supplementation, it is unsurprising that the prospect of implementing iron supplementation in blood donors is a matter of some controversy [209, 210, 211], as discussed by Browne and colleagues [208]. Our current review confirms the findings of previous systematic reviews and other good RCT data, but amply illustrates the need for further research in this field.

AUTHORS' CONCLUSIONS

Implications for practice

Blood services who are seeking a reduction in the levels of deferral due to low haemoglobin (Hb) must consider any reasonable methods to prevent iron deficiency, weighing cost against benefits and feasibility.

This review confirms previous reviews that iron supplementation may reduce deferral due to low Hb, and may also lead to higher Hb and measures of iron stores. The apparently modest increase in Hb is important for blood services because the key outcome measure is the rate of low-Hb deferral, and this may change significantly with modest changes in a blood donor's Hb. However, the different outcomes of iron supplementation in specific demographic and clinical groups, and of using specific methods and doses of iron supplementation, have not been defined. The side effects of iron supplements in a healthy population and the logistical and medical support needed to administer iron supplements would have to be considered before embarking on extensive operational change or pilot studies.

Implications for research

We suggest a number of areas where the quality and quantity of relevant data available for this review could be improved.

Participants (potential risk modifiers)

Subgrouping by sex and baseline iron status indicated there may be some potential differences in the outcomes as a result of these characteristics. However, these analyses were deemed low credibility due to lack of within-study data. In the future, researchers could stratify and report data within these groups. Additionally, further granularity of the study groups (e.g. female donors with low baseline iron status, female donors with healthy baseline iron status, male donors with low baseline iron status, male donors with healthy baseline iron status) may show different effect sizes in the respective categories of donors. We have currently grouped data by all males, all females, all low iron, and all healthy iron statuses.

The rates of low Hb deferral in low- and middle-income countries (LMICs) have been reported to be much higher than in higher-income countries. Iron supplementation trials in iron deficient donors in LMICs would be of great interest.

We also grouped all those with low baseline iron status together, without further separation: some of these participants were defined as having sufficient Hb to donate but low ferritin, and were thus grouped together with those with low Hb. If future trials could provide further details on donors and/or study define subsets of donors, one could determine whether some groups of donors have different responses to others. This could also aid in the assessment of the use of ferritin-guided deferral compared to (or alongside) Hb deferral.

We did not assess the impact of age in this review, but potentially assessing older men and older women (postmenopausal compared to those of child-bearing age/still having menses) would be of interest in future studies. For example, the effect of routine iron supplementation for some 'more vulnerable' donors (like premenopausal female donors) could be compared to the effect of iron supplements in other groups of donors.

An important consideration for future studies in this, as in other clinical trials, is to have an adequately powered study to be able to detect a difference in all subgroups or target groups, not just for the entire cohort, using calculations based on both measures of iron (Hb, ferritin, etc.) and adverse events for these groups.

Interventions and comparators

Many of the subgroups that we investigated for having potentially credible differences (using the test for subgroup differences, then investigating the credibility of these differences using the ICEMAN criteria) had low credibility due to the small number of contributing studies that had stratified by the subgroup criteria within their own study. For example, where the subgrouping of interest was daily dose (low/normal/high), each included study should be sufficiently powered for each subgroup, and then each outcome reported for each group (low-dose data, normal-dose data, and high-dose data).

Future research could examine questions where evidence is weak:

- **iron preparation/formulation:** different types of iron (FE2+/FE3+) and with or without additives like vitamin C;
- **duration of treatment:** analysis of the duration of iron supplementation (before the next donation) using logistic regression may be able to identify an ideal duration for peak uptake of iron alongside minimal side effects;
- **elemental iron daily dose:** most studies used doses within what is considered a 'normal' range for correcting anaemia (25 to 105 mg/day elemental iron). Studies using lower doses may find reduced side effects alongside a sufficient iron repletion rate.

A study assessing dose and duration, utilising the same total dose but over different periods, would be of interest to also ascertain how iron absorption differs over time, or according to iron stores. Similarly, investigating whether dosing once or twice per day compared to alternate day may illuminate the relationship between dosing and adverse events (side effects) or iron absorption, or both.

Finally, assessment of adherence and recording the methods used to optimise adherence would be helpful in future trials.

Outcomes

Blood indices, iron stores, and deferral

Hb and ferritin (serum/plasma) were most commonly reported. Hb was used as a guide to deferral. In the future it may be possible to use indirect measures of iron stores from detailed analysis of blood cell parameters. Some countries are already using ferritin-guided deferral as well as Hb-based deferral. This may impact the definition of deferral, and future analyses may have to incorporate this. It would be valuable to continue to report Hb (and number achieving/not achieving the Hb threshold) if studies use ferritin or other parameters as an additional indication for deferral or intervention.

Reporting ferritin (serum or plasma) as both the geometric mean (or log-transformed data) and an arithmetic mean and standard deviation (non-log-transformed data), when they expect ferritin (or other) data to be skewed, would allow more direct comparison between trials, as well as highlighting the extent of the skewness.

Adverse events (side effects) and adherence

Future studies could clearly define how and when they are recording adverse events to reduce potential bias from misremembering or misreporting, for example the use of daily diaries with checklists and space for free text to describe severity of symptoms, as well as what is normal for each participant.

Research into the impact of iron supplementation on the gut microbiome over time would be of interest, as the majority of reported side effects are gastrointestinal.

Study adherence to a regimen of tablets should ideally be measured using MEMS (Medication Event Monitoring System) bottles where possible – so recording that a pill was dispensed alongside the time it was dispensed. Pill counting is useful, but is only unbiased in a truly blinded study. The impact of blinding could also be measured and reported, to ascertain whether participants (and investigators) remained unaware of their assignment throughout.

If adverse events and adherence were reported stratified by population characteristics and treatment characteristics, then it would be possible to evaluate whether certain populations experience more/less (or different types) of adverse events or are more compliant, or both. These analyses could indicate how the implementation of interventions may be targeted to maximise compliance or adherence to a supplementation regimen.

Quality of life and donor satisfaction

Very few studies reported quality of life, and those that did assess it often did not report by group, so there is a paucity of data on how (if) iron supplementation impacts day-to-day living, including: fatigue and sleep disturbances, physical or mental functioning, breathlessness, and restless leg syndrome (also a common side effect of anaemia), overall donor satisfaction and their decision to return to donate in future.

Timing and follow-up

Studies are generally followed up for the supplementation period only, or until the final donation/testing point in multiple-donation studies. Ideally, further follow-up, beyond this period, could assess the longer-term impact on all outcomes of interest.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: 10.1002/14651858.CD009532.pub3.

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Differences between the protocol (2012), previous review (2014), and review update (2025)

Supplementary material 10 Summary of findings: Comparison 1 (ii) Oral iron vs none (placebo/SoC) at/before 2nd donation since treatment

Supplementary material 11 Summary of findings: Comparison 1 (iii) Oral iron vs none (placebo/SoC) after multiple donations since treatment

Supplementary material 12 Summary of findings: Comparison 2 (ii) IV iron vs none (placebo/SoC) at/before 2nd donation since treatment

Supplementary material 13 Summary of findings: Comparison 3 (ii) Oral iron vs IV iron at/before 2nd donation since treatment

Supplementary material 14 Summary of findings: Comparison 3 (iii) Oral iron vs IV iron after multiple donations since treatment

ADDITIONAL INFORMATION

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We would like to thank all those who were authors on the previous versions of this review (protocol (2012) [212] and (2014) [41]). See Contributions of authors for more information.

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Editorial and peer-reviewer contributions

Cochrane Haematology supported the authors in the development of this review update.

The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): Professor Lise Estcourt, Senior Editor, Cochrane;
- Managing Editor (provided editorial guidance to authors, edited the article): Luisa Fernandez Mauleffinch, Cochrane Central Editorial Service;
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Andrew Savage, Cochrane Central Editorial Service;

- Copy Editor (copy editing and production): Lisa Winer, Cochrane Central Production Service;
- Peer reviewers (provided comments and recommended an editorial decision): Bernard Appiah, Department of Public Health, Maxwell School of Citizenship and Public Affairs, Syracuse University, Syracuse, New York, USA (clinical/content review); Tomohiko Sato, Division of Transfusion Medicine and Cell Therapy, Jikei University Hospital (clinical/content review); Radheshyam Meher (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); and Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

The author contributions for the 2026 update are listed below. All authors have contributed to the review, and have read and checked the manuscript prior to submission.

Louise J Geneen (LJG): screening and full-text assessment, retrieved full-text publications, data extraction (including checking and amending data from previous versions of the review), risk of bias assessment using the Cochrane RoB 2 tool, entered data into RevMan, performed all analyses including subgroup/sensitivity analyses and GRADE assessments, interpreted the results, wrote the manuscript.

Karyna Atha (KA): risk of bias assessment using the Cochrane RoB 2 tool, interpreted the results.

Catherine Kimber (CK): screening and selection of trials, data extraction, interpreted the results.

WIlliam McClune (WM): data extraction, interpreted the results.

Carolyn Dorée (CD): developed and performed all search strategies and de-duplication, retrieved full-text publications, contributed to the development of the manuscript.

Chiara Vendramin (CV): calculated elemental iron doses, interpreted the results, contributed to the development of the manuscript.

Lian Wea Chia (LWC): calculated elemental iron doses, interpreted the results, contributed to the development of the manuscript.

David J Roberts (DJR): interpreted the results, contributed to the development of the manuscript.

The author contributions to the previous (2014) publication and protocol (2012), who were not involved in the current update, are listed below.

Graham A Smith (GAS): screening and selection of trials, data extraction and assessment of risk of bias, analysis of results, and preparation of the protocol and final report.

Sheila A Fisher (SAF): screening and selection of trials, data extraction and assessment of risk of bias, analysis of results, and preparation of the final report.

Emanuele Di Angelantonio (EDA): content expert.

Sally Hopewell (SH): preparation of the protocol.

Declarations of interest

None of the authors have any conflicts of interest related to this review.

Five authors (KA, WM, CV, LWC, DJR) are medical doctors working in the National Health Service (NHS), UK.

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External sources

- No sources of support provided

Registration and protocol

Protocol: Smith (2012)[212] DOI: 10.1002/14651858.CD009532

Original review: Smith (2014) [41] DOI: 10.1002/14651858.CD009532.pub2.

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see Supplementary material 8):

- full search strategies for each database;
- full citations of each unique report for all studies (included, ongoing or awaiting classification, or excluded at the full text screen) in the final review;
- consensus risk of bias assessments;
- analysis data, including overall estimates and settings, subgroup estimates, and individual data rows.

Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods.

What's new

Date	Event	Description
4 June 2026	New citation required and conclusions have changed	Inclusion of 8 new studies. Comparisons reformatted to focus on different forms of iron supplementation (oral or IV separately) and to focus on subgroups: sex, baseline iron status, elemental iron daily dosage, duration of dosage (between testing/dona-

Date	Event	Description
		tion), and iron preparation (FE2+, FE3+, with/out additives, and haem iron).
4 June 2026	New search has been performed	Searches updated 28 January 2025. New authors included (6): Louise J Geneen; Karyna Atha; Catherine Kimber; William McClune; Chiara Vendramin; Lian Wea Chia. No change (2): Carolyn Dorée; David J Roberts. Previous authors who are no longer part of this review (3): Graham A Smith; Sheila A Fisher; Emanuele Di Angelantonio. All studies (previously included and newly identified) were re-assessed for risk of bias by two authors (LJG and KA) using Cochrane's RoB 2 tool.

History

Protocol first published: Issue 1, 2012
Review first published: Issue 7, 2014

Date	Event	Description
20 September 2023	Amended	Changed to new format

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ADDITIONAL TABLES

Table 1. Overview of syntheses and included studies

Study details	Population	Intervention (iron)	Comparator (control)	Outcomes reported (time points)	Time between donations/ samples
Oral iron versus no iron (placebo/standard care)					

Table 1. Overview of syntheses and included studies *(Continued)*

Blot 1980 Parallel RCT France Registration NR	Regular/repeat donors 57% male	N = 91 (73 analysed) 105 mg iron + 500 mg ascorbic acid (3150 mg/month)	N = 80 (70 analysed) No intervention	Hb, MCV, total serum iron, TIBC, serum ferritin (at second donation)	Mean 108 days (male), 124 days (female)
Brittenham 1996 (abstract only) Parallel RCT USA Registration NR	Regular female donors	N = NR (assumed 98) Carbonyl iron, 100 mg daily, at bedtime, for 56 days after each donation (scheduled visits)	N = NR (assumed 98) No supplementation (scheduled visit)	Hb, serum ferritin (at 30 months, after multiple donations)	4 donations per year (every 3 months)
Bucher 1973* Parallel RCT (4 arms) Switzerland Registration NR	Donors (Hb 12.5 to 13.5 g/dL) 22.5% male (all 4 groups, no detail per group)	Group 1 "R" (N = 43) and 3 "RR" (N = 38) Resoferon, 3 times/day for 28 days. Daily iron intake 111 mg (total dose 3108 mg)	Group 2 "P" (N = 38) Placebo, 3 times/day for 28 days	Hb, Hct, adherence, plasma iron, transferrin saturation (28 days)	> 28 days (no donation after treatment)
Busch 1972* Unclear RCT Germany Registration NR	Blood donors	N = unclear Iron sulphate, 152 mg + 222 mg ascorbic acid + 84 mg bicarb, 2/day for 30 days	N = unclear Placebo	Side effects: unable to use data due to inappropriate analysis	NR
Cable 1988 Parallel RCT USA Registration NR	Female deferred donors Hct 33% to 41%	N = 100 75 mg elemental (ferrous gluconate) iron, daily	N = 100 Placebo	Hb, Hct, serum ferritin, transferrin saturation, deferrals (at up to 4 further visits/attempts to donate)	Mean 66 to 68 days (for those completing study)
Dara 2017 Parallel RCT India Registration NR	Repeat/regular donors Hb > 12.5 g/dL 95.5% male	N = 98 300 mg ferrous fumarate (equivalent to 98.6 mg elemental iron), daily for 21 days	N = 102 Placebo	Hb, Hct, MCV, MCH, serum ferritin, side effects, adherence (at visit 3, next blood donation)	3- to 6-month interval
Ehn 1968* Parallel RCT (3 arms) Sweden Registration NR	Male military conscripts, first-time donors	N = 24 (Group A and B) Ferromyn S, containing 37 mg iron as ferrous succinate and 0.11 g succinic acid, once or twice a day for 2 weeks (total dose 1000 mg or 2000 mg per donation)	N = 18 (Group C) Placebo	Hb, serum iron, TIBC *No usable data: presented graphically, without SDs*	2 months (6 donations expected, study stopped at 4)
Gordeuk 1987a	Female, regular blood donors	N = 50 Carbonyl iron, 600 mg (N = 24)	N = 25 Placebo	Side effects (7 days), Hb, serum ferritin, serum iron, TIBC,	56 days

Table 1. Overview of syntheses and included studies *(Continued)*

Parallel RCT (3 arms) USA Registration NR	Hct > 38%	Ferrous sulphate, 300 mg (60 mg FE2+) (N = 26) 3/day for 1 week after donation		MCV, transferrin saturation (56 days)	
Gordeuk 1990 Parallel RCT USA Registration NR	Female, regular blood donors Hct > 38% or Hb > 12.5 g/dL	N = 50 Carbonyl iron, 100 mg (low dose), daily at night, for 56 days after donation	N = 49 Placebo	Side effects, adherence, Hb, MCV, serum ferritin, serum iron, TIBC, transferrin saturation (56 days)	56 days
Kiss 2015 (REDS-III HEIRS) Parallel RCT USA Registered prospectively	Regular blood donors Hb > 12.5 g/dL	N = 108 Ferrous gluconate, 325 mg (37.5 mg elemental iron), daily for 24 weeks (168 days)	N = 104 No treatment (standard care)	Deferrals (8 weeks), adherence, Hb, ferritin (24 weeks)	56 days (8 weeks) or 24 weeks (168 days)
Linpisarn 1986 Parallel RCT Thailand Registration NR	Regular (professional) blood donors Hb > 12 g/dL	N = 72 56 mg elemental iron, 3/day for 3 months	N = 69 Placebo	Hb, Hct, transferrin saturation, serum ferritin	3 months (mean 95 days)
Mackintosh 1988 Parallel RCT South Africa Registration NR *2 randomisations by population: HSF, LSF*	Regular male blood donors, Hb >/= 13.5 g/dL HSF: ferritin 50 to 150 µg/L LSF: ferritin < 20 µg/L	HSF group: N = 11 LSF group: N = 11 Same intervention: 100 mg elemental iron, as ferric polymaltose (Ferrimed DS chewable tablets), 2/day for 56 days	HSF group: N = 12 LSF group: N = 12 Same intervention: placebo	Hb, serum ferritin, adverse events (56 days)	56 days
Maghsudlu 2008 Parallel RCT Iran Registration NR	Female blood donors, initial successful donation	N = 207 Ferrous sulphate, 150 mg (elemental iron 50 mg), 3/day for 7 days after each donation	N = 205 Placebo	Hb, Hct, serum ferritin, TIBC, transferrin saturation	Every 4 months for repeat donation
Marks 2013 (FIRST) Parallel RCT Australia Registered retrospectively	Regular female blood donors, initial successful donation	N = 141 Carbonyl iron (Feosol), 45 mg (elemental iron), daily (bedtime) for 8 weeks (56 days)	N = 141 Placebo	Deferral, Hb, serum ferritin, adherence, side effects (12 weeks)	12 weeks

Table 1. Overview of syntheses and included studies *(Continued)*

Mast 2016 (STRIDE) Parallel RCT (3 relevant arms) USA Registered retrospectively	Regular blood donors 50.5% male	N = 178 Group 1 (N = 139) 38 mg or Group 2 (N = 139) 19 mg elemental ferrous gluconate, for 60 days after each donation	N = 138 Placebo	Plasma ferritin, Hb, adverse events, adherence (2 years)	Minimum 56-day donation interval, min of 2 per year (female) or 3 per year (male)
Mirrezaie 2008 Parallel RCT Iran Registration NR	Regular female blood donors Hb >/= 12 g/dL	N = 49 Ferrous sulfate (50 mg elemental iron), daily (bedtime) for 56 days	N = 46 Placebo	Serum ferritin, adherence, adverse events (56 days)	56 days
Pachikian 2020 Parallel RCT (3 relevant arms) Belgium Registered prospectively	First-time male blood donors Ferritin > 12 µg/L	N = 24 Ferrous gluconate, effervescent “Losferron”, elemental iron 20 mg (N = 12) or 80 mg (N = 12), daily (morning) for 28 days after each donation	N = 12 Placebo	Hb, plasma ferritin, MCV, adverse events, adherence (1 week before and after 3 donations)	3 months between donations (for 3 donations)
Radtke 2004a Parallel RCT Germany Registration NR	Regular blood donors Hb > 12.5 g/dL (female) or 13.5 g/dL (male) 55% male	N = 351 Ferrous gluconate, FE2+, 40 mg (N = 176) or 20 mg (N = 175) with ascorbic acid, daily (as 2 capsules) for 6 months	N = 175 Placebo	Deferrals, serum ferritin, adverse events, adherence (6 months)	3 (males) or 2 (females) further donations in 6 months
Radtke 2004b Cross-over RCT (only using first period) Germany Registration NR	Regular aphaeresis blood donors (2-unit RBC) Hb > 14.5 g/dL 98.8% male	N = 131 Iron (II)-glycine-sulfate-complex, ferro sanol duodenal, 100 mg FE2+, daily until 3rd visit (56 to 70 days)	N = 129 Placebo	Deferral, Hb, serum ferritin, serum iron, adherence, adverse events (every visit for 6 visits)	8 to 10 weeks between donations for 3 further donations
Rosvik 2010 Parallel RCT Norway Registered prospectively	Regular blood donors Hb > 12.5 g/dL (female) or 13.5 g/dL (male) Serum ferritin > 20 µg/L 50.1% male	N = 198 Standard Niferex ferroglycin sulphate complex, 100 mg, daily	N = 201 No intervention (standard care)	Hb, serum ferritin, side effects (8 to 10 days)	8 to 10 days

Table 1. Overview of syntheses and included studies *(Continued)*

Rybo 1971 Parallel RCT Sweden Registration NR	Regular blood donors 56.5% male	N = 920 Iron, 100 mg as ferrous sulphate, as rapid disintegrating (N = 460) or sustained release (N = 460), 2/day for 14 days	N = 456 Placebo	Adverse events, adherence (14 days)	14 days
Simon 1984 Parallel RCT USA Registration NR	Regular female donors Hct >/= 38%	N = 104 Ferrous sulphate, elemental iron, 39 mg (N = 55) or with 75 mg ascorbic acid (N = 49), daily for 56 days	N = 57 100 mg ascorbic acid	Hb, Hct, TIBC, serum ferritin (each visit for a year: 2,3,4,5)	Donation every 8 weeks for a year (mean 9.5 weeks)
Waldvogel 2012 Parallel RCT Switzerland Registered prospectively	Female donors Hb >/= 12 g/dL Serum ferritin < 30 ng/mL 1 week after donation	N = 78 Ferrous sulphate, 80 mg elemental iron, daily for 28 days	N = 76 Placebo	Hb, ferritin, fatigue (4 weeks)	5 weeks from donation
IV iron versus no iron (placebo/standard care)					
Fontana 2014a (ISUB) Parallel RCT Switzerland Registered retrospectively	Regular donor Serum ferritin > 50 µg/L, non-anaemic iron deficiency 53.8% male	N = 203 Iron 800 mg ferric carboxymaltose (Ferinject) in 200 mL 0.9% sodium chloride solution over 15 min	N = 202 Placebo	Hb, serum ferritin, HRQoL, adverse events (6 to 8 weeks later)	6 to 8 weeks
Gybel-Brask 2018 Parallel RCT Denmark Registered prospectively	First-time female donors P-ferritin < 60 ng/mL (2 weeks following first donation)	N = 43 Iron isomaltoside, 1000 mg (Monofer) single infusion over 15 min	N = 42 Placebo	Hb, plasma iron transferrin saturation, adverse events (before 2nd and 3rd blood donation)	12 weeks
Hod 2022 (DIDS) Parallel RCT USA Registered prospectively	Regular donors Hb to donate, Hct > 38% (F) 39% (M) 31.6% male	N = 39 Iron dextran 1000 mg (INFeD) or ferric carboxymaltose, single infusion over 40 min, 42 days after donation	N = 40 Placebo	Hb, serum ferritin, transferrin saturation, SF36, Beck depression score (6 months)	6 months
Oral iron versus IV iron					
Birgegard 2010 Parallel RCT	Regular donors	N = 60	N = 60	Serum ferritin, adverse events, adher-	Maximum 3 donations in 12 months (fe-

Table 1. Overview of syntheses and included studies *(Continued)*

Sweden Registration NR	42% male	Oral iron (FE2+) sulphate (Duroferon): 20 days x 100 mg: 200 mg after each donation	IV iron (III) sucrose (Venofer): 200 mg after each donation	ence (at 2nd, 3rd, 4th, 5th (male) donation)	male), 4 for males
Macher 2016 (IronWoMan) Parallel RCT Austria Registered prospectively	Regular whole blood or aphaeresis donors Ferritin ≤ 30 ng/mL	N = 90 Oral iron, Ferretab (100 capsules of 152 mg iron fumarate each), 8 to 12 weeks (10 tablets a week: alternate 2 then 1 per day), morning with orange/lemon juice	N = 86 IV iron high dose, Ferrinject (ferric carboxymaltose, 1000 mg) over 30 min	Hb, MCV, ferritin, transferrin saturation, adverse effects, adherence	8- to 12-week treatment. Donation was 4 to 6 weeks before treatment.
Oral iron (different durations)					
Bucher 1973* Parallel RCT (4 arms) Switzerland Registration NR	Donors (Hb 12.5 to 13.5 g/dL) 22.5% male (all 4 groups, no detail per group)	Standard/longer duration (4 weeks) Group 3 “RR” (N = 38) Resoferon, 3 times/day for 28 days. Daily iron intake 111 mg (total dose 3108 mg)	Shorter duration (4 days) Group 4 “RP” (N = 42) Resoferon, 3 times/day for 4 days (then placebo for remaining 24 days) Daily iron intake 111 mg (total dose 444 mg)	Hb, Hct, adherence, plasma iron, transferrin saturation (28 days)	> 28 days (no donation after treatment)
Oral iron (different doses)					
Ehn 1968* Parallel RCT (3 arms) Sweden Registration NR	Male military conscripts, first-time donors	N = 12 (Group A) Ferromyn S, containing 37 mg iron as ferrous succinate and 0.11 g succinic acid, twice/day for 2 weeks (total dose 2000 mg per donation)	N = 12 (Group B) Ferromyn S, containing 37 mg iron as ferrous succinate and 0.11 g succinic acid, once or twice a day for 2 weeks (total dose 1000 mg per donation)	Hb, serum iron, TIBC *No usable data: presented graphically, without SDs*	2 months (6 donations expected, study stopped at 4)
Frykman 1994 Parallel RCT Sweden Registration NR	Regular blood donors	N = 49 (26F, 23M) Hemofer: haem iron combination (1.2 mg haem iron from porcine blood plus 8 mg FE2+ as iron fumarate), 2/day for 60 days in 90-day period. 18 mg iron/day	N = 48 (25F, 23M) Erco-fer: non-haem iron (60 mg FE2+ as iron fumarate), daily for 60 days in 90-day period. 60 mg iron/day	Hb, serum ferritin, side effect (3 months)	3 months
Gordeuk 1987a Parallel RCT (3 arms)	Female, regular blood donors	N = 24 Carbonyl iron, 600 mg	N = 26	Hb, serum ferritin, serum iron, TIBC, MCV, transferrin satu-	56 days

Table 1. Overview of syntheses and included studies *(Continued)*

USA Registration NR	Hct > 38%	3/day for 1 week after donation (high-dose iron, total dose 12,600 mg)	Ferrous sulphate, 300 mg (60 mg FE2+) 3/day for 1 week after donation (total dose 1260 mg)	ration, side effects – recorded for 7 days	
Gordeuk 1987b Parallel RCT USA Registration NR	Female, repeat, deferred blood donors Hct < 38%	N = 25 Carbonyl iron, 600 mg 3/day for 3 weeks (high-dose iron, total dose 37,800 mg)	N = 25 Ferrous sulphate, 300 mg (60 mg FE2+) 3/day for 3 weeks (total dose 3780 mg)	Side effects (weeks 1 and 3), Hb, serum ferritin, serum iron, TIBC, transferrin saturation (16 weeks) *Lab data only available graphically*	16 weeks
Jacobs 1993 Parallel RCT (3 arms) South Africa Registration NR	Deferred donors Hb < normal: Hb < 13.6 g/dL (F); 15.3 (M) or SF < 20 ng/mL or TS < 17%	N = 55 (Group 3) Chewable ferric polymaltose 100 mg, 2/day for 12 weeks	N = 53 (Group 2) Chewable ferric polymaltose 100 mg, daily (morning) for 12 weeks	Adherence, side effects, serum iron, TIBC, ferritin (week 12)	12 weeks (no donation after treatment)
Lieden 1975 Parallel RCT Sweden Registration NR	Male military conscripts, first-time donors	N = 10 Ferrous carbonate (equivalent to ferrous sulphate), 100 mg daily for 1 year	N = 7 Ferrous carbonate (equivalent to ferrous sulphate), 20 mg daily for 1 year	Serum iron, TIBC (after 4th and 6th donation)	Donated every 2 months (6 donations expected)
Lindholm 1981* Parallel RCT Sweden Registration NR	Regular male donors, no iron deficiency in recent years	N = 250 Ferrous sulphate, 100 mg FE2+, daily for 30 days after each donation	N = 250 Ecofer, 60 mg FE2+, daily for 30 days after each donation	Side effects, deferrals, adherence, Hb, serum iron, TIBC (test 2)	Average 51 to 75 days
Mast 2016 (STRIDE) Parallel RCT (3 relevant arms) USA Registered retrospectively	Regular blood donors 50.5% male	N = 139 (Group 1) 38 mg elemental ferrous gluconate, for 60 days after each donation	N = 139 (Group 2) 19 mg elemental ferrous gluconate, for 60 days after each donation	Plasma ferritin, Hb, adverse events, adherence (2 years)	Minimum 56-day donation interval, min of 2 per year (female) or 3 per year (male)
Pachikian 2020 Parallel RCT (3 relevant arms) Belgium	First-time male blood donors Ferritin > 12 µg/L	N = 12 Ferrous gluconate, effervescent “Losferron”, 695 mg (80 mg elemental iron), daily (morning) for 28 days after each donation	N = 12 Ferrous gluconate, effervescent “Losferron”, 173.73 mg (20 mg elemen-	Hb, plasma ferritin, MCV, adverse events, adherence (1 week before and 2 days after 3 donations)	3 months between donations (for 3 donations)

Table 1. Overview of syntheses and included studies *(Continued)*

Registered prospectively			tal iron), daily (morning) for 28 days after each donation		
Radtke 2004a Parallel RCT Germany Registration NR	Regular blood donors Hb > 12.5 g/dL (female) or 13.5 g/dL (male) 55% male	N = 176 Ferrous gluconate, FE2+, 40 mg, with ascorbic acid, daily (as 2 capsules) for 6 months	N = 175 Ferrous gluconate, FE2+, 20 mg, with ascorbic acid, daily (as 2 capsules) for 6 months	Deferrals, serum ferritin, adverse events, adherence (6 months)	3 (males) or 2 (females) further donations in 6 months
Oral iron (different preparations)					
Borch-Iohnsen 1993 Parallel RCT Norway Registration NR	Regular female donors with ferritin < 20 µg/L and Hb > 12 g/dL	N = 18 (Group A) 20 mg iron fumarate with 120 mg ascorbic acid	N = 16 (Group B) 2 mg haem iron plus 16 mg iron fumarate	Change in Hb, serum ferritin, transferrin (at 5 months)	5 months
Busch 1972 Unclear RCT Germany Registration NR	Blood donors	N = unclear Iron sulphate, 152 mg + 222 mg ascorbic acid + 84 mg bicarb, 2/day for 30 days	N = unclear Iron sulphate, 152 mg + 222 mg ascorbic acid, 2/day for 30 days	Side effects: unable to use data due to inappropriate analysis	NR
Buzi 1980* Parallel RCT Switzerland Registration NR	Deferred donors Hb < 13 g/dL, Hct < 37% 3.1% male	Group A (N = 32) Iron sulphate, Tardyferon (slow release) 80 mg elemental iron, daily, 30 days (total dose 2400 mg FE2+)	Group B (N = 32) Iron fumarate, 66 mg, 2 capsules/day to reach a total dose of 2376 mg FE2+ (18 days)	Hb, Hct, serum iron, TIBC, side effects	> 32 days (no donation after treatment)
Devasthali 1991 Parallel RCT USA Registration NR	Deferred female donors Hct < 35%	N = 24 100 mg carbonyl iron, daily (at night) for 84 days	N = 25 500 mg ferrous sulphate (containing 100 mg FE2+), daily (at night) for 84 days	Serum iron, TIBC, transferrin saturation, serum ferritin, Hb *All blood data were extracted by visual approximation of published graphs.*	Study period: 16 weeks
Frykman 1994 Parallel RCT Sweden Registration NR	Regular blood donors	N = 49 (26F, 23M) Hemofer: haem iron combination (1.2 mg haem iron from porcine blood plus 8 mg FE2+ as iron fumarate), 2/day for 60 days in 90-day period. 18 mg iron/day	N = 48 (25F, 23M) Erco-fer: non-haem iron (60 mg FE2+ as iron fumarate), daily for 60 days in	Hb, serum ferritin, side effect (3 months)	3 months

Table 1. Overview of syntheses and included studies *(Continued)*

			90-day period. 60 mg iron/day		
Jacobs 1993 Parallel RCT (3 arms) South Africa Registration NR	Deferred donors Hb < normal: Hb < 13.6 g/dL (F); 15.3 (M) or SF < 20 ng/mL or TS < 17%	N = 55 (Group 3) Chewable ferric polymaltose 100 mg, 2/day for 12 weeks	N = 51 (Group 1) Ferric sulphate 60 mg, 2/day (fasted) for 12 weeks	Adherence, deferral, side effects, serum iron, TIBC, ferritin (week 12)	12 weeks (no donation after treatment)
Jacobs 2000 Parallel RCT (4 arms) South Africa Registration NR	Deferred donors Hb < normal: Hb < 13.6 g/dL (F); 15.3 (M) or SF < 20 ng/mL or TS < 17%	N = 46 (Group 1) Iron polymaltose 100 mg, 2/day for 12 weeks	N = 48 (Group 4) Ferrous sulphate "equivalent amount of iron" 2/day for 12 weeks	Adherence, tolerance, side effects, Hb, serum iron, transferrin saturation, serum ferritin (week 12)	12 weeks (no donation after treatment)
Jacobs 2000 Parallel RCT (4 arms) South Africa Registration NR	Deferred donors Hb < normal: Hb < 13.6 g/dL (F); 15.3 (M) or SF < 20 ng/mL or TS < 17%	N = 46 (Group 1) Iron polymaltose 100 mg, 2/day for 12 weeks	N = 79 Iron polymaltose 100 mg with glycerophosphate 0.9 mmol/L (Group 2, N = 36) or 1.8 mmol/L (Group 3, N = 43), 2/day for 12 weeks	Adherence, tolerance, side effects, Hb, serum iron, transferrin saturation, serum ferritin (week 12)	12 weeks (no donation after treatment)
Landucci 1987 Parallel RCT Italy Registration NR	Donors with low stored iron: serum ferritin < 30 ng/100 mL 27.5% male	N = 20 Iron protein succinylate "Legofer" (FE3+ 80 mg), daily for 30 days	N = 20 Iron sulphate (FE2+ 10.5 mg), daily for 30 days	Hb, Hct, MCV, serum iron, serum ferritin, transferrin concentration, tolerance (30 days)	30 days
Lindholm 1981* Parallel RCT Sweden Registration NR	Regular male donors, no iron deficiency in recent years	N = 250 Ferrous sulphate ACO, 100 mg FE2+, daily for 30 days after each donation	N = 250 Ecofer, 60 mg FE2+, daily for 30 days after each donation	Side effects, deferrals, adherence, Hb, serum iron, TIBC (test 2)	Average 51 to 75 days
Rybo 1971 Parallel RCT Sweden Registration NR	Regular blood donors 56.5% male	N = 460 Iron, 100 mg as ferrous sulphate, rapid disintegrating, 2/day for 14 days	N = 460 Iron, 100 mg as ferrous sulphate, sustained release, 2/day for 14 days	Adverse events, adherence (14 days)	14 days
Simon 1984 Parallel RCT	Regular female donors	N = 49	N = 55	Hb, Hct, TIBC, serum ferritin (each visit for a year: 2,3,4,5)	Donation every 8 weeks for a year

Table 1. Overview of syntheses and included studies *(Continued)*

USA Registration NR	Hct >/= 38%	Ferrous sulphate, elemental iron, 39 mg with 75 mg ascorbic acid, daily for 56 days	Ferrous sulphate, elemental iron, 39 mg, daily for 56 days	*All blood data were extracted by visual approximation of published graphs.*	(mean 9.5 weeks)

A study may be listed under multiple comparisons where there are more than two relevant arms.
***translated**
F: female
FE2+: ferrous salts
FE3+: ferric complex
Hb: haemoglobin
Hct: haematocrit
HRQoL: health-related quality of life
HSF: high serum ferritin
IV: intravenous
LSF: low serum ferritin
M: male
MCH: mean corpuscular haemoglobin
MCV: mean cell volume
NR: not reported
RBC: red blood cell
RCT: randomised controlled trial
SD: standard deviation
SF: serum ferritin
SF36: short-form 36 (quality of life measurement tool)
SF12: short-form 12 (quality of life measurement)
TIBC: total iron binding capacity
TS: transferrin saturation

Table 2. Overview of studies awaiting classification

Study details (study design, country, trial registration, total N)	Reason for classification	Population	Intervention (iron supplementation)	Control (placebo/standard care)	Outcomes reported (time points)
Ekermo 2013 [213] Parallel RCT, 2 arms, no control **Sweden** Registration NR N = 210	Abstract only: no data per group (e.g. sample size), no outcome data provided. No full-text publication identified.	Adults with >/= 5 previous whole blood donations	**IV iron infusion** (Dextriferron) 1000 mg (first donation) + 200 mg after each other donation (4 or 5 donations, sex-dependent) N = NR (likely 105) **Oral iron (tablets)** (FE2+) 20 x 100 mg after each donation N = NR (likely 105)	N/A	• Hb • Iron status • Health status: SF36 & RLS • Fatigue • Tolerance • Adherence • Adverse events
EUCTR 2010-023790-19-DK [214] Parallel RCT, 2 arms **Denmark**	Trial ended prematurely (19 Mar 2012); no results posted. No further information	Adults, female, first-time donors, P-ferritin < 30 µg/L	**IV iron infusion** (isomaltoside 1000 (Monofer)) N = NR	**IV placebo** N = NR	• Hb (1st, 2nd, 3rd donation) • Tolerance of 3 donations

Table 2. Overview of studies awaiting classification *(Continued)*

Registered, unclear study dates N = NR					• Quality of life • Exercise tolerance • Adverse events
Faed 2015 (ACTRN12612000911897) [215, 216] Parallel RCT, 3 arms, only 2 relevant **New Zealand** Prospective registration N = 5190	Abstract and trial registration only, no results by arm	Adults, female, first-time donors, ferritin level of 12 to 25 µg/L	**Oral iron** (carbonyl 18 mg) for 30 days "Group 1" N = NR (5190 total recruited, unclear per group)	**No intervention** (no information provided) N = NR (5190 total recruited, unclear per group)	• Iron status (1st, 2nd, 3rd donation) • Donor return-rate • Inter-donation interval
Feustel 1975 [217] Unclear design, 2 arms **Germany** Registration NR N = 108	Abstract only. In German, translation may have missed key words: states double-blind, but doesn't state it was random.	Adults, first-time donors	**Oral iron** (tablets) (divalent iron sulfate) N = 48	**Oral placebo** N = 60	• Pain • Tolerance • Side effects • Discontinuation • Hb
Grosz 2011 [218] Parallel RCT, 4 arms, 2 comparisons **Norway** Registration NR N = NR	Abstract only. Limited information regarding sample sizes. Appears to be 30-day follow-up, not impact on next donation	Active blood donors Excluded all previous donors	**Oral iron (tablets)** "Group 1" FE2+ (Niferex) 100 mg/day for 20 days N = NR	**No iron** "Group 4" no supplementation N = NR	• S-ferritin (30 days) • Hb • S-iron
			Oral iron (tablets) with vitamin C "Group 2" FE2+ (Niferex) 100 mg/day for 20 days N = NR	**No iron with vitamin C** (tablets) "Group 3" vitamin C supplementation only N = NR	
Kaltwasser 1982 [219] Unclear design, 2 arms **Germany** Registration NR	Abstract only. Refers to a double-blind, placebo-controlled trial, but unclear if this was randomised	Female blood donors Excluded "registered blood donors"	**Oral iron** (ferrous iron) 105 mg/day for 3 months (length of study, assumed length of supplementation) N = NR	**Oral placebo** N = NR	• Blood count • Serum iron • TIBC • Serum ferritin

Table 2. Overview of studies awaiting classification *(Continued)*

N = NR

NCT04778072 [220] Parallel RCT, 3 arms, no control **Ireland** Retrospective registration N = NR	Trial registration only. No full text identified. Unclear if participants are donors: described as treatment resistant, so unlikely to be.	Ferritin levels < 30 µg/L	**Oral iron** (tablets) "Group 1" 14 mg/day elemental iron for 12 weeks N = NR **Oral iron** (tablets) "Group 2" 25 mg/day elemental iron for 12 weeks N = NR **Oral iron** (tablets) "Group 3" 25 mg x 2/day for 12 weeks N = NR	N/A	• Adherence (12 weeks) • Side effects • Hb • Ferritin • Quality of life
Stotzer 2013 [221] Parallel RCT, 3 arms, possibly only 2 relevant for direct comparison **Germany** Registration NR N = 142	Abstract only. Unclear whether the intervention was properly controlled, as vitamin C arm may not be appropriate to use as control. Only comparison is therefore high-dose iron vs low-dose + vitamins vs high-dose vitamin C.	Blood donors. No information on age, sex, regular/first-time status	**Oral iron?** "Group 1" 50 mg/day elemental iron (Plastufer mite) Cumulative dose: 1200 mg N = 50 **Oral iron (food suppl)** "Group 2" Food supplement (vitamins + 20 mg elemental iron/day) (NBV Vital Kapseln) Cumulative dose: 1200 mg N = 46	**No iron** "Group 3": vitamin C supplementation: 100 mg ascorbic acid/day N = 46	• Adherence (at donations for 2 years) • Adverse events (side effects) • Deferrals • Hb, MCV, ferritin
Wong 2012 [222] Parallel RCT, 2 arms **Hong Kong** Registration NR N = 223	Abstract only. Limited information	Repeat (regular) donors 50% male	**Oral iron** (iron fumarate (fortifier) equivalent to 100 mg elemental iron) 100 mg/day for 2 weeks N = 112	Standard care (no supplementation) N = 111	• Adherence (at next donation: 3 to 4 months) • Ferritin levels • Deferral

F: female
FE2+: ferrous salts
FE3+: ferric complex
Hb: haemoglobin
Hct: haematocrit
HRQoL: health-related quality of life
HSF: high serum ferritin
IV: intravenous

LSF: low serum ferritin
M: male
MCV: mean cell volume
N/A: not applicable
NR: not reported
RCT: randomised controlled trial
RLS: restless leg syndrome
SF: serum ferritin
SF36: short-form 36 (quality of life measurement tool)
SF12: short-form 12 (quality of life measurement)
TIBC: total iron binding capacity

Table 3. Overview of ongoing studies

Study details (study design, country, trial registration, total N)	Reason for classification	Population	Intervention (iron supplementation)	Control (placebo/standard care)	Outcomes reported (time points)
CTRI/2017/02/007860 [223] Parallel RCT, 3 arms **India** Retrospective registration N = 300	Start date: Jan 2017 Expected period: 2 years, 6 months	Blood donors aged 18 to 60 years	**Oral iron** (carbonyl, 100 mg elemental iron) for 35 days after each donation **Oral iron** (50 mg elemental iron) for 35 days after each donation	**No intervention**	• Serum ferritin • Soluble transferring receptor • Hepcidin • C-reactive protein
CTRI/2024/09/073361 [224] Parallel RCT, 2 arms **India** Prospective registration N = 160	Start date: Sept 2024 Expected period: 3 months, 30 days	Regular blood donors aged 18 to 65 years Previous donation >/= 3 Hb 12.5 to 16.5 g	Ferrous ascorbate N = 80 *Assumed to be taken orally, but not explicitly stated*	**No intervention** N = 80	Ferritin (45, 60, 90 days)
Karregat 2022 (FORTE) [225, 226, 227, 228] Parallel RCT, 6 arms **The Netherlands** Prospective registration N = 2400 to 3000	Start date: July 2021 Expected end date: Dec 2022	Successful baseline donation Age 18+ years Non-anaemic iron deficiency Ferritin level ≤ 30 µg/L	**Oral iron** (ferrous bisglycinate) Daily low-dose iron (30 mg elemental iron) Group 4, N = 200 For 56 days **Oral iron** (ferrous bisglycinate) Daily high-dose iron (60 mg elemental iron) Group 6, N = 200 For 56 days	**Daily oral placebo** Group 2 N = 200	• Ferritin & Hb (56, 122, and 182 days) • Adverse events • Mental health (SF12) • Physical health (SF12)
			Oral iron (ferrous bisglycinate)	**Alternate day oral placebo**	

Table 3. Overview of ongoing studies *(Continued)*

			Alternate day low-dose iron (30 mg elemental iron) Group 3, N = 200 For 56 days	Group 1 N = 200	
			Oral iron (ferrous bisglycinate) Alternate day high-dose iron (60 mg elemental iron) Group 5, N = 200 For 56 days		
NCT03014921 (FERDOP) [229] Parallel RCT, 2 arms **Switzerland** N = 19 Retrospective registration	Start date: Dec 2016 Expected end date: April 2017	Men aged 20 to 35 years Ferritin < 50 µg/L Transferrin saturation < 20 if ferritin between 20 and 50 µg/L	**IV iron** (Ferinject 500 mg)	**IV placebo** (250 mL saline)	• Serum ferritin (8 weeks) • Hb • Reticulocytes
NCT05678647 [230] Parallel RCT, 2 arms, no control **Sweden** N = 120 Prospective registration	Start date: Jan 2023 Expected end date: Jan 2026	Regular blood donors aged 18+ years	**Oral iron** (capsules) (sucrosomial iron, SiderAl Forte 30 mg), 20 capsules N = 60	N/A	• Side effects (1 year) • RLS (1 year) • Iron balance (1 year)
			Oral iron (tablets) (iron sulfate, Duroferon, 100 mg), 20 tablets N = 60		
NCT06101238 (BLOODSAFE) [231] Parallel RCT, 2 arms **Ghana/USA/UK** N = 528 Prospective registration	Start date: Jan 2024 Expected end date: April 2026	Deferred donors for low Hb aged 18 to 60 years	**Oral iron** (daily 65 mg elemental iron) N = 264	**Standard care** N = 264	• Successful donations (12 months) • Attempted donations (12 months)

F: female
FE2+: ferrous salts
FE3+: ferric complex
Hb: haemoglobin
Hct: haematocrit
HRQoL: health-related quality of life
HSF: high serum ferritin
IV: intravenous
LSF: low serum ferritin
M: male

MCV: mean cell volume
NR: not reported
N/A: not applicable
RCT: randomised controlled trial
RLS: restless leg syndrome
SF: serum ferritin
SF36: short-form 36 (quality of life measurement tool)
SF12: short-form 12 (quality of life measurement)
TIBC: total iron binding capacity

Table 4. RoB 2 overview: health-related quality of life

Study ID	Experimental	Comparator	Randomisation process	Deviations from intended interventions	Missing outcome data	Measurement of the outcome	Selection of the reported result	Overall bias
Fontana 2014a (ISUB)	IV iron carboxymaltose	IV placebo	Low	Low	Low	Low	Low	Low
			"centrally randomized 1:1...using a locked, concealed web-based system (WebSpirit, 2mt Sofware GmbH, Ulm, Germany). Randomization was computer-generated, stratifed… Trial treatment assignment was double-blinded, so that neither study physicians and nurses involved in donor care and outcome assessment, nor participants were aware of treatment assigned. " All outcomes reported at baseline, no differences between groups. Participants and personnel were blinded to allocation, and infusions were masked to hide the colour of the infusion, given through non-transparent tubing: described as "true blinding" in publication, and in trial registration as Triple blinded (Participant, Investigator, Outcomes Assessor). Methods of blinding detailed within publication. 6 to 8 weeks after treatment: ITT analysis, despite 12/405 protocol violations or withdrew Validated assessment tools used: EQ-5D, MFSI, Jenkins sleep score Trial registration available online, and checked. Primary and secondary outcomes reported as predefined.					
Gybel-Brask 2018	IV iron oligosaccharide complex	IV placebo	Low	Low	Low	Low	Low	Low
			"Permuted block randomization with a block size of 6 was used to randomize the subjects. The randomization list was prepared centrally by a contract research organization, Max Neeman International Data Management Center, using a validated computer program (Statistical Analysis Software [SAS] 9.1.3, SAS Institute, Inc.) PROC PLAN procedure. An interactive Web response system method was used to randomize the eligible subjects to the treatment groups. When the subject data had been entered into the interactive Web response system, a unique randomization number was generated for the subject, identifying which treatment the subject was allocated to. The screening and enrollment of the subjects were performed by the investigator at the site, whereas the entering of the subject data into the interactive Web response system generating the randomization number was performed by the trial nurse or trial coordinator." Presented data appear balanced. Described as double-blind, placebo-controlled Full analysis set (mITT), PP, and safety set (ITT) presented. Validated fatigue visual numeric scale and fatigue severity score Described as double-blinded, also lists roles with blinding as: Subject, Investigator, Monitor. It is possible that outcome assessors were aware of allocation. Trial prospectively registered on two databases. Clear statistical analysis planned.					
Hod 2022 (DIDS)	IV iron dextran	IV placebo	Low	Low	Low	Low	Low	Low
			"Randomization will be performed using a computerized system with equal allocation (1:1) to iron repletion or placebo. Randomization will be stratified by gender; randomly permuted block sizes of 4, 6, or 8 will be used" "The research pharmacy will provide a standard 1-gram dose of low molecular weight iron dextran, ferric carboxymaltose, or saline in special bags designed to conceal group assignment." Full analysis population (mITT) For the secondary outcome analyses, the full analysis population will be defined as all participants who were randomised.					

Table 4. RoB 2 overview: health-related quality of life *(Continued)*

			They state analysis would include all those who were randomised (40 and 39), but HRQoL data only available for 33 and 35. Under flowchart they state: "Participants who completed at least 1 cognitive performance evaluation after randomization" with numbers of 39 and 37. Withdrawals explained. Validated HRQoL scales (e.g. RAND 36-item Short-Form Health Survey) "Neuropsychological tests will be administered to subjects on an iPad or Tablet. Subjects will be supervised at all times during this portion of the evaluation and will be encouraged to complete the tests within one session (i.e. without interruption). Total time for test administration is approximately 30 minutes and the time of start and completion of the test battery will be recorded. When the neuropsychological battery has been completed, the subject will be provided with a clipboard containing printed copies of four questionnaires. The subject will be asked to complete the surveys on his/her own without interruption. The time of start and completion of the surveys will be recorded." Described in trial registration as triple-blinded: participants, personnel/carers, outcome assessors Protocol, trial registration, and analysis plan available.					
Mast 2016 (STRIDE)	Oral ferrous gluconate	Oral placebo	Low	Low	Low	Low	Some concerns	Some concerns
			"The lists were generated using standard block randomization methods, implemented with computer software (nQuery Advisor, 2014, Statistical Solutions, Boston, MA). The randomized numbers from the algorithm correspond to one of the five study arms. While they appeared to be random numbers, they coded for the group assignment. Each of the five study arms had three to four codes so that patterns could not be identified by investigators, coordinators, and staff at study sites" Uneven medical de-enrolment from study, appears that placebo group were "healthier", suggesting some baseline imbalance. Bias in favour of placebo (reduced effect size of iron) When focusing on only 3 interventions (iron or placebo pills), described as double-blind (participants and investigator). Blood tests – mITT (those who presented) Adverse events – ITT (safety data) Deferral – unclear, as presented as a percentage over 2 years Baseline characteristics available for those completing, versus abandoning the study (Spencer 2020, table 1), then HRQoL (fatigue) data for only those who completed the study. Multivariable analysis appears to include withdrawals. Validated questionnaire based on FACIT Described as double-blind: participants and investigators Trial registration identified. Does not include a detailed analysis plan, but checked and reported as planned.					
Pachikian 2020	Oral ferrous gluconate	Oral placebo	Some concerns	Some concerns	Low	High	Some concerns	High
			Randomisation process not described. Baseline comparison data for iron stores (table 1) show no significant differences between groups. Age and fitness level data in trial registry show no differences. Described as single-blinded, placebo-controlled mITT, all those completing each follow-up point Objective measures of aerobic power Knowledge of allocation could impact how investigators "encourage" participants during maximal exertion.					

Table 4. RoB 2 overview: health-related quality of life *(Continued)*

			Trial registration available: prespecified analyses. Primary outcome (VO2 peak) only reported for final endpoint (in trial registration) or graphically, or in comparison to other groups. Not reported as raw data at each time point as with other outcomes.					
Waldvogel 2012	Oral ferrous sulphate	Oral placebo	Low	Low	Low	Low	Low	Low
			"simple random allocation sequence without restriction was generated by an independent pharmacy according to a pre-established computer-generated list. Each drug package was identified with a unique number according to the randomization schedule and given to the nurse in charge of the participant. The code was held by the pharmacist and remained unbroken until the end of the trial. The allocation remained concealed from participants, care providers, investigators and the statistician until the end of the statistical analysis." Quadruple-blinded (participants, personnel, outcome assessors, statistical analysis) mITT – "Missing data from dropouts were not replaced and only donors who were followed-up after four weeks' treatment were included in the analysis." Very small number of dropouts. All explained and unrelated to trial. Standard blood analysis Fully masked study Protocol and trial registration available, and reported as stated.					

FACIT: Functional Assessment of Chronic Illness Therapy
HRQoL: health-related quality of life
ITT: intention to treat
IV: intravenous
MFSI: Multidimensional Fatigue Symptom Inventory
mITT: modified intention to treat
PP: per protocol

Table 5. Sensitivity analyses: Cochrane RoB 2, domain 1 (randomisation process)

Outcome (time point)	Total (all studies)* RR/MD [95% CI]	Low risk of bias for domain 1 (randomisation process) studies only RR/MD [95% CI]	Studies at low risk of bias for domain 1
Comparison 1: Oral iron versus no iron			
Deferral (1)	RR 0.39 [0.20, 0.76] 7 RCTs, N = 1647 *Favours oral iron*	RR 0.34 [0.22, 0.53] 3 RCTs, N = 700 *Favours oral iron*	Kiss 2015 (REDS-III HEIRS) Marks 2013 (FIRST) Radtke 2004b
Deferral (2)	RR 0.38 [0.16, 0.91] 4 RCTs, N = 936 *Favours oral iron*	RR 0.24 [0.12, 0.48] 1 RCT, N = 220 *Favours oral iron*	Radtke 2004b
Deferral (3)	RR 0.43 [0.15, 1.24] 4 RCTs, N = 702 *No difference*	**RR 0.26 [0.15, 0.46]** **1 RCT, N = 211** ***Favours oral iron***	Radtke 2004b
Hb (1)	MD 2.52 [0.77, 4.26] 14 RCTs, N = 1940 *Favours oral iron*	MD 5.15 [3.42, 6.88] 4 RCTs, N = 606 *Favours oral iron*	Kiss 2015 (REDS-III HEIRS) Marks 2013 (FIRST) Simon 1984 Waldvogel 2012
Hb (2)	MD 1.30 [−6.05 8.65] 4 RCTs, N = 454 *No difference*	**MD 10.50 [5.72, 15.28]** **1 RCT, N = 63** ***Favours oral iron***	Simon 1984
Hb (3)	MD 4.91 [0.66, 9.16] 5 RCTs, N = 622 *Favours oral iron*	**MD 7.40 [−0.25, 15.04]** **2 RCTs, N = 246** ***No difference***	Mast 2016 (STRIDE) Simon 1984
Adverse events (1)	RR 1.69 [1.15, 2.47] 8 RCTs, N = 2641 *Favours no iron*	**RR 1.71 [0.63, 4.66]** **3 RCTs, N = 639** ***No difference***	Marks 2013 (FIRST) Kiss 2015 (REDS-III HEIRS) Waldvogel 2012
Adverse events (2)	RR 3.15 [1.54, 6.46] 1 RCT, N = 314 *Favours no iron*	**No studies (all excluded)**	N/A
Adverse events (3)	RR 1.46 [0.94, 2.25]	RR 2.23 [0.77, 6.47]	Mast 2016 (STRIDE)

Table 5. Sensitivity analyses: Cochrane RoB 2, domain 1 (randomisation process) *(Continued)*

	4 RCTs, N = 1216 *No difference*	1 RCT, N = 416 *No difference*	
Comparison 2: IV iron versus no iron			
Deferral (1 & 2)	Not reported		N/A
Hb (1)	MD 8.02 [3.22, 12.83] 3 RCTs, N = 558 *Favours IV iron*	Same studies	Fontana 2014a (ISUB) Gybel-Brask 2018 Hod 2022 (DIDS)
Hb (2)	MD 13.90 [9.50, 18.30] 1 RCT, N = 74 *Favours IV iron*	Same study	Gybel-Brask 2018
Adverse events (1)	RR 0.85 [0.17, 4.14] 2 RCTs, N = 484 *No difference*	Same studies	Fontana 2014a (ISUB) Hod 2022 (DIDS)
Adverse events (2)	RR 0.90 [0.69, 1.18] 1 RCT, N = 82 *No difference*	Same study	Gybel-Brask 2018
Comparison 2: Oral iron versus IV iron			
Deferral (1,2,3)	Not reported		N/A
Hb (1,2,3)	Not reported		N/A
Adverse events (1)	RR 1.00 [0.68, 1.56] 1 RCT, N = 172 *No difference*	**No studies (all excluded)**	N/A
Adverse events (2,3)	Not reported		N/A

*None of the studies contributing data to these comparisons and outcomes were at high risk of bias for domain 1 (randomisation process).
Bolded cells show a change in direction of the effect when focusing on only studies at low risk of bias for randomisation (from favouring iron to no difference/crossing the line of no effect, or vice versa), or where all studies were removed, as none were at low risk of bias for this domain.
Abbreviations: CI: confidence interval; Hb: haemoglobin; MD: mean difference; N: number of participants; N/A: not applicable; RCT: randomised controlled trial; RR: risk ratio
Time points:
(1) before/at 1st donation since treatment
(2) before/at 2nd donation since treatment
(3) after multiple donations since treatment

Table 6. Overview of results: Comparison 1. Oral iron versus none (subgrouping: donor characteristics)

Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Before/at 1st donation since treatment							
Deferrals	RR 0.39 [0.20, 0.76]** 7 RCTs, N = 1647 #○○○	RR 0.32 [0.17, 0.59]** 3 RCTs, N = 538	RR 0.43 [0.19, 0.98]** 6 RCTs, N = 1109		RR 0.35 [0.20, 0.61]** 6 RCTs, N = 1283	RR 0.47 [0.14, 1.57] 3 RCTs, N = 364	
Hb (g/L)	MD 2.52 [0.77, 4.26]** 14 RCTs, N = 1940 #○○○	MD 1.97 [−1.86, 5.80] 6 RCTs, N = 571	MD 2.69 [0.53, 4.85]** 10 RCTs, N = 1250	MD 3.0 [0.19, 5.81]** 1 RCT, N = 119	MD 2.28 [0.04, 4.52]** 12 RCTs, N = 1487	MD 4.9 [0.86, 8.93]** 4 RCTs, N = 504	
Serum/plasma ferritin (ng/mL)	MD 12.39 [8.16, 16.62]** 8 RCTs, N = 1335 #○○○	MD 9.67 [1.75, 17.59]** 4 RCTs, N = 462	MD 13.34 [8.08, 18.60]** 6 RCTs, N = 873		MD 12.07 [5.45, 18.68]** 6 RCTs, N = 974	MD 15.27 [7.94, 22.60]** 3 RCTs, N = 360	
Serum/plasma ferritin (geometric mean) (ng/mL) *$ - iron status (healthy/low only)*	RoGM 4.20 [3.34, 5.28]** 3 RCTs, N = 371 #○○○		RoGM 4.20 [3.34, 5.28]** 3 RCTs, N = 371		RoGM 4.22 [3.60, 4.95]** 3 RCTs, N = 227	RoGM 6.30 [5.17, 7.67]** 1 RCT, N = 144	
MCV (fl) *$ - sex (male/female only)*	MD 0.56 [−0.33, 1.45] 5 RCTs, N = 430	MD −0.23 [−1.47, 1.01] 3 RCTs, N = 245	MD 1.39 [0.11, 2.67]** 3 RCTs, N = 185		MD 0.50 [−0.40, 1.39] 5 RCTs, N = 430		
Serum/plasma iron (µg/dL)	MD 5.99 [−1.81, 13.78] 6 RCTs, N = 675	MD −4.53 [−18.59, 9.54] 2 RCTs, N = 119	MD 4.15 [−2.95, 11.25] 4 RCTs, N = 437	MD 23.12 [9.14, 37.10]** 1 RCT, N = 119	MD 6.54 [−1.89, 14.96] 6 RCTs, N = 675		

Table 6. Overview of results: Comparison 1. Oral iron versus none (subgrouping: donor characteristics) *(Continued)*

TIBC (µg/dL) *$ - sex (male/female only)*	**MD −24.39 [−40.43, −8.35]**** **5 RCTs, N = 585**	MD −2.24 [−22.00, 17.52] 1 RCT, N = 85	**MD −33.28 [−45.69, −20.86]**** **5 RCTs, N = 500**		**MD −31.50 [−43.59, −19.42]**** **5 RCTs, N = 585**		
Transferrin saturation (%) *$ - iron status (healthy/low only)*	**MD 3.60 [1.41, 5.80]**** **7 RCTs, N = 860**	MD 0.36 [−3.58, 4.30] 2 RCTs, N = 183	**MD 4.24 [1.03, 7.45]**** **5 RCTs, N = 558**	**MD 4.61 [2.04, 7.19]**** **1 RCT, N = 119**	**MD 3.16 [1.55, 4.76]**** **6 RCTs, N = 739**	**MD 10.00 [5.62, 14.38]**** **1 RCT, N = 121**	
Adverse events *$- sex (male/female only)*	**RR 1.69 [1.15, 2.47]^^** **8 RCTs, N = 2641** #○○○	**RR 3.62 [1.56, 8.38]^^** **2 RCTs, N = 189**	RR 1.54 [0.96, 2.48] 4 RCTs, N = 864	**RR 1.46 [1.15, 1.85]^^** **2 RCTs, N = 1588**	**RR 1.64 [1.22, 2.21]^^** **6 RCTs, N = 2090**	**RR 2.54 [1.39, 4.64]^^** **3 RCTs, N = 269**	RR 1.02 [0.91, 1.14] 1 RCT, N = 282
Adherence (n/N)	**RR 0.93 [0.90, 0.97]^^** **5 RCTs, N = 1685** ##○○		RR 0.96 [0.88, 1.04] 4 RCTs, N = 1166	**RR 0.93 [0.89, 0.97]^^** **1 RCT, N = 1166**	**RR 0.92 [0.89, 0.96]^^** **4 RCTs, N = 1403**		RR 0.98 [0.89, 1.07] 1 RCT, N = 282
Adherence (mean % tablets)	MD 0.00 [−5.04, 5.04] 1 RCT, N = 145 ⊕⊕⊕⊕		MD 0.00 [−5.04, 5.04] 1 RCT, N = 145			MD 0.00 [−5.04, 5.04] 1 RCT, N = 145	
Before/at 2nd donation since treatment							
Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Deferrals *$ - iron status (healthy/low only)*	**RR 0.38 [0.16, 0.91]**** **4 RCTs, N = 936** #○○○	**RR 0.24 [0.12, 0.46]**** **2 RCTs, N = 382**	RR 0.52 [0.19, 1.43] 3 RCTs, N = 554		**RR 0.28 [0.09, 0.84]**** **3 RCTs, N = 830**	RR 0.81 [0.54, 1.22] 1 RCT, N = 106	
Hb (g/L)	MD 1.30 [−6.05, 8.65] 4 RCTs, N = 454	MD −3.5 [−10.65, 3.65]	MD 2.69 [−6.63, 12.01]		MD 0.35 [−9.84, 10.53] 3 RCTs, N = 348	**MD 4.0 [1.23, 6.77]****	

Table 6. Overview of results: Comparison 1. Oral iron versus none (subgrouping: donor characteristics) *(Continued)*

	⊕○○○	1 RCT, N = 33	3 RCTs, N = 421		**1 RCT, N = 106**
Serum/plasma ferritin (ng/ml) *$ - iron status (healthy/low only)*	**MD 8.92 [3.86, 13.98]**** **4 RCTs, N = 694** **#○○○**	MD 6.20 [−1.99, 14.40] 2 RCTs, N = 195	**MD 9.54 [3.07, 16.00]**** **3 RCTs, N = 499**	**MD 5.99 [0.92, 11.06]**** **3 RCTs, N = 588**	**MD 13.00 [8.62, 17.38]**** **1 RCT, N = 106**
Serum/plasma ferritin (geometric mean) (ng/mL)	**RoGM 7.70 [5.76, 10.28]**** **1 RCT, N = 63** **#○○○**		**RoGM 7.70 [5.76, 10.28]**** **1 RCT, N = 63**	**RoGM 7.70 [5.76, 10.28]**** **1 RCT, N = 63**	
MCV (fl)	MD 0.40 [−1.88, 2.68] 1 RCT, N = 33	MD 0.40 [−1.88, 2.68] 1 RCT, N = 33		MD 0.40 [−1.88, 2.68] 1 RCT, N = 33	
Serum/plasma iron (µg/dL)	**MD 8.49 [2.03, 14.95]**** **2 RCTs, N = 285**	MD −2.50 [−45.24, 40.24] 1 RCT, N = 33	**MD 8.75 [2.21, 15.29]**** **1 RCT, N = 252**	**MD 8.49 [2.03, 14.95]**** **2 RCTs, N = 285**	
TIBC (µg/dL)	**MD −30.25 [−44.44, −16.07]**** **2 RCTs, N = 315**		**MD −30.25 [−44.44, −16.07]**** **2 RCTs, N = 315**	**MD −30.25 [−44.44, −16.07]**** **2 RCTs, N = 315**	
Transferrin saturation (%) *$ - iron status (healthy/low only)*	MD 8.26 [−1.86, 18.37] 2 RCTs, N = 358		MD 8.26 [−1.86, 18.37] 2 RCTs, N = 358	MD 3.62 [1.17, 6.07] 1 RCT, N = 252	MD 14.00 [6.93, 21.07] 1 RCT, N = 106
Adverse events	**RR 3.15 [1.54, 6.46]^^** **1 RCT, N = 314** **##○○**		**RR 3.15 [1.54, 6.46]^^** **1 RCT, N = 314**	**RR 3.15 [1.54, 6.46]^^** **1 RCT, N = 314**	
Adherence (n/N)					

Table 6. Overview of results: Comparison 1. Oral iron versus none (subgrouping: donor characteristics) *(Continued)*

Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
Adherence (mean % tablets)							
After multiple donations since treatment							
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Deferrals *$ - sex (male/female only)* *$ - iron status (healthy/low only)*	RR 0.43 [0.15, 1.24] 4 RCTs, N = 702 ⊕○○○	**RR 0.25 [0.15, 0.43]**** **2 RCTs, N = 359**	RR 0.89 [0.41, 1.96] 2 RCTs, N = 343		**RR 0.27 [0.17, 0.45]**** **3 RCTs, N = 611**	RR 1.11 [0.75, 1.66] 1 RCT, N = 91	
Hb (g/L)	**MD 4.91 [0.66, 9.16]**** **5 RCTs, N = 622** #○○○	MD 1.73 [−5.72, 9.18] 2 RCTs, N = 122	**MD 6.28 [0.90, 11.67]**** **4 RCTs, N = 500**		MD 5.57 [−4.22, 15.35] 3 RCTs N = 348	**MD 5.0 [2.23, 7.77]**** **1 RCT, N = 91**	MD 3.0 [−0.99, 6.99] 1 RCT, N = 183
Serum/plasma ferritin (ng/mL)	**MD 8.28 [4.76, 11.81]**** **4 RCTs, N = 524** ##○○	MD 7.16 [−1.43, 15.75] 2 RCTs, N = 181	**MD 8.76 [4.11, 13.42]**** **2 RCTs, N = 343**		**MD 7.05 [2.99, 11.12]**** **3 RCTs, N = 433**	**MD 12.00 [4.93, 19.07]**** **1 RCT, N = 91**	
Serum/plasma ferritin (geometric mean) (ng/mL)	**RoGM 7.76 [5.00, 12.02]**** **2 RCTs, N = 251** ##○○	**RoGM 7.95 [5.27, 11.98]**** **1 RCT, N = 251**	**RoGM 7.65 [3.93, 14.89]**** **2 RCTs, N = 155**		**RoGM 13.45 [9.09, 19.91]**** **1 RCT, N = 63**		**RoGM 5.78 [4.48, 7.45]**** **1 RCT, N = 188**
MCV (fl)	MD 0.85 [−1.54, 3.24] 1 RCT, N = 33	MD 0.85 [−1.54, 3.24] 1 RCT, N = 33			MD 0.85 [−1.54, 3.24] 1 RCT, N = 33		
Serum/plasma iron (µg/dL)	**MD 8.59 [2.00, 15.18]**** **2 RCTs, N = 285**	MD 21.50 [−7.52, 50.52]	**MD 7.89 [1.13, 14.65]****		**MD 7.89 [1.13, 14.65]****		

Table 6. Overview of results: Comparison 1. Oral iron versus none (subgrouping: donor characteristics) *(Continued)*

		1 RCT, N = 33	**1 RCT, N = 252**		**1 RCT, N = 252**		
TIBC (µg/dL)	**MD −40.24 [−54.05, −26.43]**** **2 RCTs, N = 315**		**MD −40.24 [−54.05, −26.43]**** **2 RCTs, N = 315**		**MD −40.24 [−54.05, −26.43]**** **2 RCTs, N = 315**		
Transferrin saturation (%)	**MD 4.84 [2.78, 6.90]**** **2 RCTs, N = 343**		**MD 4.84 [2.78, 6.90]**** **2 RCTs, N = 343**		**MD 4.81 [2.59, 7.03]**** **1 RCT, N = 252**	MD 5.00 [−0.54, 10.54] 1 RCT, N = 91	
Adverse events	RR 1.46 [0.94, 2.25] 4 RCTs, N = 1216 ⊕◯◯◯	RR 0.75 [0.15, 3.85] 1 RCT, N = 33	**RR 2.15 [1.06, 4.34]^^** **1 RCT, N = 241**	RR 1.34 [0.72, 2.50] 2 RCTs, N = 942	RR 1.36 [0.81, 2.29] 3 RCTs, N = 800		RR 2.23 [0.77, 6.47] 1 RCT, N = 416
Adherence (n/N)							
Adherence (mean % tablets)	MD 0.10 [−0.13, 0.33] 1 RCT, N = 91 ⊕⊕◯◯		MD 0.10 [−0.13, 0.33] 1 RCT, N = 91			MD 0.10 [−0.13, 0.33] 1 RCT, N = 91	

Abbreviations: CI: confidence interval; Hb: haemoglobin; MCV: mean cell volume; MD: mean difference; N: number of participants contributing to the analysis; NR: not reported; RCT: randomised controlled trial; RoGM: ratio of geometric means; RR: risk ratio; TIBC: total iron binding capacity
Certainty of the evidence: ⊕⊕⊕⊕ high; ⊕⊕⊕◯ moderate; ⊕⊕◯◯ low; ⊕◯◯◯ very low
Bolded cells indicate a difference between oral iron group and no iron group. In each case: **favours oral iron, and ^^no iron.
Blank cells are where there were no data available for this comparison, time point, outcome, and subgroup/characteristic.
$ indicates that the test for subgroup differences reached P < 0.1, and was investigated for credibility using the ICEMAN assessment tool (see Table 10).

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration)

Outcome	Subgroup: daily elemental iron dose				Subgroup: duration of supplementation			
	Very high	High	Normal	Low	< 2 weeks	2 weeks to 1 month	1 to 3 months	3+ months
Before/at 1st donation since treatment								

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration) *(Continued)*

Deferrals		RR 0.79 [0.21, 2.88] 1 RCT, N = 361	**RR 0.37 [0.16, 0.85]**** **6 RCTs, N = 1152**	RR 0.33 [0.09, 1.22] 1 RCT, N = 252	RR 0.79 [0.21, 2.88] 1 RCT, N = 361		**RR 0.35 [0.14, 0.91]**** **5 RCTs, N = 1079**	**RR 0.40 [0.21, 0.77]**** **1 RCT, N = 207**
Hb (g/L) *$ - daily dose*	MD −1.0 [−7.2, 5.2] 1 RCT, N = 34	MD 1.5 [−0.09, 3.10] 4 RCTs, N = 505	**MD 3.53 [0.93, 6.13]**** **10 RCTs, N = 1408**	**MD −8.0 [−15.07, −0.93]^^** **1 RCT, N = 23**	MD 0.35 [−1.21, 1.91] 3 RCTs, N = 617	MD 1.75 [−4.57, 8.06] 4 RCTS, N = 424	**MD 4.08 [1.31, 6.84]**** **5 RCTs, N = 564**	MD 1.94 [−4.39, 8.27] 2 RCTs, N = 335
Serum/plasma ferritin (ng/mL) *$ - duration*		**MD 6.59 [1.95, 11.23]**** **1 RCT, N = 252**	**MD 13.40 [8.34, 18.46]**** **7 RCTs, N = 936**	**MD 6.76 [0.35, 13.17]**** **2 RCTs, N = 291**	**MD 6.59 [1.95, 11.23]**** **1 RCT, N = 252**	**MD 14.75 [11.91, 17.60]**** **3 RCTs, N = 305**	**MD 10.54 [4.92, 16.17]**** **3 RCTs, N = 601**	**MD 25.00 [15.84, 34.16]**** **1 RCT, N = 192**
Serum/plasma ferritin (geometric mean) (ng/mL) *$ - daily dose* *$ - duration*	**RoGM 3.32 [2.57, 4.29]**** **1 RCT, N = 34**	**RoGM 3.32 [2.57, 4.29]**** **1 RCT, N = 36**	**RoGM 4.83 [4.15, 5.63]**** **2 RCTs, N = 320**		**RoGM 3.32 [2.77, 3.98]**** **1 RCT, N = 51**		**RoGM 4.83 [4.15, 5.63]**** **2 RCTs, N = 320**	
MCV (fl)	MD 0.00 [−2.77, 2.77] 1 RCT, N = 34	MD 1.00 [−1.77, 3.77] 1 RCT, N = 36	MD 0.50 [−0.51, 1.51] 4 RCTs, N = 367	MD −0.20 [−2.43, 2.03] 1 RCT, N = 23	MD 0.48 [−1.92, 2.88] 1 RCT, N = 51	MD −0.33 [−1.82, 1.16] 2 RCTs, N = 160	MD 2.00 [−0.03, 4.03] 1 RCT, N = 76	MD 0.50 [−1.13, 2.13] 1 RCT, N = 143
Serum/plasma iron (µg/dL) *$ - duration*	MD 17.00 [−12.92, 46.92] 1 RCT, N = 34	MD 11.15 [−2.01, 24.30] 3 RCTs, N = 407	MD 0.45 [−7.65, 8.54] 3 RCTs, N = 241	MD −3.00 [−49.75, 43.75] 1 RCT, N = 23	MD 5.59 [−4.77, 15.96] 2 RCTs, N = 303	**MD 21.03 [7.70, 34.36]**** **2 RCTs, N = 153**	MD 3.00 [−8.05, 14.05] 1 RCT, N = 76	MD −2.84 [−15.08, 9.40] 1 RCT, N = 143
TIBC (µg/dL)	MD 5.00 [−42.20, 52.20]	**MD −34.53 [−54.10, −14.96]****	**MD −30.63 [−48.86, −12.39]****		**MD −30.67 [−50.30, −11.04]****		**MD −35.82 [−52.22, −19.43]****	MD 1.98 [−45.55, 49.51]

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration) *(Continued)*

	1 RCT, N = 34	**2 RCTs, N = 288**	**3 RCTs, N = 282**		**2 RCTs, N = 303**		**2 RCTs, N = 139**	1 RCT, N = 143
Transferrin saturation (%)	MD 3.00 [−5.32, 11.32] 1 RCT, N = 34	**MD 3.71 [1.97, 5.45]**** **4 RCTs, N = 505**	MD 4.27 [−1.97, 10.50] 3 RCTs, N = 340		**MD 3.31 [0.72, 5.90]**** **2 RCTs, N = 303**	**MD 4.61 [2.04, 7.19]**** **1 RCT, N = 119**	**MD 5.40 [0.72, 10.09]**** **3 RCTs, N = 295**	MD −1.21 [−5.64, 3.22] 1 RCT, N = 143
Adverse events *$ - duration*	**RR 1.40 [1.01, 1.94]^^** **1 RCT, N = 46**	**RR 1.44 [1.13, 1.84]^^** **4 RCTs, N = 1836**	RR 2.17 [0.78, 6.02] 4 RCTs, N = 782		**RR 1.48 [1.05, 2.09]^^** **2 RCTs, N = 437**	**RR 2.13 [1.22, 3.73]^^** **3 RCTs, N = 1664**	RR 1.02 [0.91, 1.14] 2 RCTs, N = 328	RR 4.82 [0.23, 99.14] 1 RCT, N = 212
Adherence (n/N)	RR 0.83 [0.59, 1.16] 1 RCT, N = 45	**RR 0.93 [0.89, 0.97]^^** **2 RCTs, N = 1212**	RR 0.97 [0.88, 1.06] 3 RCTs, N = 451		RR 0.86 [0.66, 1.12] 1 RCT, N = 68	**RR 0.93 [0.89, 0.97]^^** **1 RCT, N = 1166**	RR 0.97 [0.88, 1.06] 3 RCTs, N = 451	
Adherence (mean % tablets)			MD 0.00 [−5.04, 5.04] 1 RCT, N = 145			MD 0.00 [−5.04, 5.04] 1 RCT, N = 145		
Before/at 2nd donation since treatment								
Outcome	Subgroup: daily elemental iron dose				Subgroup: duration of supplementation			
	Very high	High	Normal	Low	< 2 weeks	2 weeks to 1 month	1 to 3 months	3+ months
Deferrals		RR 0.74 [0.26, 2.07] 1 RCT, N = 307	RR 0.35 [0.11, 1.14] 3 RCTs, N = 526	**RR 0.06 [0.00, 0.96]**** **1 RCT, N = 191**	RR 0.74 [0.26, 2.07] 1 RCT, N = 307		**RR 0.30 [0.10, 0.88]**** **3 RCTs, N = 639**	

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration) *(Continued)*

Hb (g/L) *$ - daily dose* *$ - duration*	**MD −5.9 [−8.12, −3.68]^^** **1 RCT, N = 252**	MD 4.09 [−3.43, 11.62] 3 RCTs, N = 191	MD −3.0 [−11.32, 5.32] 1 RCT, N = 22	**MD −5.9 [−8.12, −3.68]^^** **1 RCT, N = 252**	MD −3.53 [−10.74, 3.68] 1 RCT, N = 33	**MD 6.95 [0.6, 13.29]**** **2 RCTs, N = 169**
Serum/plasma ferritin (ng/mL) *$ - daily dose* *$ - duration*	MD 3.33 [−0.56, 7.22] 1 RCT, N = 252	**MD 11.95 [8.52, 15.38]**** **3 RCTs, N = 329**	**MD 7.41 [0.31, 14.51]**** **2 RCTs, N = 212**	MD 3.33 [−0.56, 7.22] 1 RCT, N = 252	MD 11.93 [−11.95, 35.80] 1 RCT, N = 33	**MD 11.41 [8.00, 14.82]**** **2 RCTs, N = 410**
Serum/plasma ferritin (geometric mean) (ng/mL)		**RoGM 7.70 [5.76, 10.28]**** **1 RCT, N = 63**				**RoGM 7.70 [5.76, 10.28]**** **1 RCT, N = 63**
MCV (fl)		MD 0.20 [−2.19, 2.59] 1 RCT, N = 22	MD 0.60 [−2.17, 3.37] 1 RCT, N = 22		MD 0.37 [−1.91, 2.64] 1 RCT, N = 33	
Serum/plasma iron (µg/dL)	**MD 8.75 [2.21, 15.29]**** **1 RCT, N = 252**	MD −7.00 [−53.75, 39.75] 1 RCT, N = 22	MD 2.00 [−43.71, 47.71] 1 RCT, N = 22	**MD 8.75 [2.21, 15.29]** **1 RCT, N = 252****	MD −2.78 [−45.71, 40.14] 1 RCT, N = 33	
TIBC (µg/dL)	**MD −25.07 [−42.63, −7.51]**** **1 RCT, N = 252**	**MD −40.00 [−64.08, −15.92]**** **1 RCT, N = 63**		**MD −25.07 [−42.63, −7.51]**** **1 RCT, N = 252**		**MD −40.00 [−64.08, −15.92]**** **1 RCT, N = 63**
Transferrin saturation (%) *$ - daily dose* *$ - duration*	**MD 3.62 [1.17, 6.07]**** **1 RCT, N = 252**	**MD 14.00 [6.93, 21.07]**** **1 RCT, N = 106**		**MD 3.62 [1.17, 6.07]**** **1 RCT, N = 252**		**MD 14.00 [6.93, 21.07]**** **1 RCT N = 106**
Adverse events	**RR 3.15 [1.54, 6.46]^^**			**RR 3.15 [1.54, 6.46]^^**		

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration) *(Continued)*

Outcome	Subgroup: daily elemental iron dose				Subgroup: duration of supplementation			
	Very high	High	Normal	Low	< 2 weeks	2 weeks to 1 month	1 to 3 months	3+ months
		1 RCT, N = 314			**1 RCT, N = 314**			
Adherence (n/N)								
Adherence (mean % tablets)								
After multiple donations since treatment								
Deferrals		RR 0.45 [0.12, 1.78] 1 RCT, N = 252	RR 0.44 [0.12, 1.65] 3 RCTs, N = 396	RR 0.19 [0.02, 1.59] 1 RCT, N = 94	RR 0.45 [0.12, 1.78] 1 RCT, N = 252		RR 0.39 [0.12, 1.30] 3 RCTs, N = 450	
Hb (g/L)		**MD 3.7 [1.48, 5.92]**** **1 RCT, N = 252**	MD 4.99 [−2.38, 12.36] 4 RCTs, N = 300	MD 2.06 [−1.97, 6.09] 2 RCTs, N = 132	**MD 3.7 [1.48, 5.92]**** **1 RCT, N = 252**	MD −2.75 [−9.54, 4.03] 1 RCT, N = 33	**MD 6.57 [1.01, 12.13]**** **3 RCTs, N = 337**	
Serum/plasma ferritin (ng/mL)		**MD 7.02 [2.40, 11.64]**** **1 RCT, N = 252**	**MD 11.61 [5.87, 17.35]**** **3 RCTs, N = 207**	MD 3.40 [−5.52, 12.32] 2 RCTs, N = 116	**MD 7.02 [2.40, 11.64]**** **1 RCT, N = 252**	MD 5.13 [−14.00, 24.26] 1 RCT, N = 33	**MD 10.40 [4.71, 16.09]**** **2 RCTs, N = 239**	
Serum/plasma ferritin (geometric mean) (ng/mL)			**RoGM 9.24 [5.20, 16.45]**** **2 RCTs, N = 190**	**RoGM 5.39 [3.70, 7.84]**** **1 RCT, N = 113**			**RoGM 7.76 [5.00, 12.02]**** **2 RCTs, N = 251**	
MCV (fl)			MD 0.60 [−1.86, 3.06] 1 RCT, N = 22	MD 1.10 [−1.69, 3.89] 1 RCT, N = 22		MD 0.81 [−1.57, 3.20] 1 RCT, N = 33		

Table 7. Overview of results: Comparison 1. Oral iron versus none (subgrouping: dose and duration) *(Continued)*

Serum/plasma iron (µg/dL)	**MD 7.89 [1.13, 14.65]** 1 RCT, N = 252**	MD 31.00 [−3.68, 65.68] 1 RCT, N = 22	MD 12.00 [−16.94, 40.94] 1 RCT, N = 22	**MD 7.89 [1.13, 14.65]** 1 RCT, N = 252**	MD 21.14 [−7.82, 50.10] 1 RCT, N = 33	
TIBC (µg/dL)	**MD −35.45 [−52.31, −18.59]** 1 RCT, N = 252**	**MD −50.00 [−74.08, −25.92]** 1 RCT, N = 63**		**MD −35.45 [−52.31, −18.59]** 1 RCT, N = 252**		**MD −50.00 [−74.08, −25.92]** 1 RCT, N = 63**
Transferrin saturation (%)	**MD 4.81 [2.59, 7.03]** 1 RCT, N = 252**	MD 5.00 [−0.54, 10.54] 1 RCT, N = 91		**MD 4.81 [2.59, 7.03]** 1 RCT, N = 252**		MD 5.00 [−0.54, 10.54] 1 RCT, N = 91
Adverse events	RR 2.15 [1.06, 4.34] 1 RCT, N = 241	RR 1.32 [0.56, 3.12] 3 RCTs, N = 650	RR 1.23 [0.75, 2.03] 3 RCTs, N = 649	**RR 2.15 [1.06, 4.34]^^ 1 RCT, N = 241**	RR 0.74 [0.14, 3.86] 1 RCT, N = 33	RR 1.24 [0.78, 1.97] 2 RCTs, N = 942
Adherence (n/N)						
Adherence (mean % tablets)		MD 0.10 [−0.13, 0.33] 1 RCT, N = 91				MD 0.10 [−0.13, 0.33] 1 RCT, N = 91

Data presented as **(MD/RR) [95% CI]**.
Abbreviations: CI: confidence interval, Hb: haemoglobin; MCV: mean cell volume; MD: mean difference; N: number of participants contributing to the analysis; RCT: randomised controlled trial; RoGM: ratio of geometric means; RR: risk ratio; TIBC: total iron binding capacity
Bolded cells indicate a difference between oral iron group and no iron group. In each case: **favours oral iron, and ^^no iron.
Blank cells are where there were no data available for this comparison, time point, outcome, and subgroup/characteristic.
$ indicates that the test for subgroup differences reached P < 0.1, and was investigated for credibility using the ICEMAN assessment tool (see Table 10).

Table 8. Overview of results: Comparison 1. Oral iron versus none (subgrouping: iron preparation)

Outcome	Subgroup: iron preparation					
	FE2+	FE3+	FE2+ with additives	FE3+ with additives	"iron" with additives	Haem iron (combi)
Before/at 1st donation since treatment						
Deferrals	RR 0.74 [0.31, 1.76] 3 RCTs, N = 689	**RR 0.21 [0.09, 0.50]**** **2 RCTs, N = 328**	**RR 0.31 [0.17, 0.55]**** **2 RCTs, N = 630**			
Hb (g/L)	MD 2.54 [−0.09, 5.17] 9 RCTs, N = 1057	MD 3.58 [−0.13, 7.28] 4 RCTs, N = 414	MD 2.64 [−4.24, 9.52] 2 RCTs, N = 345		MD −1.47 [−5.96, 3.02] 1 RCT, N = 143	
Serum/plasma ferritin (ng/mL) *$ - preparation*	**MD 13.39 [8.47, 18.32]**** **7 RCTs, N = 940**		**MD 6.67 [1.49, 11.84]**** **1 RCT, N = 410**			
Serum/plasma ferritin (geometric mean) (ng/mL)	**RoGM 3.89 [2.65, 5.71]**** **2 RCTs, N = 77**	**RoGM 4.11 [2.77, 6.10]**** **2 RCTs, N = 291**	**RoGM 4.06 [2.68, 6.13]**** **1 RCT, N = 41**			
MCV (fl)	MD −0.03 [−1.35, 1.28] 3 RCTs, N = 196	MD 1.23 [−0.67, 3.14] 2 RCTs, N = 110			MD 0.50 [−1.13, 2.13] 1 RCT, N = 143	
Serum/plasma iron (µg/dL)	MD 10.40 [−1.63, 22.43] 4 RCTs, N = 441	MD 4.68 [−5.68, 15.04] 2 RCTs, N = 110			MD −2.84 [−15.08, 9.40] 1 RCT, N = 143	
TIBC (µg/dL)	**MD −35.32 [−48.34, −22.30]**** **4 RCTs, N = 405**	MD 5.00 [−42.20, 52.20] 1 RCT, N = 43	**MD −30.00 [−57.88, −2.12]**** **1 RCT, N = 41**		MD 1.98 [−45.55, 49.51] 1 RCT, N = 143	
Transferrin saturation (%) *$ - preparation*	**MD 5.03 [1.94, 8.12]**** **4 RCTs, N = 528**	**MD 3.31 [0.46, 6.16]****			MD −1.21 [−5.64, 3.22]	

Table 8. Overview of results: Comparison 1. Oral iron versus none (subgrouping: iron preparation) *(Continued)*

		3 RCTs, N = 208			1 RCT, N = 143	
Adverse events	**RR 1.67 [1.14, 2.47]^^** **7 RCTs, N = 2572**	**RR 1.40 [1.01, 1.94]^^** **2 RCTs, N = 92**				
Adherence (n/N)	**RR 0.93 [0.89, 0.97]^^** **3 RCTs. N = 1282**	RR 0.96 [0.88, 1.05] 3 RCTs, N = 426				
Adherence (mean % tablets)	MD 0.00 [−5.04, 5.04] 1 RCT, N = 145					
Before/at 2nd donation since treatment						
Outcome	Subgroup: iron preparation					
	FE2+	FE3+	FE2+ with additives	FE3+ with additives	“iron” with additives	Haem iron (combi)
Deferrals *$ - preparation*	RR 0.80 [0.55, 1.17] 2 RCTs, N = 413		**RR 0.21 [0.11, 0.40]**** **2 RCTs, N = 533**			
Hb (g/L) *$ - preparation*	MD 0.57 [−5.65, 6.78] 4 RCTs, N = 432		**MD 110 [5.46, 16.54]**** **1 RCT, N = 41**			
Serum/plasma ferritin (ng/mL)	**MD 8.49 [0.53, 16.45]**** **3 RCTs, N = 391**		**MD 8.98 [3.55, 14.40]**** **1 RCT, N = 304**			
Serum/plasma ferritin (geometric mean) (ng/mL)	**RoGM 8.73 [5.79, 13.17]**** **1 RCT, N = 41**		**RoGM 6.80 [4.52, 10.22]**** **1 RCT, N = 41**			
MCV (fl)	MD 0.37 [−1.91, 2.64] 1 RCT, N = 33					
Serum/plasma iron (µg/dL)	**MD 8.49 [2.03, 14.95]**** **2 RCTs, N = 285**					
TIBC (µg/dL)	**MD −29.31 [−44.17, −14.45]**** **2 RCTs, N = 293**		**MD −40.00 [−67.88, −12.12]**** **1 RCT, N = 41**			

Table 8. Overview of results: Comparison 1. Oral iron versus none (subgrouping: iron preparation) *(Continued)*

Transferrin saturation (%)	MD 8.26 [−1.86, 18.37] 2 RCTs, N = 358					
Adverse events	**RR 3.15 [1.54, 6.46]^^** **1 RCT, N = 314**					
Adherence (n/N)						
Adherence (mean % tablets)						
After multiple donations since treatment						
Outcome	Subgroup: iron preparation					
	FE2+	FE3+	FE2+ with additives	FE3+ with additives	“iron” with additives	Haem iron (combi)
Deferrals *$ - preparation*	RR 0.89 [0.41, 1.96] 2 RCTs, N = 343		**RR 0.25 [0.15, 0.43]**** **2 RCTs, N = 359**			
Hb (g/L) *$ - preparation*	**MD 4.16 [0.53, 7.78]**** **5 RCTs, N = 600**		**MD 16.0 [10.46, 21.54]**** **1 RCT, N = 41**			
Serum/plasma ferritin (ng/mL)	**MD 8.45 [4.41, 12.49]**** **3 RCTs, N = 376**		MD 7.45 [−2.14, 17.03] 1 RCT, N = 148			
Serum/plasma ferritin (geometric mean) (ng/mL)	**RoGM 7.21 [4.33, 12.01]**** **2 RCTs, N = 229**		**RoGM 11.02 [6.80, 17.87]**** **1 RCT, N = 41**			
MCV (fl)	MD 0.81 [−1.57, 3.20] 1 RCT, N = 33					
Serum/plasma iron (µg/dL)	**MD 8.58 [1.99, 15.16]**** **2 RCTs, N = 285**					
TIBC (µg/dL)	**MD −39.35 [−53.77, −24.92]**** **2 RCTs, N = 293**		**MD −50.00 [−77.88, −22.12]**** **1 RCT, N = 41**			
Transferrin saturation (%)	**MD 4.84 [2.78, 6.90]**** **2 RCTs, N = 343**					
Adverse events	**RR 1.91 [1.09, 3.32]^^**		MD 1.10 [0.66, 1.83]			

Table 8. Overview of results: Comparison 1. Oral iron versus none (subgrouping: iron preparation) *(Continued)*

	3 RCTs, N = 690	1 RCT, N = 526
Adherence(n/N)		
Adherence (mean % tablets)	MD 0.10 [−0.13, 0.33] 1 RCT, N = 91	

Data presented as **(MD/RR) [95% CI]**.
Abbreviations: CI: confidence interval; Hb: haemoglobin; MCV: mean cell volume; MD: mean difference; N: number of participants contributing to the analysis; RCT: randomised controlled trial; RoGM: ratio of geometric means; RR: risk ratio; TIBC: total iron binding capacity
Bolded cells indicate a difference between oral iron group and no iron group. In each case: **favours oral iron, and ^^no iron.
Blank cells are where there were no data available for this comparison, time point, outcome, and subgroup/characteristic.
$ indicates that the test for subgroup differences reached P < 0.1, and was investigated for credibility using the ICEMAN assessment tool (see Table 10).

Table 9. Oral iron versus no iron: quality of life and physical function data as presented by each study

Assessment time point*		**Assessment tool** range of scores, direction of tool	**Iron** Mean (SD) N	**No iron** Mean (SD) N	**Reported difference (95% CI)** *as reported by trialists, unless otherwise stated in Notes*	**Notes**
Exercise capacity						
Pachikian 2020 80 mg iron vs placebo	3	VE max L/min (aerobic capacity) *higher is better*	154 (SEM 11) N = 11	160 (SEM 8) N = 11	MD −6.00 (−32.66 to 20.66)	*Difference calculated by review authors.*
Pachikian 2020 20 mg iron vs placebo	3	VE max L/min (aerobic capacity) *higher is better*	169 (SEM 11) N = 11		MD 9.00 (−17.66 to 35.66)	*Difference calculated by review authors.*
Pachikian 2020 80 mg iron vs placebo	3	HR max bpm (aerobic capacity) *higher is better*	180 (SEM 3) N = 11	184 (SEM 3) N = 11	MD −4.00 (−12.32 to 4.32)	*Difference calculated by review authors.*
Pachikian 2020 20 mg iron vs placebo	3	HR max bpm (aerobic capacity) *higher is better*	191 (SEM 3) N = 11		MD 7.00 (−1.32 to 15.32)	*Difference calculated by review authors.*
Waldvogel 2012	1	Chester step test (aerobic capacity) *higher is better*	40.5 (14.5) N = 74	40.1 (17) N = 70	MD 0.02 (−4.8 to 4.8)	Adjusted for baseline menstrual bleeding

Table 9. Oral iron versus no iron: quality of life and physical function data as presented by each study *(Continued)*

Patient-reported fatigue scores						
Mast 2016 (STRIDE) 38 mg iron vs placebo	3	Fatigue	N = 66	Reference N = 45	RR 1.57 (0.86 to 2.89)	Adjusted models for fatigue at final visit weighted for inverse probability of censoring (using modified Poisson regression with generalised estimating equations).
Mast 2016 (STRIDE) 19 mg iron vs placebo	3	Fatigue	N = 51		RR 1.59 (0.94 to 2.69)	Adjusted models for fatigue at final visit weighted for inverse probability of censoring (using modified Poisson regression with generalised estimating equations).
Waldvogel 2012	1	Fatigue VAS 0 to 10; *lower is better*	3.4 (2.4) N = 74	3.5 (2.5) N = 71	MD −0.18 (−0.9 to 0.6)	Adjusted for baseline menstrual bleeding
Waldvogel 2012	1	FSS mean of 1 to 7 per domain; *lower is better*	2.5 (1.3) N = 74	2.6 (1.5) N = 70	MD −0.05 (−0.4 to 0.3)	Adjusted for baseline menstrual bleeding
Patient-reported physical health						
Waldvogel 2012	1	SF12 vitality 0% to 100%; *higher is better*	53.6 (12.7) N = 72	55.3 (12.3) N = 71	MD −0.13 (−3.8 to 3.6)	Adjusted for baseline menstrual bleeding
Waldvogel 2012	1	SF12 physical 0% to 100%; *higher is better*	54.8 (3.3) N = 65	52.4 (5.2) N = 68	MD 2.4 (1.1 to 3.7)	Adjusted for baseline menstrual bleeding
Patient-reported mental health						
Waldvogel 2012	1	SF12 mental 0% to 100%; *higher is better*	40.1 (4.8) N = 65	40.7 (4.8) N = 68	MD −0.5 (−2 to 1.1)	Adjusted for baseline menstrual bleeding
Waldvogel 2012	1	PHQ-9 depression 0 to 27; *lower is better*	3/74	4/71	RR 0.72 (0.17 to 3.10)	Number of people diagnosed with depression (PHQ-9 score > 15/27) *Difference calculated by review authors.*

Abbreviations: bpm: beats per minute; CI: confidence interval; FSS: Fatigue Severity Scale; HR max: maximum heart rate; MD: mean difference; PHQ-9: Patient Health Questionnaire-9; RR: risk ratio; SD: standard deviation; SEM: standard error of the mean; SF12: short-form 12; VAS: visual analogue scale; VE (max or peak): ventilatory equivalent
***Assessment time point:**
1: before/at first donation since treatment commenced
2: before/at 2nd donation since treatment commenced

3: after multiple donations

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none

Outcome	Test for subgroup differences	1: Within or between studies?	2: Effect modification within trials	3: Number of trials	4: Direction hypothesised a priori	5: Chance an explanation	6: Only a small number of modifiers tested	7: Random-effects model used	8: Arbitrary cutpoints avoided	9: Other considerations	Overall subgroup credibility
Oral versus none @ 1st donation – sex (male and female, ignoring mixed-sex data)											
MCV Analysis 1.5	$Chi^2 = 3.16$, df = 1 ($P = 0.08$), $I^2 = 68.3\%$	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Rather small** (3 to 4 in smallest subgroup) **LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a very likely explanation** (P > 0.05) **VERY LOW**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used in all analyses **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
TIBC Analysis 1.7	$Chi^2 = 6.80$, df = 1 ($P = 0.009$), $I^2 = 85.3\%$	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Adverse events Analysis 1.9	$Chi^2 = 2.99$, df = 1 ($P = 0.08$), $I^2 = 66.5\%$	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a very likely explanation** (P > 0.05) **VERY LOW**					**LOW:** 2 or more domains reducing credibility
Oral versus none after multiple donations – sex (male and female, ignoring mixed-sex data)											
Deferrals Analysis 3.1	$Chi^2 = 6.82$, df = 1 ($P = 0.009$), $I^2 = 85.3\%$	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**	**Definitely no** (exploratory analysis)	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none *(Continued)*

								VERY LOW				
Oral versus none @ 1st donation – baseline iron status (healthy, low Hb/ferritin, IDA, or NAID, ignoring mixed data)												
Serum ferritin (log-transformed data only) Analysis 4.4	Chi^2 = 9.58, df = 1 (P = 0.002), I^2 = 89.6%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility	
Transferrin saturation Analysis 4.8	Chi^2 = 8.26, df = 1 (P = 0.004), I^2 = 87.9%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**					**LOW:** 2 or more domains reducing credibility	
Oral versus none @ 2nd donation – baseline iron status (healthy, low Hb/ferritin, IDA, or NAID, ignoring mixed data)												
Deferrals Analysis 5.1	Chi^2 = 3.15, df = 1 (P = 0.08), I^2 = 68.2%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a very likely explanation** (P > 0.05) **VERY LOW**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used in all analyses **HIGH**	**Probably yes** (pre-specified by each study author) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility	
Ferritin (non-log data only) Analysis 5.3	Chi^2 = 4.20, df = 1 (P = 0.04), I^2 = 76.2%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a likely explanation** (P > 0.01 to </= 0.05) **LOW**					**LOW:** 2 or more domains reducing credibility	
Transferrin saturation	Chi^2 = 7.40, df = 1 (P = 0.007), I^2 = 86.5%	Completely between studies	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains re-	

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none *(Continued)*

Analysis 5.8		**VERY LOW**									ducing credibility
Oral versus none after multiple donations – baseline iron status (healthy, low Hb/ferritin, IDA, or NAID, ignoring mixed data)											
Deferrals Analysis 6.1	Chi^2 = 18.89, df = 1 (P < 0.001), I^2 = 94.7%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used in all analyses **HIGH**	**Probably yes** (pre-specified by each study author) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Oral versus none @ 1st donation – daily dose of elemental iron (low, normal, high, very high)											
Hb Analysis 7.2	Chi^2 = 9.90, df = 3 (P = 0.02), I^2 = 69.7%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a likely explanation** (P > 0.01 to </= 0.05) **LOW**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used in all analyses **HIGH**	**Probably yes** (taken from prior literature and previous RCTs, and defined prior to any analysis) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Serum ferritin (log-transformed data only) Analysis 7.4	Chi^2 = 9.73, df = 2 (P = 0.008), I^2 = 79.5%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Oral versus none @ 2nd donation – daily dose of elemental iron (low, normal, high, very high)											
Hb Analysis 8.2	Chi^2 = 6.46, df = 2 (P = 0.04), I^2 = 69.1%	Mostly between studies (some within)	**Unclear:** only 1 within-RCT	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a likely explanation** (P > 0.01 to </= 0.05) **LOW**	**Definitely no** (explorato-	Yes, random-effects model used in	**Probably yes** (taken from prior litera-	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none *(Continued)*

		LOW	comparison **LOW**				ry analysis) **VERY LOW**	all analyses **HIGH**	ture and previous RCTs, and defined prior to any analysis) **MODERATE**		
Ferritin (non-log data only) Analysis 8.3	Chi^2 = 10.65, df = 2 (P = 0.005), I^2 = 81.2%	Mostly between studies (some within) **LOW**	**Unclear:** 2 studies, but too imprecise to tell **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**					**LOW:** 2 or more domains reducing credibility
Transferrin saturation Analysis 8.8	Chi^2 = 7.40, df = 1 (P = 0.007), I^2 = 86.5%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Oral versus none @ 1st donation – treatment duration (< 2 weeks, 2 weeks to 1 month, 1 to 3 months, 3+ months)											
Ferritin (non-log data only) Analysis 10.3	Chi^2 = 16.12, df = 3 (P = 0.001), I^2 = 81.4%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used in all analyses **HIGH**	**Probably yes** (determined by expert consensus, and defined prior to any analysis) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Serum ferritin (log-transformed data only) Analysis 10.4	Chi^2 = 9.73, df = 1 (P = 0.002), I^2 = 89.7%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**					**LOW:** 2 or more domains reducing credibility
Serum/plasma iron	Chi^2 = 7.17, df = 3 (P = 0.07), I^2 = 58.2%	Completely between studies	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a very likely explanation** (P > 0.05) **VERY LOW**					**LOW:** 2 or more domains re-

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none *(Continued)*

Analysis 10.6		**VERY LOW**									ducing credibility
Adverse events Analysis 10.9	Chi² = 10.89, df = 3 (P = 0.01), I² = 72.4%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Oral versus none @ 2nd donation – treatment duration (< 2 weeks, 2 weeks to 1 month, 1 to 3 months, 3+ months)											
Hb Analysis 11.2	Chi² = 14.11, df = 2 (P < 0.001), I² = 85.8%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**Probably yes** (determined by expert consensus, and defined prior to any analysis) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Ferritin (non-log data only) Analysis 11.3	Chi² = 9.48, df = 2 (P = 0.009), I² = 78.9%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Transferrin saturation Analysis 11.8	Chi² = 7.40, df = 1 (P = 0.007), I² = 86.5%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Oral versus none @ 1st donation – iron preparation (FE2+, FE3+, FE2+ with additives, FE3+ with additives, haem iron combination)											
Ferritin (non-log data only) Analysis 13.3	Chi² = 3.41, df = 1 (P = 0.06), I² = 70.6%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a likely explanation** (P > 0.01 to </= 0.05) **LOW**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility

Table 10. ICEMAN subgroup credibility assessment: oral iron versus none *(Continued)*

Transferrin saturation Analysis 13.8	Chi^2 = 5.15, df = 2 (P = 0.08), I^2 = 61.1%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance a very likely explanation** (P > 0.05) **VERY LOW**					**LOW:** 2 or more domains reducing credibility
Oral versus none @ 2nd donation – iron preparation (FE2+, FE3+, FE2+ with additives, FE3+ with additives, haem iron combination)											
Deferrals Analysis 14.1	Chi^2 = 12.59, df = 1 (P < 0.001), I^2 = 92.1%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Hb Analysis 14.2	Chi^2 = 6.03, df = 1 (P = 0.01), I^2 = 83.4%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**					**LOW:** 2 or more domains reducing credibility
Oral versus none after multiple donations – iron preparation (FE2+, FE3+, FE2+ with additives, FE3+ with additives, haem iron combination)											
Deferrals Analysis 15.1	Chi^2 = 6.82, df = 1 (P = 0.009), I^2 = 85.3%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Hb Analysis 15.2	Chi^2 = 12.29, df = 1 (P < 0.001), I^2 = 91.9%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**					**LOW:** 2 or more domains reducing credibility

ICEMAN [55] **question/domain**
1: Is the analysis of effect modification based on comparison within rather than between trials?
2: For within-trial comparisons, is the effect modification similar from trial to trial?
3: For between-trial comparisons, is the number of trials large?
4: Was the direction of effect modification correctly hypothesised a priori?
5: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification?
6: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?
7: Did the authors use a random-effects model?
8: If the effect modifier is a continuous variable, were arbitrary cut points avoided?
9: Optional: Are there any additional considerations that may increase or decrease credibility?
10: How would you rate the overall credibility of the proposed effect modification?
Abbreviations: Hb: haemoglobin; IDA: iron deficiency anaemia; MCV: mean cell volume; NAID: non-anaemic iron deficiency; N/A: not applicable; RCT: randomised controlled trial; TIBC: total iron binding capacity

Table 11. Overview of results: Comparison 2. Intravenous iron versus none (subgrouping: donor characteristics)

Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Before/at 1st donation since treatment							
Deferrals							
Hb (g/L) *$ - iron status*	MD 8.02 [3.22, 12.83]** 3 RCTs, N = 558 ###○	MD 14.2 [4.84, 23.56]** 1 RCT, N = 25	MD 8.04 [−0.96, 17.04] 2 RCTs, N = 128	MD 5.5 [3.27, 7.73]** 1 RCT, N = 405	MD 12.9 [6.96, 18.84]** 1 RCT, N = 74	MD 5.04 [3.11, 6.97]** 2 RCTs, N = 484	
Serum/plasma ferritin (ng/mL) *$- sex* *$ -iron status*	MD 78.16 [37.20, 119.12]** 2 RCTs, N = 479 #○○○	MD 42.38 [24.63, 60.13]** 1 RCT, N = 25	MD 76.70 [57.37, 96.03]** 1 RCT, N = 49	MD 114.00 [102.70, 125.30]** 1 RCT, N = 405	MD 65.49 [50.57, 80.41]** 1 RCT, N = 74	MD 114.00 [102.70, 125.30]** 1 RCT, N = 405	
Serum/plasma ferritin (geometric mean) (ng/mL)							
MCV (fl)	MD 5.81 [3.66, 7.95]** 1 RCT, N = 74	MD 7.00 [3.23, 10.77]**	MD 5.23 [2.62, 7.84]** 1 RCT, N = 49		MD 5.98 [3.57, 8.39]** 1 RCT, N = 74		

Table 11. Overview of results: Comparison 2. Intravenous iron versus none (subgrouping: donor characteristics) *(Continued)*

		1 RCT, N = 25					
Serum/plasma iron (µg/dL)	**MD 24.30 [11.55, 37.05]**** **2 RCTs, N = 149**	MD 25.96 [−4.58, 56.50] 1 RCT, N = 25	**MD 23.95 [9.92, 37.99]**** **2 RCTs, N = 124**		**MD 25.65 [8.92, 42.38]**** **1 RCT, N = 74**	**MD 22.36 [3.01, 41.71]**** **1 RCT, N = 75**	
TIBC (µg/dL)	**MD 25.79 [8.84, 42.75]^^** **1 RCT, N = 74**	MD 25.96 [−4.58, 56.50] 1 RCT, N = 25	**MD 25.72 [5.33, 46.11]^^** **1 RCT, N = 49**		**MD 25.65 [8.92, 42.38]^^** **1 RCT, N = 74**		
Transferrin saturation (%)	**MD 12.21 [8.70, 15.72]**** **2 RCTs, N = 149**	**MD 12.91 [3.48, 22.34]**** **1 RCT, N = 25**	**MD 12.15 [7.70, 16.60]**** **2 RCTs, N = 124**		**MD 13.97 [9.31, 18.63]**** **1 RCT, N = 74**	**MD 10.00 [5.12, 14.88]**** **1 RCT, N = 75**	
Adverse events	RR 0.85 [0.17, 4.14] 2 RCTs, N = 484 ⊕○○○			RR 0.85 [0.17, 4.14] 2 RCTs, N = 484	RR 0.26 [0.03, 2.19] 1 RCT, N = 79	RR 1.46 [0.94, 2.26] 1 RCT, N = 405	
Adherence (n/N)							
Before/at 2nd donation since treatment							
Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Deferrals							
Hb (g/L)	**MD 13.9 [9.5, 18.3]**** **1 RCT, N = 74** **####**		**MD 13.9 [9.5, 18.3]**** **1 RCT, N = 74**			**MD 13.9 [9.5, 18.3]**** **1 RCT, N = 74**	
Serum/plasma ferritin (ng/mL)							

Table 11. Overview of results: Comparison 2. Intravenous iron versus none (subgrouping: donor characteristics) *(Continued)*

Serum/plasma ferritin (geometric mean) (ng/mL)			
MCV (fl)			
Serum/plasma iron (µg/dL)	**MD 36.81 [18.49, 55.13]**** **1 RCT, N = 73**	**MD 36.81 [18.49, 55.13]**** **1 RCT, N = 73**	**MD 36.81 [18.49, 55.13]**** **1 RCT, N = 73**
TIBC (µg/dL)			
Transferrin saturation (%)	**MD 11.30 [6.21, 16.39]**** **1 RCT, N = 73**	**MD 11.30 [6.21, 16.39]**** **1 RCT, N = 73**	**MD 11.30 [6.21, 16.39]**** **1 RCT, N = 73**
Adverse events	RR 0.90 [0.69, 1.18] 1 RCT, N = 82 ⊕⊕◯◯	RR 0.90 [0.69, 1.18] 1 RCT, N = 82	RR 0.90 [0.69, 1.18] 1 RCT, N = 82
Adherence (n/N)			

Abbreviations: CI: confidence interval; Hb: haemoglobin; MCV: mean cell volume; MD: mean difference; N: number of participants contributing to the analysis; NR: not reported; RCT: randomised controlled trial; RR: risk ratio; TIBC: total iron binding capacity
Certainty of the evidence: ⊕⊕⊕⊕ high; ⊕⊕⊕◯ moderate; ⊕⊕◯◯ low; ⊕◯◯◯ very low
Bolded cells indicate a difference between intravenous (IV) iron group and no iron group. In each case: **favours IV iron, and ^^no iron.
Blank cells are where there were no data available for this comparison, time point, outcome, and subgroup/characteristic.
$ indicates that the test for subgroup differences reached P < 0.1, and was investigated for credibility using the ICEMAN assessment tool (see Table 12).

Table 12. ICEMAN subgroup credibility assessment: intravenous iron versus none

Outcome	Test for subgroup differences	1: Within or between studies?	2: Effect modification within trials	3: Number of trials	4: Direction hypothesised a priori	5: Chance an explanation	6: Only a small number of modifiers tested	7: Random-effects model used	8: Arbitrary cutpoints avoided	9: Other considerations	**10:** Overall subgroup credibility
IV versus none @ 1st donation – sex (male and female, ignoring mixed-sex data)											

Table 12. ICEMAN subgroup credibility assessment: intravenous iron versus none *(Continued)*

Serum ferritin (non-log data only) Analysis 16.2	Chi^2 = 6.57, df = 1 (P = 0.01), I^2 = 84.8%	Mostly between studies (some within) **LOW**	**Unclear:** only 1 within-RCT comparison **LOW**	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**N/A** – not a continuous effect modifier	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
IV versus none @ 1st donation – baseline iron status (healthy, low Hb/ferritin, IDA, or NAID, ignoring mixed data)											
Hb Analysis 18.1	Chi^2 = 6.08, df = 1 (P = 0.01), I^2 = 83.6%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance may not explain** (P > 0.005 to </= 0.01) **MODERATE**	**Definitely no** (exploratory analysis) **VERY LOW**	Yes, random-effects model used **HIGH**	**Probably yes** (pre-specified by each study author) **MODERATE**	**N/A** – none noted	**LOW:** 2 or more domains reducing credibility
Serum ferritin (non-log data only) Analysis 18.2	Chi^2 = 25.80, df = 1 (P < 0.001), I^2 = 96.1%	Completely between studies **VERY LOW**	**N/A:** no within-RCT comparison	**Very small** (1 to 2 in smallest subgroup) **VERY LOW**	**Definitely no** (post hoc only) **VERY LOW**	**Chance unlikely** (P </= 0.005) **HIGH**					**LOW:** 2 or more domains reducing credibility

ICEMAN [55] **question/domain**
1: Is the analysis of effect modification based on comparison within rather than between trials?
2: For within-trial comparisons, is the effect modification similar from trial to trial?
3: For between-trial comparisons, is the number of trials large?
4: Was the direction of effect modification correctly hypothesised a priori?
5: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification?
6: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?
7: Did the authors use a random-effects model?
8: If the effect modifier is a continuous variable, were arbitrary cut points avoided?
9 Optional: Are there any additional considerations that may increase or decrease credibility?
10: How would you rate the overall credibility of the proposed effect modification?
Abbreviations: Hb: haemoglobin; IDA: iron deficiency anaemia; IV: intravenous; MCV: mean cell volume; NAID: non-anaemic iron deficiency; N/A: not applicable; RCT: randomised controlled trial; TIBC: total iron binding capacity

Table 13. Intravenous iron versus no iron: quality of life and physical function data as presented by each study

Assessment time point*		Assessment tool range of scores, direction of tool	Iron Mean (SD) N	No iron Mean (SD) N	Reported difference (95% CI) *as reported by trialists, unless otherwise stated in Notes*	Notes
Patient-reported fatigue scores						
Fontana 2014a (ISUB)	1	Fatigue level 0 to 10; *lower is better*	3.9 (1.8) N = 203	4 (2.2) N = 202	MD 0.0 (−0.4 to 0.3)	
Hod 2022 (DIDS)	1	Global fatigue index 1 to 50; *lower is better*	N = 33	N = 35	MD −0.2 (−5 to 4.7)	
Gybel-Brask 2018	1	Fatigue Visual Numeric Scale 0 to 10; *lower is better*	LS mean: −1.0028 N = 36	LS mean: −0.1400 N = 35	Difference of LS means −0.8627 (−1.8137 to 0.08819)	Change from baseline in fatigue symptoms scores
Gybel-Brask 2018	2	Fatigue Visual Numeric Scale 0 to 10; *lower is better*	LS mean: −0.8307 N = 32	LS mean: −0.5367 N = 25	Difference of LS means −0.2940 (−1.4167 to 0.8287)	Change from baseline in fatigue symptoms scores
Fontana 2014a (ISUB)	1	Jenkins sleep scale 0 to 20; *lower is better*	8.5 (3.3) N = 203	8.3 (2.9) N = 202	MD −0.2 (−0.6 to 0.3)	
Patient-reported physical health						
Fontana 2014a (ISUB)	1	EQ-5D index 0 to 10; *higher is better*	9.5 (1.1) N = 203	9.6 (1.1) N = 202	MD −0.1 (−0.3 to 0.1)	
Fontana 2014a (ISUB)	1	EQ-5D VAS 0 to 10; *higher is better*	8.5 (1.1) N = 203	8.4 (1.1) N = 202	MD 0.1 (−0.1 to 0.3)	
Patient-reported mental health						
Hod 2022 (DIDS)	1	Beck anxiety (BAI) 0 to 63; *lower is better*	N = 33	N = 35	MD −2.2 (−5.7 to 1.3)	
Hod 2022 (DIDS)	1	Beck depression (BDI-II) 0 to 63; *lower is better*	N = 33	N = 35	MD −0.0 (−3.4 to 3.3)	
Fontana 2014a (ISUB)	1	SCL27 mean of 0 to 4 per domain; *lower is better*	1.2 (0.0) N = 203	1.2 (0.4) N = 202	MD 0 (−0 to 0)	
Pain and general well-being						

Table 13. Intravenous iron versus no iron: quality of life and physical function data as presented by each study *(Continued)*

Hod 2022 (DIDS)	1	SF36 pain score 0% to 100%; *higher is better*	N = 33	N = 35	MD 5.4 (−2.1 to 12.9)
Hod 2022 (DIDS)	1	SF36 general health 0% to 100%; *higher is better*	N = 33	N = 35	MD 3.3 (−5.1 to 11.6)
Hod 2022 (DIDS)	1	RLS rating scale 0 to 40; *lower is better*	N = 33	N = 35	MD −0.4 (−1.6 to 0.9)

Abbreviations: CI: confidence interval; LS mean: least squares mean; MD: mean difference; RLS: restless leg syndrome; RR: risk ratio; SCL27: Symptom Checklist-27; SD: standard deviation; SF36: short form-36; VAS: visual analogue scale
***Assessment time point:**
1: before/at first donation since treatment commenced
2: before/at 2nd donation since treatment commenced
3: after multiple donations

Table 14. Overview of results: Comparison 3. Oral iron versus intravenous iron (subgrouping: donor characteristics)

Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Before/at 1st donation since treatment							
Deferrals							
Hb (g/L)							
Serum/plasma ferritin (ng/mL)	MD −1.18 [−9.12, 6.77] 1 RCT, N = 120 ⊕⊕⊕◯	MD 1.40 [−12.25, 15.05] 1 RCT, N = 50	MD −2.50 [−12.27, 7.27] 1 RCT, N = 70		MD −0.80 [−8.87, 7.27] 1 RCT, N = 120		
Serum/plasma ferritin (geometric mean) (ng/mL)							
MCV (fl)	MD −0.30 [−1.43, 0.83] 1 RCT, N = 143			MD −0.30 [−1.43, 0.83] 1 RCT, N = 143		MD −0.30 [−1.43, 0.83] 1 RCT, N = 143	
Serum/plasma iron (µg/dL)							
TIBC (µg/dL)							
Transferrin saturation (%)							
Adverse events	RR 1.00 [0.68, 1.46] 1 RCT, N = 172 ⊕◯◯◯			RR 1.00 [0.68, 1.46] 1 RCT, N = 172		RR 1.00 [0.68, 1.46] 1 RCT, N = 172	
Adherence (n/N)	**RR 0.86 [0.79, 0.94]^^** **1 RCT, N = 172** **##**◯◯			**RR 0.86 [0.79, 0.94]^^** **1 RCT, N = 172**		**RR 0.86 [0.79, 0.94]^^** **1 RCT, N = 172**	
Before/at 2nd donation since treatment							

Table 14. Overview of results: Comparison 3. Oral iron versus intravenous iron (subgrouping: donor characteristics) *(Continued)*

Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Deferrals							
Hb (g/L)							
Serum/plasma ferritin (ng/mL)	MD −6.62 [−13.57, 0.33] 1 RCT, N = 120 ⊕⊕⊕◯	MD −6.10 [−17.82, 5.62] 1 RCT, N = 50	MD −6.90 [−15.53, 1.73] 1 RCT, N = 70		MD −6.50 [−13.49, 0.49] 1 RCT, N = 120		
Serum/plasma ferritin (geometric mean) (ng/mL)							
MCV (fl)							
Serum/plasma iron (µg/dL)							
TIBC (µg/dL)							
Transferrin saturation (%)							
Adverse events							
Adherence (n/N)							
After multiple donations since treatment							
Outcome	Total (MD/RR) [95% CI]	Subgroup: sex			Subgroup: baseline iron status		
		Male	Female	Mixed/NR	Healthy	Low	NR/no restriction
Deferrals							
Hb (g/L)							

Table 14. Overview of results: Comparison 3. Oral iron versus intravenous iron (subgrouping: donor characteristics) *(Continued)*

Serum/plasma ferritin (ng/mL)	**MD −7.89 [−15.17, −0.62]^^** **1 RCT, N = 120** ###○	MD −3.40 [−15.13, 8.33] 1 RCT, N = 50	**MD −10.70 [−19.97, −1.43]^^** **1 RCT, N = 70**		MD −7.20 [−15.02, 0.62] 1 RCT, N = 120
Serum/plasma ferritin (geometric mean) (ng/mL)					
MCV (fl)					
Serum/plasma iron (µg/dL)					
TIBC (µg/dL)					
Transferrin saturation (%)					
Adverse events					
Adherence (n/N)					

Abbreviations: CI: confidence interval; Hb: haemoglobin; MCV: mean cell volume; MD: mean difference; N: number of participants contributing to the analysis; NR: not reported; RCT: randomised controlled trial; RR: risk ratio; TIBC: total iron binding capacity
Certainty of the evidence: ⊕⊕⊕⊕ high; ⊕⊕⊕◯ moderate; ⊕⊕◯◯ low; ⊕◯◯◯ very low
Bolded cells indicate a difference between oral iron group and intravenous (IV) iron group. In each case: **favours oral iron, and ^^favours IV iron.
Blank cells are where there were no data available for this comparison, time point, outcome, and subgroup/characteristic.
$ indicates that the test for subgroup differences reached P < 0.1, and was investigated for credibility using the ICEMAN assessment tool (none for this comparison).